

Genealogical arithmetic of prime numbers and spectral structure of the Riemann zeta function

Arithmetic transport, spectral moments, and Weil's positivity criterion

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Abstract

An equivalent positivity criterion for Weil's arithmetic form on each fixed window is proved through a finite-dimensional Schur complement that retains the effect of the infinite-dimensional detail space. Its application establishes $\mathcal{W}(f, f) \geq \frac{1}{2} \|f\|_2^2$ on the joint readout domain supported in $(-1/18, 1/18)$, and on its translations and interior refinements. A readout recovering the test function from the gamma tail and reversible transports preserving the complete form, including its prime, gamma and polar contributions, are also constructed. The bound is transmitted to entire collections of test functions with exponential tails.

These results are developed from a formal mathematical core called Triadic Modular Holography, henceforth HMT. *All Possible Paths* (APP) is the arithmetic path structure on the toroidal support $(\mathbb{Z}/9\mathbb{Z})^2$, whose cardinal translations form an additive Cayley graph; its sheets preserve sum and product with residue and quotient. TRIT is the oriented signature $\{-1, 0, +1\}$ distinguishing local regimes. The *Topological Phase Kernel* (TPK) composes selection, transport, return and memory update on the enriched state. The arithmetic and spectral realizations use the joint discrete structure of the continuum produced by this development.

The native product on nonadic words has a unique normal form; its multiplicative decompositions select the irreducibles that evaluation identifies with prime numbers. Occupations of these modes generate the Euler product and prime-power moments. The limit of cyclic heat traces and the Mellin transform produce the completed function and its functional equation. Reconstruction of the APP interior from the lifted boundary, continuation sheets and vacancies specifies the information preserved by refinement.

The double circle relates the spectral coordinate to return memory. Nonadic compression determines the 8/81 balance and the central modes determine the ratio 5/9, whose moments admit a positive factorization in every dimension. For the arithmetic form, control of details and cross terms provides the Schur reduction and the stated coercivity. Weil's global criterion is formulated through joint identification of the arithmetic moments with a positive Gram form, including all mixed moments and coverage of the test-function domain. This global identification remains a hypothesis of the spectral results using it: the coercivity proofs established here do not constitute a proof of the Riemann hypothesis.

Keywords: Weil form; coercivity; Schur complement; prime numbers; Riemann zeta function; Triadic Modular Holography; native product; boundary reconstruction; memory; mixed moments.

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1 Introduction

The sign problem for the Weil form brings together two requirements: preservation of all its arithmetic contributions and control of the form on complete function spaces. The global criterion relates its positivity to the Riemann hypothesis. A spectral construction seeking to apply that criterion must therefore specify the product, weights and domains determining the form and prove how its cross terms are transmitted. This article develops that relation from an arithmetic of paths with memory and obtains exact reconstruction, transport and coercivity results.

The principal functional contribution is a Schur reduction for each fixed window. A sufficiently fine partition separates cell averages from an infinite-dimensional space of zero-mean details whose form is coercive. Exact elimination of these details preserves their entire coupling with the averages and produces a finite matrix with the same negative index as the complete form. The result supplies an exact criterion, not a replacement of the function space by samples. Its application on $(-1/18, 1/18)$ proves the lower bound $\frac{1}{2}\|f\|_2^2$ on the entire joint readout domain. A reversible transport extends this bound to complete collections of test functions with exponential tails.

Arithmetic construction and preserved structure

The starting point is a formal mathematical core called Triadic Modular Holography, henceforth HMT. Its object *All Possible Paths* (APP) combines paths and arithmetic evaluations on the support $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The cardinal generators $\{(1, 0), (-1, 0), (0, 1), (0, -1)\}$ define the Cayley graph of its additive group, a two-dimensional cyclic lattice with toroidal realization. Sum and product evaluations act on the same positions, each with its residue and quotient. The support specifies where a route moves; the arithmetic sheets specify what is computed and preserved during that movement.

TRIT is the oriented ternary signature $\{-1, 0, +1\}$, obtained through balanced representatives modulo 3. It distinguishes the hyperbolic, parabolic and elliptic types in its quadratic realization. The *Topological Phase Kernel*, henceforth TPK, composes phase selection, transport and memory update. Its state preserves positions, sheets, orientation, quotients, prefixes and boundary. Nonadic prolongation transports this information between depths: phase return advances history instead of restarting it.

The joint discrete structure of the continuum is developed on these same states and their compatibilities. Its correlative constructions preserve restrictions, incidences, forms and memory within a single refinement system. The Archimedean readout obtains values from compatible cylinders, and the boundary readout retains the incidences used by subsequent realizations. The present arithmetic and spectral problem preserves this common origin, with the operations and domains actually entering each result.

The center–boundary relation provides a first instance of reconstruction. In the multiaffine APP chart, the lifted boundary determines the interior; residual values are completed by the quotients distinguishing their integer values. Continuation signatures require the sheets to be preserved, and the vacancy calendar retains the discrepancy between capacity and depth. Section 10 places the center (5, 5), the regional signature 555555, the readouts 72, 27, 77, 22 and the intervals 21/22 and 6/7 within this construction. Their role specifies transportable information before any spectral identification.

From the product to the completed function

The native product composes the two APP sheets through a transducer of digits and quotients. Its evaluation invariant proves exactness of the operation on nonadic words, and uniqueness of normal form subsequently identifies integer multiplication. Product fibres determine its decompositions; the absence of a nontrivial decomposition selects the irreducibles. Theorem 11.6 identifies their evaluations with positive primes.

Factorization and arithmetic clocks describe different properties of these objects. The former determines their multiplicative components; the latter records returns of residual orbits. The mode-occupation representation preserves the multiplicity of every irreducible and leads to the Euler product. The trace over all occupations and the trace over individual modes are related by the prime–power moments of Proposition 12.3.

Cyclic refinement subsequently supplies a heat trace whose limit is the theta series. The Mellin transform produces $\pi^{-s/2}\Gamma(s/2)Z(s)$; scale inversion determines the polar contribution and the functional equation of the completed form. Propositions 15.1 and 15.2 preserve the normalization and convergence of this step. The finite-part and residual-readout developments extend its evaluations with controlled remainders. The Weil form jointly preserves the prime, gamma and polar terms arising from this composition.

Double circle, reconstruction and transport

The double circle distinguishes the spectral coordinate from winding memory. The nine-phase decomposition produces an exact balance with coefficient $8/81$; the central-block modes determine the ratio $q = 5/9$. The memory moments $q^{|n|}$ have a positive triangular factorization in every dimension. The loss register additionally preserves the terminal term and cross products, so visible contraction can be accompanied by isometric preservation of the recorded history.

The analytic realization constructs a reversible operator C_a on exponentially decreasing test functions, with $a > b > 1/2$. Its integral action simultaneously preserves correlations and polar cross products. The identity in Subsection 17.11 proves

$$\mathcal{W}(C_a f, C_a g) = \mathcal{W}(f, g).$$

The margin $a > 1/2$ belongs to the domain of this analytic readout and ensures convergence of its polar contributions. The regular memory circle and a representation receiving an atomic limit retain their respective domains.

The gamma tail contains an explicit inverse readout: it recovers the test function and reconstructs the negative column from the positive one. Equations (18.30) and (18.33) describe its action, and (18.35) preserves the complete difference of forms. This reconstruction provides the operator to be studied; boundedness and contractivity are different properties, whose estimates are developed next.

Coercivity and criterion for global extension

Theorem 19.2 proves that, once a positive bound for the detail block D has been obtained, positivity of the form on V_R is equivalent to that of $S = A - B^* D^{-1} B$. The correction retains the effect of all details and their cross terms with the averages. Theorem 19.3 bounds this matrix through residuals and an operator error estimate. On the window of length $1/9$, the estimates $D > I$, $A > 97/100$ and $\|B\|^2 < 1/5$ yield Theorem 19.4. The local window satisfies $1/9 < \log 2$; its transported collections may have tails and then receive the complete prime sum through the proved invariance.

The global criterion also requires cross terms between distinct collections and coverage of the entire test-function domain. Its precise formulation is given in Subsections 17.12.2, 17.13 and 18.8: identification of the complete arithmetic moments with a positive Gram form amounts to joint preservation of energy and transport, with their domains and limits. This condition remains a hypothesis of the global results; proved coercivity belongs to the explicit domains above. The article's contribution is to connect arithmetic generation to reconstruction identities and functional estimates precise enough to establish that scope and its exact condition for extension.

Map of results and dependencies

The argument contains five levels that should be distinguished when reading. The arithmetic construction supplies the modes; their readouts define the complete form; the estimates establish positivity on precise domains; the Weil criterion determines which additional identification permits RH to be concluded. The order of presentation makes the principal result visible, while the body preserves the order of its dependencies.

Level	Result and role in the argument	Location
Common construction	Histories, sheets, memory, and incidences remain within a single refinement system. The center, lifted boundary, and vacancies specify the information transmitted by each readout.	Sections 9 and 10.
Arithmetic and functional	Native product, normal forms, and irreducibles; occupations, Euler product, prime–power moments, finite parts, and theta–Mellin continuation.	Sections 11–13; beginning of 15.
Reconstruction and transport	The columns jointly preserve the prime, gamma, and polar terms. The gamma tail reconstructs the test function. Compression, memory, and Cayley preserve their balances and cross products.	Sections 15, 17, and 18.
Sign of the form	Coercivity of the specified assembly, control of details, and Schur reduction. The complete window of length $1/9$ satisfies $\mathcal{W}(f, f) \geq \frac{1}{2} \ f\ _2^2$.	Section 15 and Theorems 19.2, 19.4.
Global criterion	Identification of the complete moments, with all their cross terms and coverage of the test domain, would permit application of the Weil criterion. It is not replaced by the preceding local bounds.	Section 15; subsections 17.5 and 17.13.

Two distinctions accompany this map. In arithmetic, the period of a residual action does not replace irreducibility of the product. In analysis, a positive factorization of the memory moments is not automatically identified with that of the arithmetic moments. The constants $8/81$ and $5/9$ belong, respectively, to the compression balance and the modal memory ratio; their proofs preserve these distinct roles.

The center $(5, 5)$, the regional signature 555555 , its three continuation sheets, and the schedules $21/22$ and $6/7$ are explained in their constructive positions. The readouts $72, 27, 77, 22$, the lifted boundary, and the completion $729 + 271$ are not grouped by numerical coincidence: each preserves the operation and domain that produces it. Their role in this article is to specify the transportable information before passing to the spectral problem.

Composition of readouts also preserves two distinct operations: the diagonal product of residual tables and the tensor product of their moments, which produces divisor convolution. Stieltjes transport includes the affine contribution of the pole. The Bendersky–Glaisher hierarchy relates regularized derivatives at negative even levels to positive odd values of the same functional. Enclosures from irreducible-mode cutoffs complement the depth bounds, with their respective errors explicit (Sections 12.6, 14 and 16).

The definitions and proofs used are brought together in the indicated sections and Appendices A–C, with the domain of each result stated explicitly. The local proofs, functional results and conditional statements retain their own scope here.

2 Formal core of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information required to prolong them. Only subsequently are numerical realizations formed. A constant recognized at the end of this process does not replace either the rule that produced its region or the history that determines its evaluation.

The designation *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is the Feynman path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory updating, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks of a line graduated between zero and ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must not be confused with the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not retain the integer prior to reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i a total of j times produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves the two sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

The term *sheet* here denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:

$$S(i, j) = i + j, P(i, j) = ij.$$

Each row is generated by accumulation:

$$P(i, j + 1) - P(i, j) = i.$$

The chart represents $0 \pmod{9}$ by 9.

The marked block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix},$$

with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Nonunit rows and columns are distinguished by grey shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representational edges do not constitute an intrinsic topological boundary of the torus.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone determine neither these integer evaluations nor their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, the positions (5, 5) and (8, 2) give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residual projection identifies data that the lifted evaluation distinguishes. $\square$

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southward and the second eastward. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement prior to reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Return to the same residual position therefore does not require the winding to vanish. This is the first instance of a distinction that will recur in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of the two evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Suppressing it would change the arithmetic being transported, even if the residual table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readouts: its direct reduction modulo three is 0, 0, 0, whereas extraction of the factor three followed by reduction of its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as direct reduction of the original number.

2.1.4 Latin realization of the additive evaluation

The distinction between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in each row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a+b$ permutes the nine residues. This yields a rank-one-projector realization of a Latin square, in the terminology of quantum Latin squares [1, 3]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different information: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Accordingly, the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[2]. This reference provides precise terminology for comparing APP realizations.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1), (0, 0), (0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A residual zero does not mean the absence of information about orientation, phase, or quotient. Nor does it by itself introduce a physical quantity. Its function is to distinguish algebraically the transport classes to be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + tJ_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, every term of degree at least two vanishes. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau) \exp(-tJ_\tau) = I$. The same algebraic principle preserves unity across realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors, and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction occur jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parameterization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction makes it possible to express an integer APP evaluation in a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the writing of an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes explicit the compatibility of that rule with sign reversal.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 lying in the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it constitutes its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values $+1, -1, 0$. The TPK selector defined next uses these classes to determine the operational mode. Reading an evaluation and selecting a mode thus have the same ternary reduction, but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define*

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity guarantees that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component removes that accounting: $1 + 1 = -1 + 3$, whereas the residue readout retains only -1 . The discarded information is determined, in this case, by $\chi(1, 1) = 1$.

The relation to modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well-defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3, and then $\bar{c}$. The addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels are not composed by independent coordinate addition: normalizing the first level modifies the second.

In particular, the classes 3, 6, and 0 in $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially turned into a single position. The modulus 1000, used later to organize decimal blocks, serves another positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas agreement of residues in different bases does not by itself identify their transport states.

2.2.3 Reversal, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum magnitude 1, they are the three possibilities in the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, mode sequence, or winding. Thus the decomposition $\mathcal{B}$ is integrated into the transport state instead of replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reverse order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while maintaining the threshold 0. Transport of the remaining trajectory data is specified through the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed next, acts within the type already fixed. The distinction prevents attributing to sign reversal an algebraic equivalence between different regimes.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{zw} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

All three types can be faithfully realized by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. Thus the complex structure, the dual-number structure, and the split-real structure are represented by the same matrix expression. Each preserves its law and zero divisors. The form of signature $(1, 1)$ in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau) \exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution in the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the norm-one circle. In the hyperbolic type, the eigen-coordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotence eliminates terms of order at least two, but preserves the linear term transporting the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with the absence of a state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component is nonzero, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the complete algebra; a coordinate singularity does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional readout. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus the signature change organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, domains, and effective operator acting at each transition.

In the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution squaring to I and two nilpotent directions. The joint realization will be proved on that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear in the same matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. The relation equips the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its link to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each preserves its characteristic, dimension, and incidence information. The development of the respective operators establishes the link, not equality of their cardinalities.

Preservation of norm one therefore describes the quadratic realizations of evolutions. Preservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties enter the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations giving rise to them.

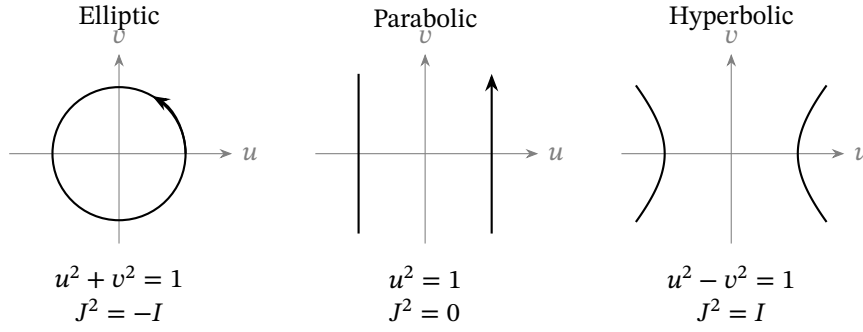


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. The parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation acts. Cardinal transport displaces the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore distinct functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object with a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection by M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields through Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The next section fully specifies its finite two-cursor realization. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and dependence of the operations

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content by specifying which coordinates each of its factors receives and modifies. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before its reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_\times, d_\times, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary registers. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may preserve different registers.

The dependence of the operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and the update calculates the new arithmetic coordinates. At the trajectory level, these transitions are composed and preserve the registers of earlier steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of its predecessor. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Trit selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operational phase and dodecaphasic sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)\nu_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)\nu_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event is part of the chronology and its register. Cardinal directions determine the vectors ν_d , while the phase signature determines which of those vectors is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + a\nu_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + a\nu_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records earlier crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + a\nu_d.$$

The integer transport datum remains reconstructible after reducing the position on the torus.

Proposition 2.12 (Preservation of lifted displacement). *For a cardinal trajectory of n steps, whose activation indicators are a_t , we have*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible trajectories are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Dividing the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the winding number defined earlier. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its register simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation maintains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After (2.15) is applied, both directions are reversed whenever a block of 27 steps ends. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The readout forming the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission are therefore composed in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. In a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, one first recovers the previous phase by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The next block preserves the activations and reverses the directions, so the integer displacements cancel in 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory register, its evaluations, and new prefixes are preserved in the dynamic state on which $\mathcal{U}_t$ acts; the observable equality does not reset them.

2.3.4 Affine memory transport and state composition

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q and S_q denote, respectively, the sets of residue–quotient data, the abelian group of transport memory, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified with their components in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue-and-quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate truncation maps and the cylinders developed in 4 and 5.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed by maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

The symbol k_{tr} here denotes the transport cocycle; the rational coordinate κ of the register K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the previous memory, and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$, and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating (2.19) by γ_1 , followed by updating by γ_2 , coincides with updating by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves both arithmetic transport and boundary prolongability. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphasic register, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The register preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12},$$

where τ_m is the window subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence register. We adopt the chart to be developed in 8.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal registers. The composition of segments and their increments is governed by (2.19). The group C_{12} , by contrast, describes translation of the window index; Θ_{108} preserves the step calendar. This distinction makes it possible to specify what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancement by 9 steps preserves r and increases b by one. Phase addition introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar retains $[b]_{12}$; a winding register retains $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its role is to describe the relation between these coordinates before they are used in constructing the signed state.

The signed register receives phase and memory data through the composition developed in Section 8:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between that signed state and K , as well as the rational readout of K , are proved in the corresponding chapter. The registration operations precede those transformations. Reconstructing a register from its cyclic differences verifies recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_i) preserved by the register, the event readout counts the codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different coordinates of the register. Chapter 8 develops this readout, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data that chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases, and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

The cardinality 271 here has a realization through phase incidence. The cubic completion in Section 4 has its own construction of the shell $1000 - 729$. The correspondence between both realizations must preserve the maps and markings that identify them, in addition to cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories, before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The origin of the reverse trajectory is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities are obtained by reversing the order of steps and replacing each cardinal vector by its opposite. For a loop it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On registers, reversal also requires transporting the order of their components and the orientation of their events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete register. In particular, the regional involution interchanging the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 5.8. This regional operation acts on a different domain from reversal of a cardinal trajectory.

Extensions over a route $r : q \rightarrow q'$ are expressed through relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , one requires

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

One phase winding composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \cdots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain one retains $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed below. The cyclic base has nine phases, whereas the fiber preserves the trajectory, quotients, and boundary. Monodromy is the action of one winding of that base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization by the effective TPK transitions.

If g observes the phase and M the number of retained windings,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure, and the second identifies the new temporal depth. Coinductive prolongation additionally preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its predecessors without replacing them by the last observation. The development of this arithmetic of histories and its unit occupies 4.5–4.7; the construction of regions and their readouts is carried out in Section 5.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, with cardinalities 18 and 26. Their product identifies the catalogue $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = R_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. Transport of histories remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the role of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic register determines the data entering K and the closure of α . The numerical and incidence realizations receive that prior structure. This order makes it possible to present each consequence in its chapter while preserving, from the formal kernel, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not eliminate the increments produced during the traversal. To express this difference in operator terms, an accumulative register of trajectory events is constructed first, followed by its generating series. Memory variables will be reduced on that transport, while preserving the source and output readout as well.

2.4.1 Oriented events and register update

Let (d_t) be the sequence of integer direction codes preserved by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification between their alphabets is assumed. Number the windows from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphasic register. Its orientation is $\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1)$. For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign preserved in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts the events in the trace; it does not choose directions through a value to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$, and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated register in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at another origin, z_0 preserves the previous register. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The register in the moving frame satisfies*

$$y_{m+1} = R y_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1} y_m + a_m S^{-1} \mathbf{e}_0$, because $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, proving (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^T D_3 + D_4^T D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^T B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the register; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Balance produced by the event). *The update (2.24) satisfies*

$$\begin{aligned}\mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m(B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m.\end{aligned}\tag{2.27}$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The sum of coordinates is invariant under the permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform control of memory

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)).\tag{2.28}$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right).\tag{2.29}$$

The derivative series converges uniformly on each disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with the variable of another readout requires preserving that clock and the map relating the two readouts.

2.4.3 Elimination of memory: operator, source, and readout

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and readout). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete readout satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C)\mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1 w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting into the first gives (2.30); applying ℓ_0 and ℓ_1 to both components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2 B_1 D_u C = \sum_{k \geq 0} u^{k+2} B_1 D^k C$ describes an exit into memory, k steps within it, and a return. By contrast, $u\ell_1 D_u C$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: source and readout must also be transformed.

The difference is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n \mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example corresponds to the twelve-window register, not to the entire TPK state.

2.4.4 Composition of segments and coherence of reduction

Let there be three composable affine transports $F_i(y) = R_i y + b_i$, with compatible domains and codomains. The memory produced by the complete traversal is

$$F_3 F_2 F_1(y) = R_3 R_2 R_1 y + b_3 + R_3 b_2 + R_3 R_2 b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping either the first two segments or the last two first produces the same expression; the factors transport each increment from the point where it originated. In (2.24), $R_i = R$ and b_i is the event in its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2 Q_2 C_2, & B'_1 &= B_1 + uB_2 Q_2 D_{21}, \\ C'_1 &= C_1 + uD_{12} Q_2 C_2, & D'_{11} &= D_{11} + uD_{12} Q_2 D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$I - uA' - u^2 B'_1 (I - uD'_{11})^{-1} C'_1 = I - uA - u^2 (B_1 \ B_2) (I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting into the first two produces (2.34). Subsequently eliminating the second component gives the left-hand side of (2.35). Solving both memory equations simultaneously gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . In both procedures, source and readout are also transformed according to (2.30)–(2.31). $\square$

The balance (2.27) describes register increments. The preceding identities allow them to be transported when memory is reused in dodecaphasic closure; they do not by themselves determine additional analytic coefficients.

Proof development of the common kernel

The following constructions develop the emission rules, the calendar with memory, compatible sequences, cubic completion, regional readouts and the reversibility of the dodecaphasic register. Their definitions and proofs precede the arithmetic and spectral applications that use them. Fragments from Article I retain their statements, proofs and source locators; the source manifest records the corresponding components, line intervals and hashes.

References to alpha and the exceptional corridor specify the provenance of the objects. Subsequent identifications with codes and lattices do not enter the spectral proofs of this manuscript. References to Article I document that provenance, while the necessary arguments are presented here in full. Finite censuses and reconstructions retain the rules, domains and starting coordinates of each statement; their inclusion does not alter the scope of global positivity stated in the abstract.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t - 1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b - 1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i + 1, j + 1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = [(n - 1)/9], \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{3.9}$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \tag{3.10}$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.5 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left[\left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right]_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_\times = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The resulting word and its signatures are

$$\begin{aligned} \mathbf{U} &= (501, 614, 498, 169, 272, 272), \\ \mathbf{s} &= (5, 6, 4, 1, 2, 2), & \mathbf{e} &= (5, 5, 5, 5, 5, 5). \end{aligned}$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (3.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $\nu = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The support of eighty-one positions does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation; and the additive and multiplicative sheets have independent cursors. This distinction makes it possible to reconstruct the emission exhaustively without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so that

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The independent annex enumerates all these conditions; the main text presents the law transforming them. The rows of an emission table are fibers of this domain of initial conditions.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If it equals $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then moved; if it equals -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then moved. A zero value increments the neutral counter. Upon completion of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. The integer evaluations and their quotients are retained in the record, but the emitter uses the residual readouts specified here.

In the unit sector, the multiplicative readout uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission, it is extended with value zero to the nonunit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative unit. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ of window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus, S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and the calendar of this recurrence are an explicit part of the declared emission law. No values of π, φ, e or α are received. The distinction between an emission rule and a constant recognized from its output prevents the genealogy from being hidden in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written as

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transporting the subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with the TRIT is performed after (4.4).

Proposition 4.1 (Injective factorization of the emission). *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (4.3). Hence the coupling is injective on the product of signatures.

Enumeration of the 324 conditions of one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages; and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalog. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} \underbrace{432}_{18 \cdot 24} \text{ fibers of } 144, & 18 \text{ fibers of } 432, & 18 \text{ fibers of } 1944. \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last figure is the cardinality of the image, not that of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unbounded depth. The domain (4.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Closure of the calendar and advancement of memory

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{l} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, the sum of two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *The sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{mcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring the prefix of a prolongation. The nonadic holonomy Γ_9 denotes transport through one winding on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

4.3 Observable compression and orthogonal conservation

An isometric realization allows what a reduced observation loses to be expressed quantitatively. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one winding, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Conservation over a finite horizon). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The last summand preserves the terminal state. Suppressing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity in compression: the norm is distributed among the visible component, memory, and terminal, without being replaced by a single residue.

4.4 Cubic completion and two notions of boundary

Enlargement of the cubic support from side nine to side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.9)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The enlargement crown contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

4.4.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description that makes its operation explicit. Let E_9 be the set of the nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, enlarging E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without modifying the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.10)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.11)$$

The punctured enlargement contains 270 elements in addition to the initial cube; the full enlargement contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning to each remaining coordinate one of the nine classes of E_9 . Classification by the number of additional coordinates is exhaustive, and its classes are disjoint. The only element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation between the initial body, the enlargement strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure by restoration of the apex. The added unit therefore has an explicit incidence location.

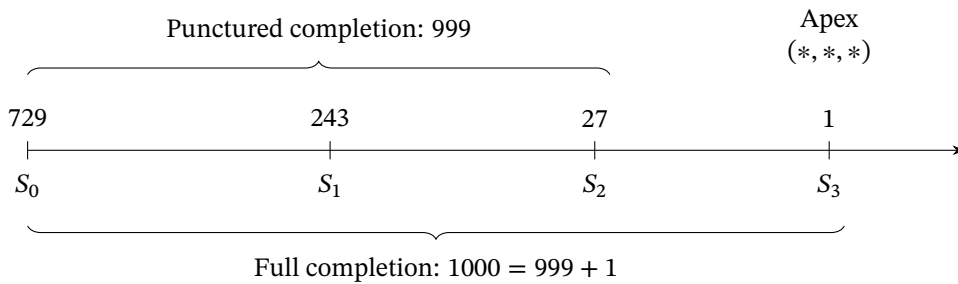


Figure 3: Strata of the cubic completion. The layout represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

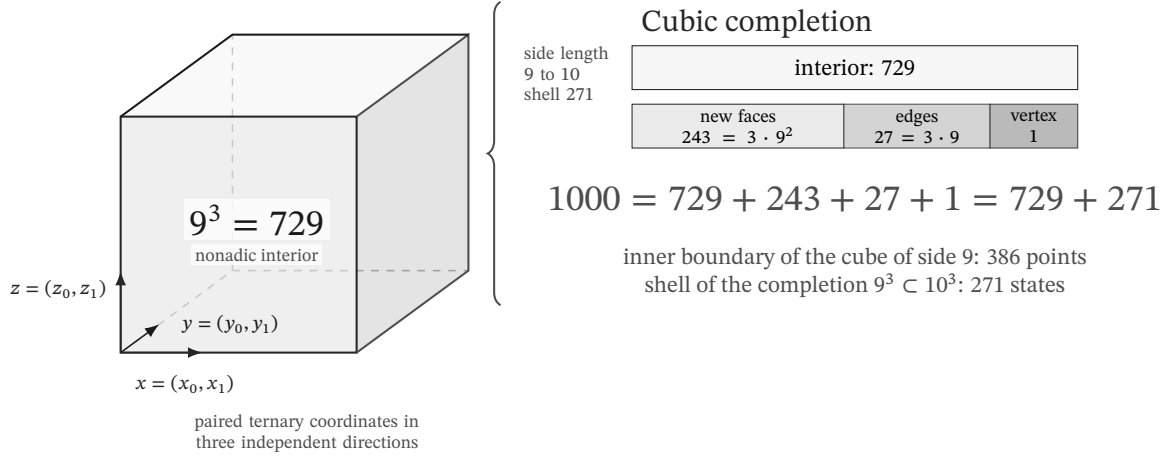


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.4.2 Preservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ removes the last coordinate.

Proposition 4.6 (Projective compatibility of the normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the sum of the masses of all strata is one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.10); the binomial $(9 + 1)^d$ gives the total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define another measure: the distinction is operationally relevant. In particular, preservation of probability mass, preservation of transport registers, and thermodynamic dissipation are statements about different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamical and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its enlargement. The cubic completion, the positional base-thousand expansion, and the exceptional incidence readout share support data; their respective maps determine which information from those data each representation uses.

4.5 Number as a compatible section and value as a readout

The distinction between state and observation makes it possible to specify which object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: preceding arithmetic evaluations and transports impose their relations. A prolongation preserves those relations in each antecedent, even as it adds new steps and new memory data. This is the meaning of the arithmetic of histories developed in (provenance preserved in the technical record).

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be the depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.7 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.12)$$

A readout is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the readout. The pair consisting of the section and its readout constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the readout associates a closed cylinder with each prefix, preserves their nesting, and makes their diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.13)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 5. Definition (4.12) does not by itself assert that the image of (4.13) is the entire line: a coverage assertion requires its own theorem. Nor does equality of values establish, without an injectivity result, equality of their histories. This separation makes it possible to speak of precision as depth while preserving the difference between the arithmetic antecedent and its scalar realization.

4.6 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are their admissible histories. Composition is written $vu = v \circ u$: u is traversed first and then v . The elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the expression $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the state being transported; a transition between regimes is not equivalent to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x is given, possibly noncommutative. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. We assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.14)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. We require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for each composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.15)$$

These are explicit assumptions on the transports and their coefficients, not a consequence of calling the dynamics holonomic. When this presentation is applied to a TPK sector, its ρ_u and $\Omega(v, u)$ must be identified. Transitions expressed through bimodules require that structure to be preserved and are not replaced by (4.14).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

is equipped with the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.16)$$

if the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. The sum brings together histories with coefficients; the product concatenates histories while preserving the record multiplier.

Proposition 4.8 (Sectorial associativity with a central multiplier). *Under (4.14) and (4.15), the product (4.16) is associative. If the set of objects is finite, its unit is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to be moved to the right of that factor; (4.15) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero by their initial and terminal states. Bilinearity completes the proof. The local identities and the normalization of Ω verify the indicated unit. $\square$

The calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

Both carry sums of a triple equal $\lfloor (a+b+c)/9 \rfloor$; hence (4.15) holds. The formal variable z records the winding without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of the calendar (4.5); retaining z without this relation distinguishes successive windings. The example realizes the memory of this calendar, without identifying it with the entirety of TPK memory.

4.7 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.17)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ brings together observations of finite depth. This algebra is not identified here with the entire route algebra: prolonging the transport operators also requires their compatibilities with refinement.

Lemma 4.9 (Unity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.18)$$

and, for every $f \in D_n$, $y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.18). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so that two compatible representatives give the same evaluation in the direct limit. $\square$

Global unity is the constant function one over all possibilities; the individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while maintaining unity and all evaluations of its antecedents.

5 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its role is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The term *generation* denotes this operation; *genealogy* denotes the reconstruction of its dependencies. Both notions are involved, but they serve different purposes.

5.1 Orbits, regions, and characters of closure, propagation, and self-scaling

For a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalogue and the multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fibre predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is exactly one orbit with these properties in the catalogue. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature retains this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, the same as for closure, uniquely selects propagation; the census $\nu = (2, 2, 2)$ uniquely selects self-scaling. The diagonal support cannot be omitted: without it there are four balanced single-preimage orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (5.1)$$

These symbols first designate regions of the catalogue. Scalar identification will be established through their readouts; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and the constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position in the support would erase the distinction between numerical region and electronic structure.

The microfibre also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions taken together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition into connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

5.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary word space act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite step of selection, transport, and update, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl, pr, au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each string of six digits represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In this row-vector chart, they yield

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The schedule (L_0, L_0, L_1, L_1) produces four new blocks of six components. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary schedules are documented in the technical supplement. The indicated schedule is the only one of these sixteen that jointly reproduces the three sequences of five blocks.

5.3 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric readout acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^T, \quad \mathbf{1} = (1, 1, 1, 1, 1)^T.$$

Invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^T \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among positions of the readout; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The elliptic chart of the readout realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule taking a normalized coordinate to a slope, rather than identifying the two notions without a map. The readout composes the four accompanying positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (5.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 5.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and yields $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second, $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, hence $q = 239$. $\square$

The resulting closure character has the realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (5.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The direction of the construction is precise: the slope and the number of compositions are specified by the penta fibre readout; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (5.3), they provide rational endpoints with explicit error bounds. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

5.4 Propagation readout and normalization of the unit

The propagation region is realized by a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (5.4)$$

The character at one unit of the parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

Normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the unit of the parameter in the readout is therefore not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 5.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric upper bound for the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

Strict inequality follows from the decrease of the successive ratios. □

5.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. On the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in this ordering of the modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

The four entries are obtained from the schedule, not from an equation that has already received the value of φ . Repeating the schedule gives counts $3F_{\text{av}}$, $9F_{\text{av}}$, $36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred and eight. These counts are not powers of the matrix.

Theorem 5.3 (Positive self-scaling character). *The incidence (5.5) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and equals $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ yields the polynomial. Only one of its two roots is positive. Since all entries of F_{av}^2 are positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in the production of F_{av} . □

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{av}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

5.6 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting from $N_0 = m$, where m is the integer part determined by the readout. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \quad (5.6)$$

The half-open convention is used to select a representative when a value lies on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This qualification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 5.4 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (5.6). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending a block preserves the previous prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the boundaries of the partition has positive distance from them. A sufficiently narrow bracket lies within a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; this decision does not follow from numerical width alone. $\square$

Corollary 5.5 (Finite determination and compatibility of regional prefixes). *For each of the realizations π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix by a finite computation. The prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ ensures their separation from the rational boundaries of every finite decimal partition. The preceding theorem applies to each character. $\square$

Arbitrary depth expresses the quantifier in the corollary: for every finite horizon there is a computation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

5.7 Specular involution and synchronization of the regions

The joint nonadic connection transports the three characters together. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under specular interchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same trajectory system.

The matching census values illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality between π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a shift of -16 per channel. The closure that will lead to α must preserve both pieces of information: relative coincidence and memory evolution.

5.8 Mirror symmetry of the regions and double realization

The mirror exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution on a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the distribution preserved by the aggregate without distinction. The inversion $\mathcal{I}$ of the time index and the channel involution $\mathfrak{c}$ consequently act on different data.

Phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 cancels their relative differences, while the diagonal component changes from (59, 59, 59) to (43, 43, 43). Equality of a relative observable does not exhaust the state determining it.

The double realization preserves this architecture. In Section 8.1 of Article I (provenance of the exceptional corridor, preserved in the technical record), the same prefix determines nested cylinders on the one hand and a sequence of encoded words with their frames on the other. The first map passes to the limit through contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 5 anticipates these two maps. The subsequent incidence progression constructs codes, lattices, and the Moonshine realization on their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

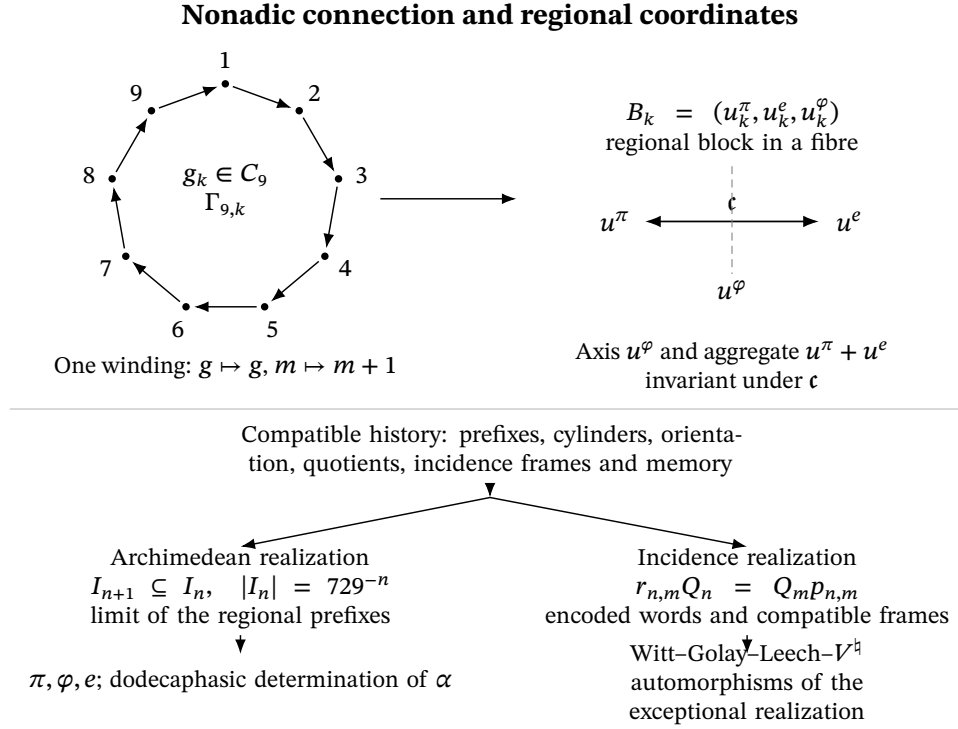


Figure 5: Nine phases of a connection and two realizations of the compatible history. The cycle represents phase; each winding increments the counter. The regional diagram represents the involution c , which interchanges closure and propagation and fixes the self-scale axis in a fibre. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit and the other preserves words and incidence frames. The maps of this second realization are specified in Section 8.1 of Article I (provenance of the exceptional corridor, preserved in the technical register); the exceptional route is developed afterwards.

6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (6.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 6.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (6.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 6.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (6.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (6.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (6.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 6.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n + m)}{b^m} \right\rfloor = P_c(b, n). \quad (6.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n + 1) \bmod b$ satisfies

$$P_c(b, n + 1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (6.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n + m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (6.5) and its residual bound. $\square$

6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t + 1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (6.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (6.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 6.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (6.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 6.2. For the second, apply (6.4) at depths n and $t + 1$, and then take the residue modulo 729. The ternary extraction in (6.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (6.8) determines its compatibility with every subsequent extension exactly.

Example 6.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (6.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (6.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 6.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 6.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (6.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 6.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (6.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (6.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (6.13)$$

Lemma 6.8 (Integer criterion for intersection). *The cylinders in (6.10) have a nonempty intersection if and only if (6.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (6.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (6.13). $\square$

Definition 6.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D), \\ T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 6.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (6.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (6.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. □

Theorem 6.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)} s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)} s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 6.2 places v_c simultaneously in both intervals in (6.10); the intersection criterion gives compatibility. The update identities are (6.5) in each base. Recovery follows from (6.4). Thus the defect recurrence and prefix recovery are consequences of the same update. □

Example 6.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 6.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$\lfloor s\ell_c(j) \rfloor \leq 729P_c(729, n) + u \leq \lceil su_c(j) \rceil - 1. \quad (6.16)$$

Theorem 6.14 (Uniqueness of the prolongation compatible with the reader). *Condition (6.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (6.16) coincide with $P_c(729, n+1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n+1)$. By (6.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (6.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (6.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (6.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (6.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 6.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (6.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (6.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (6.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (6.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (6.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (6.23). $\square$

Example 6.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 6.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (6.24)$$

Proposition 6.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (6.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. $\square$

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

6.7 Composition theorem for the block, cylinder, and signature

Theorem 6.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (6.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (6.14) and (6.15);*
2. *its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (6.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 6.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 6.11 to be applied. The deterministic application of (6.6) produces the bands used by (6.17)–(6.20). The recovery identities are those of Theorem 6.15. Finally, Theorem 6.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. $\square$

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

6.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{6.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{6.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

7 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

7.1 Regional bands, encoding, and integer signatures

7.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (7.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (7.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (7.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

7.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (7.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (7.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (7.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (7.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 7.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

7.2 Conversion clock and first zero advance

7.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (7.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 7.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (7.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 7.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

7.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \tag{7.8}$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

7.3 Oriented construction of the twenty-four-trit descriptor

7.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative–positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

7.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are (5, 7, 7, 7, 7, 7); this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (7.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase–charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (7.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

7.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\begin{aligned} \Omega_{24} = & (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \\ & \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \end{aligned} \quad (7.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 7.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022 \, 222111 \, 211201 \, 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (7.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (7.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

7.4 Functional panel and terminal repertoire

7.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{R}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (7.12)$$

Its evaluation is

$$\mathcal{R}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

7.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (7.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (7.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

7.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proposition 7.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

7.5 Comparative readers, affine reader, and temporal incidence

7.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (7.14)$$

Lemma 7.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (7.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

7.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \bmod 3.$$

This expression realizes $(t - 1) \bmod 3$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

7.5.3 Affine phase-charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

7.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu,o,s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (7.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (7.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 7.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v,u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v,u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (7.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

7.6 Enumeration of cylinders and decimal candidates

7.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau,b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (7.19)$$

7.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 7.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lfloor \frac{pD}{Q} \right\rfloor \leq N < \left\lfloor \frac{(p+1)D}{Q} \right\rfloor \right\}. \quad (7.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q)$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lfloor pD/Q \rfloor \leq N$; the second is equivalent to $N < \lfloor (p+1)D/Q \rfloor$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 7.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 7.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (7.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

7.7 Exceptional incidence and heterotypy condition

7.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (7.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (7.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (7.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

7.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (7.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (7.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (7.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

7.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 \rightarrow 4, \quad 1 \rightarrow 5, \quad 2 \rightarrow 6, \\ 3 \rightarrow 7, \quad 4 \rightarrow 8, \quad 5 \rightarrow 9, \\ 6 \rightarrow 10, \quad 7 \rightarrow 11, \quad 8 \rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

7.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{T}(N) = [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{7.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 7.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_9$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{T}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{7.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (7.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (7.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (7.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

7.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lfloor \frac{P_{78} 10^{36}}{3^{78}} \right\rfloor = 234543140729659824621058914794146601,$$

$$\left\lfloor \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rfloor = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

7.8 Regional composition and continuity of the realizations

7.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{7.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (7.13).

The register is then formed by the twelve Euclidean divisions in (7.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

7.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

7.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

7.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

7.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (7.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean's native evaluation; their reports record Lean.`ofReduceBool`, in addition to the library's customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 5, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1) - 1}{h} \right\rfloor. \quad (7.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 7.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (7.32)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (7.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (7.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (7.36)$$

The words are ordered as in (7.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 7.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (7.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h). \end{aligned} \quad (7.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (7.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196$, $0 \leq j < 197$. Substitution of the words in (7.35) and the profile (7.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

7.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_\partial(B) = (1, 2, 2, 1, 2, 0), \quad (7.39)$$

where w_{col} counts the nonzero elements in each column and q_∂ sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 7.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (7.39) give (7.40). Substituting the two final indices into (7.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the latter belongs to this line. This recovers the last step of the selector in Section 7.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_\chi]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

7.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

8 Dodecaphase record and the rational phase constant

Generation of the regional coordinates does not exhaust the information in the APP–TRIT–TPK state. One Archimedean readout determines a value through compatible cylinders; another readout preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphase record belongs to this second readout before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block catalogue word, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphase record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a simplification of notation.

This chapter preserves the causal cutoff established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite schedule. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory readout, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence of three coordinatizations of a register previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12/4}} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplying by $1000^{12} - 1$ and performing the Euclidean divisions recovers the full register. The proofs in §8.5, §8.6 and §8.7 establish these equivalences in their respective domains.

Two compositions subsequently act on this register. The positional composition combines its components with the closure, propagation and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \mapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The full register allows both compositions to be reconstructed. The support B_K specifically retains membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (8.12) of Article I (provenance of the exceptional prolongation, preserved in the technical record) connects this integer extraction to the incidence of the regional codes. The negative support of the balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional prolongation. Thus the role of K is defined by an invertible transformation and by precise subsequent maps; the additional information on depth and transport remains in the state from which the register originates.

8.1 Cyclic extension and memory grading

The observable schedule contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphase sector from the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 8.1 (The carry of the schedule). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the failure of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{mcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the identity for the section. $\square$

The result explains a distinction that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the schedule preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and the vertical memory. Closure of a finite chart does not turn this history into a memoryless cycle.

8.2 The oriented record and the integral $1 + 5 + 6$ chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the schedule forgets precisely some of the data required by the signed readout.

An initial chart of the record can be written in terms of oriented events. The reference construction (provenance preserved in the technical register) fixes the event code set $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , this gives

$$e_i = \Sigma_{12}(i+1)\#\{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is also not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary allocation of coordinates: one coordinate preserves the charge, five compare the pairs with a reference pair, and six preserve the opposition of the half-crowns.

Lemma 8.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly when the parity congruence holds. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize discarding the rest of the ledger. The complete record has a larger domain than this count chart. Extraction of the channels feeding U uses that complete record and cannot be deduced from the list (e_i) by identifying names.

8.3 Two readouts of the terminal state

Let x_{term} denote the state produced by the composition of selection, transport, and update up to the terminal cutoff. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \dots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector of the reference construction (provenance preserved in the technical register) acts on catalogues of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the stated census. It does not turn the column margin into a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed readout has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (8.1)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections begins exactly at this typed output. The material trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

8.4 From the channel record to the signed vector

The harmonic readout Π_H can be written without using either K or alpha. To avoid confusing its orbit sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (8.2)$$

Division by three belongs to the readout of the three orbits. Its integrality is checked on compatible outputs of the record; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (8.3)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (8.4)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns domain and preserved information, not merely values.

The constructive order thus far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion that follows produces the dodecaphase record. The fact that the channels can subsequently be recovered from the dodecaphase record is a reversibility check; it does not reverse this causal order.

8.5 Orbitwise Hadamard transformation and integer reconstruction

In the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation of four sectors in each orbit is what, once origin and orientation have been fixed, admits the ninety-degree angular readout. The transformation does not act on three consecutive tetrads in the natural ordering.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders by the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 8.3 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 8.4 (Integral dodecaphase transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves that both compositions are inverse. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 8.5 (Canonical integral record). *The inversion $K^{(r)} = H_4 U^{(r)}/4$ of the blocks in (8.3) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural ordering, the dodecaphase record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.5)$$

It is the unique rational preimage of U and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4 U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4 U^{(2)} &= (2172, 2636, 232, 584), \\ H_4 U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (8.5). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even and one has absolute value two; those of order three are multiples of four and one has absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor 1/4 is forced by the matrix and does not have the status of a free coefficient.

8.6 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to identification with a physical operator from another domain.

Proposition 8.6 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphase record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral statement, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The final factor is therefore twelve. $\square$

Checking (8.5) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (8.2) and (8.3), in turn, reconstruct U from those differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphase record. Equality of results does not turn their codomains into a single one; it proves that the specified maps recover the relevant information.

8.7 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart allows two different operations. The first reads a single turn; the second prolongs the dodecaphase record periodically. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (8.6)$$

The leading zeros of a block, as in 058, are part of its three-digit representation. They do not change its integer value, but they make it possible to concatenate the twelve blocks unambiguously.

Definition 8.7 (Finite and periodic readouts of the dodecaphase record). The one-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 8.8 (Exact value and period of the prolonged dodecaphase record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12}\kappa_{\text{per}}. \quad (8.7)$$

The infinite word of the dodecaphase record has minimum period twelve in base thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the minimum block period is therefore twelve. The decimal expansion is canonical, since the dodecaphase record has no tail of blocks equal to 999. If its minimum decimal period were d , it would divide 36. Grouping digits in threes would produce a block period of $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the minimum decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

8.8 Reversibility of the rational constant and phase change

The rational constant κ_{per} completely determines the register K when the base, length and readout origin are preserved. Its numerical and structural functions are related through an exact inversion.

Proposition 8.9 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1)\kappa_{\text{per}}.$$

The integer expansion of $I(K)$ in twelve base-1000 blocks recovers all K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, one has

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \quad (8.8)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Moreover,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (8.8). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the transformations D_3K, D_4K, Q already constructed. Rational evaluation preserves the signed register and the cyclic differences depending on it. A history's total depth and transport operators remain in their own coordinates: the preceding inversion has the integral dodecaphasic register as its domain.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical readout, while its rotations follow an affine law. The orbit sum is invariant under choice of origin.

Subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because that orbit first recovers the integer components. The signed support associated with α additionally uses its regional registers and boundary normalization. The relation between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. Value, transformation and domain are preserved jointly.

8.9 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphase record. It does not imply periodicity of the regional coordinates, of the closure carries, or of alpha. It is useful to make this separation precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not authorization to modify the dodecaphase record already selected by the record.

Proposition 8.10 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient come from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

9 Correlative construction of the continuum: histories, transport, incidence, and terminal

The arithmetic expressed by a normal form preserves a precise readout of the state producing it. Situating that readout within HMT also requires retaining the histories, sheets, and operations acting on them. This chapter assembles that common substrate. Its different constructions arise from the same tree of APP–TRIT–TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains selected by a subsequent application.

The starting point is the formal kernel presented earlier. An admissible emission contains the APP operation, TRIT orientation, sheet change, residue, and quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the starting object. Two executions with identical final readouts are not identified for that reason alone.

9.1 Admissible histories and memory through composition

Let X_k be the set of admissible histories of length k . Its elements are constructed by successive emissions from the kernel's initial domain. Write $h\varepsilon$ for the prolongation of h by an admissible emission ε , and $\rho_k(h\varepsilon) = h$ for deletion of the last emission. The emission label is preserved even when different labels produce the same expressed state.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate similarly admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preserving the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 9.1. Affine compatibility of concatenation

If α precedes β , then

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; in general, it is not the untransported sum of two registers.

Nonadic return preserves a second distinction. In phase and winding coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

Exactly one crossing from 8 to 0 occurs during nine steps. Phase returns and the winding coordinate increases. Enriched holonomy additionally retains the memory updates accompanying that traversal.

9.2 Tree refinement and measure on histories

Fix the finite nonempty initial set of admissible seeds, with strictly positive initial masses summing to 1. In this subsection every history has a finite positive number of prolongations. Deleting emissions defines the refinement maps. Kernel admissibility supplies the prolongations; when a residence uses a neutral emission, that emission explicitly gives a section of ρ_k . Existence of that section refers to the tree containing that emission, not to any arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h . Equiprobable choice among its successors defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon \mid h)$, summing to 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon \mid h)$. This second formulation allows the readout weights to be preserved when the numbers of prolongations are not uniform.

Proposition 9.2. Cylindrical compatibility and conditional expectation

At each level,

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

On the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$, and $\|J_k f\| = \|f\|$.

Proof. The first identity is normalization of the local probabilities. Applying it to $|f(x)|^2$ yields the isometry. Grouping the sum defining $\langle J_k f, g \rangle$ by predecessors gives the formula for E_k ; on a function constant in each fiber, that formula returns its value. $\square$

Compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and uniqueness follows because the cylinders generate the sigma-algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under deletion of emissions is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities just preserved by the construction.

9.3 Sheetwise transport and exact accounting of outputs

First consider the space with orthonormal basis consisting of the histories at level k . The transport preserving each prolongation label is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admissible at } h} \sqrt{p(\varepsilon | h)} e_{h\varepsilon}.$$

The successor sets of distinct predecessors are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation through history masses is related to this one by the diagonal basis change incorporating $\sqrt{\mu(h)}$.

Let P_k, Q_k be the complementary projectors onto the two sheets. Separate sheet preservation and sheet interchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 9.3. Sheetwise Gram identity

For a linear transport U , whether isometric or not, the preceding decomposition satisfies

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms gives $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$ and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second additionally uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross blocks of that Gram to vanish. Isometric preservation of history transport supplies that condition here; it must not automatically be attributed to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$, and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 9.4. Finite decomposition with terminal residue

For each horizon N ,

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The output sum converges strongly to $I - T_\infty$.

Proof. Proposition 9.3 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; apply this to $S = T_N - T_\infty$. $\square$

The terminal residue is preserved until its extinction on the domain in question is proved. This distinction prevents phase return alone from becoming a hypothesis that all histories are exhausted.

9.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a readout expresses the average over prolongations, define $b_c = Eb_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 9.5. Exact cancellation defect

The quantity

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposing contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine equally weighted successors; conditional expectation is the rule that remains valid for general branching and weights.

9.5 Ternary sheet incidence and the rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They yield six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus the counts 4, 6, 8 belong to different operations on the same sheet module.

On the six pair positions, define

$$(Ba)_{\{L, L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L, L'\}} q_{\{L, L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The readout $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The readout equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These averages describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already preserved in X_k .

For the effective ports at a level, let I and J be the nonempty port sets of its two sheets, and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Fix reference ports i_0, j_0 . The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0j} + w_{i_0j_0}$$

define the following sequence, valid over the finite ring A_N :

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I \times J} \xrightarrow{\delta} A_N^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 9.6. Exactness of rectangular incidence

The preceding sequence is exact, with first arrow given by the constant diagonal.

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i j_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{i j}$. Finally, any matrix in the last term extends with zero reference row and column; its image under δ is the original matrix. The proof uses only addition and subtraction, so it requires no division by a ring unit. $\square$

Reduction from A_{N+1} to A_N commutes with both maps. Reconstruction formulas are the same at every depth. Incidence and its exactness are therefore compatible with arithmetic refinement of the starting state.

9.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integral complex. Its reduction to A_N uses the same integer register before reducing it; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends generators to their new counterparts, except for

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 9.7. Naturality of the memory complex

One has $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring where appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they belong to a different domain, and their function is specified by the preceding identity. Their role can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 9.8. Homology and memory of the local complex

Over $\mathbb{Z}$, or after reduction to A_N , one has

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integral realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be projection onto the summand generated by r_y , and ι its inclusion. Checking each generator gives

$$\partial h + h \partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish values of that coordinate. Complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal readout in addition to this local complex.

9.7 Assembly with terminal paths

For a finite horizon, residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_N[\text{Path}(\nu, \text{Term})]$, preserving every terminal continuation as a generator. An initial trajectory acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs yield the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. Naturality in Proposition 9.7 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms in the square. Thus $D^2 = 0$. Compatibility is proved on path generators and is not introduced through a label declaring membership in a complex.

Terminal evaluation likewise requires no selection of a particular history. For each continuation h , the effective composition of its updates is applied. The result preserves the pair consisting of h and its readout, with its multiplicity. If $k < \ell < N$, continuations decompose as a disjoint union of prefixes through ℓ and their continuations through N . Proposition 9.1 proves that direct evaluation and evaluation in those two parts produce the same readout. This yields the terminal naturality square.

Path modules, incidences, memory transport, and their differentials arise from the same prolongations. Their assembly does not consist in adding five coordinates to a state that would be preserved by definition: their relations are constructed through sums, boundaries, and effective compositions, and verified on their generators.

9.8 Determinants, limit, and arithmetic readout

At each finite truncation, the free complex admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk} \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases, and the relevant homology identifications. Smith factors are calculated from the integer matrix produced before reduction, not from an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from existence of the determinant line.

Passage to the limit preserves compatible projective systems of residues, histories, and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylinder intervals, compatible by inclusion and with diameters tending to zero. Their nested compact closures intersect in a single point: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary-representation rule then determines its positional expression. The value does not replace the history producing it. The other representation preserves incidences of the same state. Both destinations share their origin without thereby receiving the same information.

The normal-form sector studied in the arithmetic chapter is obtained by normalized representation of TPK prefixes. Its integer evaluation and operations are subsequent readouts on the substrate constructed here. Spectral functionals, finite parts, and irreducible clocks are applied to these expressed modes. Identifying a concrete functional with an external analytic form in turn requires writing its action on generators and verifying its inner products: the conservation identities of the present chapter do not replace that composition.

9.9 Provenance, verification, and scope

This development assembles the authorial architecture of source owners 04b1, 04b2, and 04b3 of the discrete structure and the balance, incidence, and memory identities of developments 05. It jointly preserves the five constructions of the continuum as operations on the same tree. The exposition expands their local proofs and makes explicit the conditions of each specialization: isometric transport, uniform weights, terminal extinction, and nonsingularity of a determinantal block.

The accompanying checker tests exact instances of the cocycles, weighted cancellation, rectangular sequence, and memory complex. These checks accompany the general proofs in the text. Their scope is local: neither finite enumeration nor preservation of provenance files is presented as a proof of Weil positivity or as an identification of a spectral operator whose composition has not yet been carried out.

10 Center, boundary, and vacancies in the nonadic connection

10.1 The thesis and the order of construction

The relation between center and boundary acquires mathematical content when the data expressed, the transformations transporting them, and the information used to reconstruct them are specified. In APP, a residue is accompanied by its quotient; in TRIT, orientation distinguishes operations that an aggregate may bring together; in TPK, a winding returns the phase and prolongs memory. The enriched state brings these coordinates together. Its residence in the joint discrete structure of the continuum preserves the dependencies among the different subsequent readouts.

The progression is $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow \text{enriched state} \rightarrow \text{joint discrete structure of the continuum}$. This development makes local cuts in that progression: it neither reconstructs the five consubstantial aspects of the continuum from a spectral form nor turns them into five independent modules. The internal constants are received at their representation stage, without using their conventional values to select seeds, routes, or coefficients.

The author's intuition can be expressed as follows: **the representation of a value is a projection of a construction; transporting that construction requires preserving the information that distinguishes its realizations**. The following operations make this assertion concrete. The subsequent language of interpolation and orthogonal decomposition proves properties of objects already specified; it does not retrospectively select the APP state.

10.2 The local crown and reconstruction of the center

The positive chart uses

$$\rho_9(n) = 1 + [(n-1)]_9, \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n).$$

The product table is $P(i, j) = ij$, with $1 \leq i, j \leq 9$. Restriction to $Q = \{3, 4, 5, 6\}^2$ expresses

$$(\rho_9(ij))_{i,j=3}^6 = \begin{pmatrix} 9 & 3 & 6 & 9 \\ 3 & 7 & 2 & 6 \\ 6 & 2 & 7 & 3 \\ 9 & 6 & 3 & 9 \end{pmatrix}.$$

	3	4	5	6
3	9	3	6	9
4	3	7	2	6
5	6	2	7	3
6	9	6	3	9

Residual chart
corona 3, 6, 9

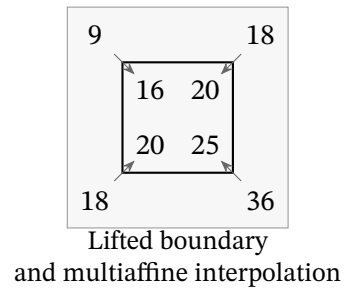


Figure 6: The same chart in two readouts. Left: the radical corona surrounds block B_c . Right: the four integer corners determine the interior values under the multiaffine law. Their corner residues coincide, but the quotients $(0, 1, 1, 3)$ distinguish the lift.

Its inner block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

The perimeter of this chart contains twelve positions: four with value 3, four with 6, and four with 9; their sum is 72. A radical crown becomes visible around the central block. The global electronic center is (5, 5), the unique fixed point of the inversion of the complete APP; its position does not change upon choosing this local chart, whose geometric center is at (4.5, 4.5).

Proposition 10.1 (Recovery from the lifted boundary). *Let $F(x, y) = a + bx + cy + dxy$ be a multiaffine law on the chart $[3, 6]^2$. For $t = (x - 3)/3$ and $u = (y - 3)/3$,*

$$F(x, y) = (1 - t)(1 - u)F(3, 3) + (1 - t)uF(3, 6) \\ + t(1 - u)F(6, 3) + tuF(6, 6).$$

Proof. For fixed y , affinity in x recovers $F(x, y)$ from $F(3, y)$ and $F(6, y)$. Affinity in y recovers each of these two values from its endpoints. Composition gives the formula. In particular, the four corners determine the entire chart within the declared space of laws. $\square$

For the product, the four lifted values are 9, 18, 18, 36. Their positive residues are all 9, but their quotients are 0, 1, 1, 3. The formula recovers 16, 20, 20, 25 at the interior positions; their residual representation gives 7, 2, 2, 7. Thus, the boundary determines the interior **while preserving the law and the lift**. Interpolation is performed before modular reduction: no division by 3 is performed in $\mathbb{Z}/9\mathbb{Z}$, where 3 is not invertible.

This specifies the meaning of “preserving the information of the center.” The residual crown alone does not individualize all lifts: F and $F + 9$ have the same residues along the entire boundary and different lifted values. The residue–quotient pair removes that ambiguity within the indicated reconstruction. The assertion is proved for multiaffine laws; extending it to complete states also requires preserving the operators, routes, and memory acting on them.

Multiaffine recovery is a **new formalization of the preexisting authorial architecture**. The charts and their readouts are results recovered from the corpus; this explicit presentation of their inverse is not credited as an independent historical discovery.

10.3 Sum, product, and the three values of discrete curvature

The multiplicative rows are constructed by additive accumulation. The difference between the two sheets appears when both coordinates are varied simultaneously. With unit differences,

$$\Delta_x \Delta_y (x + y) = 0, \quad \Delta_x \Delta_y (xy) = 1.$$

The same property is recovered on the lifted boundary:

$$\frac{P(6, 6) - P(6, 3) - P(3, 6) + P(3, 3)}{3 \cdot 3} = 1;$$

for $S(x, y) = x + y$, this quotient equals 0. The product records the cross variation that the sum does not introduce. The corpus expresses the oriented version through

$$\mathcal{F}(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x, \\ \mathcal{F}(S, S) = \mathcal{F}(P, P) = 0, \quad \mathcal{F}(S, P) = +1, \quad \mathcal{F}(P, S) = -1.$$

The order of the sheets distinguishes the sign. The formulation “the sum accumulates and the product folds or unfolds” rests here on a verifiable operation: modular projection preserves a residue; lifting recovers the quotient recording what was exceeded. Discrete curvature additionally records the order of the variations. This finite difference is not identified with a differential curvature of a physical space without its corresponding realization map.

The global radical cross

In $R = \mathbb{Z}/9\mathbb{Z}$, the ideal $\mathfrak{m} = \{0, 3, 6\}$ satisfies $\mathfrak{m}^2 = 0$. Its positive representation is $\{9, 3, 6\}$. The cross

$$(\mathfrak{m} \times R) \cup (R \times \mathfrak{m})$$

has $27 + 27 - 9 = 45$ positions. The intersection contains nine values 9, summing to 81; the remaining arms have twelve values of each class 3, 6, 9, summing to 216. The total is

$$297 = 216 + 81 = 3(72 + 27) = 3 \cdot 99.$$

Four boundaries are distinguished: the local perimeter of twelve positions, the radical cross of forty-five, the gnomon of seventeen completing 8^2 to 9^2 , and the cubic crown of 271. Each has its own domain and operation. The tritic quotient $R/\mathfrak{m}$ preserves a class; transporting the state requires accompanying it with the lifted data and their history.

10.4 The 72–27 and 77–22 readouts and the 729–271 completion

The ordered rows of B_c are expressed decimally as 72 and 27; its diagonals as 77 and 22. This gives

$$72 + 27 = 77 + 22 = 99.$$

In the cubic completion, counting the interior, faces, edges, and vertex gives

$$1000 = 729 + 243 + 27 + 1 = 729 + 271,$$

$$999 = 729 + 243 + 27 = 729 + 270.$$

The coincidence of readouts is expressed while preserving the decimal remainder:

$$729 = 10 \cdot 72 + 9, \quad 271 = 10 \cdot 27 + 1.$$

Thus, the two-dimensional readout expresses $(72, 27)$; the cubic readout expresses $((72, 9), (27, 1))$. The pair of remainders $(9, 1)$ distinguishes prolongations that would share the same quotient. The chart constructions are performed first; these representations are then compared. Identity of the integers does not reverse that causal order.

The chapter on integer channels provides another specific composition. With the internal counts $\Delta = 54$, $\nu = \Delta/2 = 27$, and $\Theta = 5\nu = 135$, it obtains

$$\mathcal{C}_\alpha = \nu^2 = 729, \quad \mathcal{C}_e = 2\Theta + 1 = 271.$$

Here ν is the new name for the local integer denoted by κ in that source, to distinguish it from the rational readout of the dodecaphase record. The channels are integer objects; their identification with the constants passes through the regional readouts and the corresponding closure, not through the first digits of an expansion. This comparison uses the integer channels and does not require the subsequent angular realization.

For the oral reference to a fraction of π , the located source contains

$$\frac{396}{126} = \frac{22}{7}, \quad 396 = 18 + 72 + 108 + 198, \quad 126 = 18 + 108.$$

This is a rational ratio of alternating rings. This ring ratio is an exact rational readout; its function is distinguished from the regional representation of π .

10.5 The electronic center and its conjugate vacancies

The inversion of the positive APP is $\rho(r, c) = (10 - r, 10 - c)$. Its unique fixed point is $(5, 5)$. In the central additive fiber,

$$\mathcal{F}_{10} = \{(5 + t, 5 - t) : t \in \mathbb{Z}, |t| \leq 4\},$$

the product equals $25 - t^2$. Preserving its central residue requires $t^2 \equiv 0 \pmod{9}$, equivalent to $3 \mid t$. The only positions are therefore

$$(5, 5), \quad (8, 2), \quad (2, 8).$$

If $\mathcal{A}$ preserves the residue–quotient pair on each sheet,

$$\begin{aligned} \mathcal{A}(5, 5) &= ((1, 1), (7, 2)), \\ \mathcal{A}(8, 2) &= \mathcal{A}(2, 8) = ((1, 1), (7, 1)). \end{aligned}$$

The two vacancies preserve the additive readout and the multiplicative residue; the product quotient decreases by exactly one unit. The mirror $t \mapsto -t$ exchanges their positions. The residual value does not distinguish their routes, whereas orientation does preserve them. This is a precise realization of the connection indicated in the note: **a vacancy is individualized by a relation to the center and by a recorded defect, not by being a cell without information.**

The initial electronic orientation $(h_0, v_0) = (1, 1)$, reversed every 27 steps, subsequently expresses the dodecaphase signature $(+++---+++---)$. This is an orientation record, distinct both from the position $(5, 5)$ and from the product signature below.

The signature 555555 and the memory enabling its prolongation

For $U_j = 100d_{1j} + 10d_{2j} + d_{3j}$, the readout

$$\mathcal{E}_\times(U) = ([d_{2j} - d_{1j}]_{10})_{j=1}^6$$

applied to the transverse region

$$U_\pi^\perp = (501, 614, 498, 169, 272, 272)$$

produces 555555; its reduction $\Psi(U) = (-U_j \bmod 3)_j$ produces 010211. The signature has twenty-four product representatives. Transport separates them into three sheets of eight: after advancement, 555177, 555717, and 555771 appear. The congruence stable under advancement and half-turn refines the twenty-six signatures into twenty-eight classes, with histogram $27 \cdot 8 + 108 = 324$. This expresses why sheet memory must be retained to continue a signature that initially coincides.

Definition 10.2 (Signature readout and transport of representatives). Let $\mathcal{X}_\times = D_9^2 \times \{N, E, S, O\}$, of cardinality 324. Advancement P moves one cell in the stored direction, and the half-turn H reverses that direction. The readout uses

$$\ell_9(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5), \quad \ell_9(3) = \ell_9(6) = \ell_9(9) = 0.$$

In each of six windows of nine steps, $\ell_9(\rho_9(ij))$ is evaluated before the product advancement selected by TRIT at times 2, 5, 8 (mod 9). The sum in each window is reduced modulo 10. After steps 27 and 54, the direction is reversed. The rule determines $E_{\text{sig}}(x)$ from the position and direction of x , with the kernel's time conventions.

Proposition 10.3 (Minimal stable refinement of the signature). *Starting from equivalence by signature $\sim_0$, the iteration*

$$x \sim_{n+1} y \iff x \sim_n y, \quad Px \sim_n Py, \quad Hx \sim_n Hy$$

terminates and produces the coarsest stable congruence refining the signature. On the preceding domain the census is $26 \rightarrow 28 \rightarrow 28$; the fiber of 555555 separates into three classes of eight representatives.

Proof. At each step, a partition of a finite set is refined. The process stabilizes after finitely many splittings. At the fixed point, equivalence is preserved under P and H . If a stable congruence refines $\sim_0$, it refines each $\sim_n$ by induction: its images under both operators remain equivalent. It therefore refines the fixed point obtained, proving its minimality in added information.

The finite enumeration applies the definition to the 81 positions and the four orientations. Their six sums modulo 10 are grouped first, and then the triple consisting of the class of x , that of Px , and that of Hx , until stabilization. The resulting histogram is $27 \cdot 8 + 108 = 324$, and the only signature splitting is the one indicated. The following representatives make the three images explicit:

x	$E_{\text{sig}}(x)$	$E_{\text{sig}}(Px)$
$(3, 5, N)$	555555	555177
$(3, 4, N)$	555555	555717
$(3, 4, S)$	555555	555771

The complete rule and its executable enumeration are preserved in the reproduction package; the witnesses in the table also show that preserving only the signature does not make its continuation unique. $\square$

The conclusion describes the minimal information of transport, not a new mass. The additive factor preserves eighteen representatives per signature (nine compatible positions and two orientations). Combining it with the product fibers of sizes 24 and 8 gives, respectively, 432 and 144. Regional incidence and sheet memory thus preserve distinct functions within the same catalog.

The link to the electronic equation passes through the regions. The transverse region has multiplicity $432 = 18 \cdot 24$; each of the other four has $144 = 18 \cdot 8$. Their five identities are inserted, in the declared gauge of Article I, into

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$$

through $(4, 0), (5, 0), (6, 0), (6, 1), (6, 2)$. Once this insertion has been fixed, its complement has $27 - 5 = 22$ positions and its three sheets per region have $5 \cdot 3 = 15$ positions. The oriented deficit and torsional memory are -22 and 15 . The proof of these counts follows from the preceding explicit injection: its five images are distinct; this development brings it together, rather than selecting its positions from a target mass.

There are therefore two linked progressions in the composition: **center** $\rightarrow$ **conjugate vacancies** $\rightarrow$ **orientation**, and **regional signature** $\rightarrow$ **transport sheets** $\rightarrow$ **incidence of the five regions** $\rightarrow$ **electronic coefficients**. Simply identifying $(5, 5)$ with 555555 would erase precisely the operations explaining their relation.

10.6 Nonadic mirror, capacity, and return with memory

On the regional fiber, the mirror acts by

$$c(u^\pi, u^e, u^\varphi) = (u^e, u^\pi, u^\varphi).$$

It preserves $s = u^\pi + u^e$ and reverses $d = u^\pi - u^e$. In $\mathbb{F}_3$,

$$u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The sum preserves the aggregate; the oriented difference preserves its distribution. The axis u^φ remains fixed under this involution, although it may evolve under transport. The regional mirror and the cellular mirror of the vacancies have different domains: their respective formulas make it possible to track what each preserves.

The capacity vacancy adds an explicit temporal link. In the normalization of refinement 729 versus 1000, let $\rho_c = \log_{10} 9$, $\delta = 1 - \rho_c$, and $\mathcal{C}_t = \lfloor (t + 1)\rho_c \rfloor$. If

$$z_t = \lfloor (t + 1)\delta \rfloor - \lfloor t\delta \rfloor,$$

then, for $t \geq 1$,

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t, \quad \mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t = b - a.$$

Proof. The irrationality of $\log_{10} 9$ follows from the impossibility of $9^n = 10^m$ for positive integers. Hence, $\mathcal{C}_t = t - \lfloor (t + 1)\delta \rfloor$. Subtraction and telescoping summation give both identities. $\square$

A vacancy retains capacity while the history advances.

If $p_j = \lfloor j/\delta \rfloor$, $v_j = p_j - j$, and $h_j = p_{j+1} - p_j$, the inherited development establishes $h_j \in \{21, 22\}$. For $s_j = 22 - h_j$,

$$[v_{j+1}]_3 = [v_j - s_j]_3, \quad \tau_j = M_{\text{ph}}(1 + [v_j]_9).$$

Vacancies thus act on the retained phase read by the tritic selector. This capacity sequence does not automatically become the electron's cellular trajectory: the manuscript preserves their respective mechanisms and their composition within the same kernel.

10.7 Vacancies of nonadic refinement and hierarchy of returns

APP-TRIT-TPK prolongation preserves, together with the expressed block, the cylinder, orientation, and history that allow it to be continued. Its capacity readout compares two cardinalities already determined by that construction: the $3^6 = 729$ possibilities of a six-position ternary block and the $10^3 = 1000$ positions of its three-digit decimal chart. This comparison produces its own schedule. Some transitions increase the capacity index; others incorporate a block into the history while retaining that index. The latter are the *capacity vacancies* studied below.

The object under consideration is therefore a representation of joint refinement. Its definition preserves its origin in the two finite supports; the logarithmic expression permits the properties of its schedule to be proved. Block time, vacancy index, and capacity index will be written separately. This separation makes clear why the returns 21/22 and 6/7 belong to successive scales of a single construction.

10.7.1 Decimal capacity and preservation of the increment

Definition 10.4 (Capacity index and vacancy). At phase origin 0, define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho = \log_{10}(10/9), \quad (10.1)$$

and, for $t \in \mathbb{Z}_{\geq 0}$,

$$\mathcal{C}_t = \lfloor (t+1)\rho \rfloor, \quad z_t = \lfloor (t+1)\delta \rfloor - \lfloor t\delta \rfloor. \quad (10.2)$$

The integer $\mathcal{C}_t$ is the decimal capacity index of the readout; $z_t = 1$ marks a vacancy. The letter K remains reserved for the dodecaphasic register and does not denote a capacity counter here.

We have $0 < \delta < 1$. Moreover, δ is irrational: if $\rho = a/b$ were rational, with a, b positive integers, the equality $9^b = 10^a$ would have zero valuation at the prime 5 on the left and positive valuation on the right. This contradiction proves the irrationality of ρ and of $1 - \rho$. In particular, no positive integer multiple of δ belongs to $\mathbb{Z}$, which fixes the endpoints of the intervals in this development unambiguously.

Proposition 10.5 (Exact capacity–vacancy balance). For $t \geq 1$, $z_t \in \{0, 1\}$ and

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t. \quad (10.3)$$

Consequently, for $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t = b - a. \quad (10.4)$$

Proof. Since $0 < \delta < 1$, the difference between the floors of two numbers separated by δ is 0 or 1. Furthermore, the established irrationality gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting this equality at t and $t - 1$ proves (10.3); telescopic summation proves (10.4). $\square$

Each transition contributes exactly one unit to the right-hand side of the balance. That unit is recorded as acquired capacity when $z_t = 0$, or as a vacancy when $z_t = 1$. Retention of the index $\mathcal{C}_t$ therefore leaves a recoverable position in the history. The equation expresses preservation of the increment through two readouts, rather than an indeterminate loss of information.

The density also follows from the same balance:

$$\frac{1}{N} \sum_{t=1}^N z_t = \frac{\lfloor (N+1)\delta \rfloor}{N} \longrightarrow \delta.$$

Thus (z_t) is not eventually periodic: an eventually periodic binary sequence has rational density. Here aperiodicity follows from the capacity rule already fixed by 729/1000.

10.7.2 Exact determination of the intervals 21/22

Lemma 10.6 (Rational bounds for the slope). The vacancy density satisfies

$$\frac{7}{153} < \delta < \frac{6}{131}. \quad (10.5)$$

If $\lambda = \delta^{-1}$, $\beta = 22 - \lambda$, and $\mu = \beta^{-1}$, then

$$21 < \lambda < 22, \quad \frac{1}{7} < \beta < \frac{1}{6}, \quad 6 < \mu < 7. \quad (10.6)$$

Proof. We provide a finite verification by rational arithmetic. For $0 < u < 1$, let

$$S_N(u) = 2 \sum_{k=0}^N \frac{u^{2k+1}}{2k+1}, \quad R_N(u) = \frac{2u^{2N+3}}{(2N+3)(1-u^2)}.$$

Integrating the geometric series for $2/(1-u^2)$ gives

$$S_N(u) < \log \frac{1+u}{1-u} < S_N(u) + R_N(u).$$

Indeed, the tail is positive and, on replacing each denominator $2k+1$ by $2N+3$, is bounded by the geometric series defining R_N . These inequalities are applied to $u = 1/19, 1/3, 1/9$, corresponding respectively to $\log(10/9), \log 2, \log(5/4)$, and we use $\log 10 = 3 \log 2 + \log(5/4)$.

With $N = 4$, writing $A = S_4(1/19)$, $B = 3S_4(1/3) + S_4(1/9)$, and $E = 3R_4(1/3) + R_4(1/9)$, we obtain

$$\frac{7}{153} < \frac{A}{B+E} < \delta < \frac{A+R_4(1/19)}{B} < \frac{6}{131}.$$

The two outer inequalities are checked by multiplying through by the positive denominators of these sums of five rational terms. This proves (10.5) using a remainder bound, without taking a decimal expansion as a premise. Inverting the bound gives $131/6 < \lambda < 153/7$. Subtracting from 22 and then inverting yields all the inequalities in (10.6). $\square$

Theorem 10.7 (Positions and gaps of the vacancies). *The positions of the events $z_t = 1$, ordered from the first event, are exactly*

$$p_j = \lfloor j\lambda \rfloor = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad j \geq 1. \quad (10.7)$$

Their gaps satisfy

$$h_j = p_{j+1} - p_j \in \{21, 22\}. \quad (10.8)$$

Both values occur infinitely often.

Proof. The event crossing the integer j occurs precisely when $t\delta < j < (t+1)\delta$. Irrationality excludes equality at the endpoints. Dividing by δ gives $t = \lfloor j/\delta \rfloor = p_j$. Since $\delta < 1$, a step crosses at most one integer, so the enumeration is exact. Writing $\lambda = 21 + \eta$, with $0 < \eta < 1$, gives

$$h_j = 21 + \lfloor (j+1)\eta \rfloor - \lfloor j\eta \rfloor,$$

which proves (10.8). Finally, $(p_{j+1} - p_1)/j \rightarrow \lambda \in (21, 22)$. If either value occurred only finitely often, this mean would converge to the other integer value, a contradiction. $\square$

10.7.3 Induction on the short intervals: the returns 6/7

Theorem 10.8 (Induced schedule of short intervals). *Let $s_j = 22 - h_j$, for $j \geq 1$. This indicator equals 1 exactly when the interval between vacancies is short, of length 21. With β and μ as defined in Lemma 10.6,*

$$s_j = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor. \quad (10.9)$$

The indices of the short intervals are

$$q_m = \lfloor m\mu \rfloor = \lfloor m/\beta \rfloor, \quad q_{m+1} - q_m \in \{6, 7\}, \quad m \geq 1. \quad (10.10)$$

Both gaps occur infinitely often, and their order is not eventually periodic.

Proof. The irrationality of δ implies that of λ , β , and μ . Since $\lambda = 22 - \beta$, for $j \geq 1$ we have

$$p_j = 22j - \lfloor j\beta \rfloor = 22j - \lfloor j\beta \rfloor - 1.$$

Subtracting consecutive terms gives (10.9). Applying the integer-crossing argument of the preceding theorem to the indicator of slope β shows that its event m occurs at $q_m = \lfloor m/\beta \rfloor$. The inequality $6 < \mu < 7$ proves the two possible gaps. Their mean converges to μ ; hence both occur infinitely often. Moreover, an eventually periodic sequence of gaps would have rational mean, whereas μ is irrational. This also rules out strict alternation, whose mean would be $13/2$. $\square$

Here the numbers 6 and 7 count *intervals between vacancy events*, not refinement blocks. The return to the primary scale is equally explicit. Between q_m and q_{m+1} there is a single short interval, located at the beginning, and all the others are long. Consequently, if $d_m = q_{m+1} - q_m$,

$$p_{q_{m+1}} - p_{q_m} = 21 + 22(d_m - 1) = 22d_m - 1 \in \{131, 153\}. \quad (10.11)$$

Thus $131 = 14 \cdot 9 + 5$ and $153 = 17 \cdot 9$ describe two return readouts *in the block clock*. The equation preserves the operation connecting the scales, rather than identifying their cardinalities merely because they occur.

10.7.4 Retained capacity and selection of the TRIT regime

Let $v_j = \mathcal{C}_{p_j}$ be the capacity recorded at vacancy j . From $p_j\delta < j < (p_j + 1)\delta$ and $\delta < 1$, it follows that $\lfloor (p_j + 1)\delta \rfloor = j$. By (10.2),

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1 = 21 - s_j, \quad [v_{j+1}]_3 = [v_j - s_j]_3. \quad (10.12)$$

A long interval preserves the class modulo 3; a short interval shifts it by one unit in the negative direction. Applying the selector of the formal kernel to the positive phase $r_j = 1 + [v_j]_9$ gives

$$\tau_j = M_{\text{ph}}(r_j), \quad M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases} \quad (10.13)$$

This composition fixes the regime induced by capacity at each event. It preserves both the order of the changes and the number of events for which each regime persists.

Corollary 10.9 (Plateaus of the induced regime). *For every $m \geq 1$, the class $[v_j]_3$, and hence τ_j , is constant on $q_m + 1 \leq j \leq q_{m+1}$. The length of each complete plateau is $q_{m+1} - q_m \in \{6, 7\}$.*

Proof. On the indicated integer interval, the next change occurs exactly when passing from $j = q_m$ to $j = q_{m+1} + 1$: this is the next position at which $s_j = 1$. At all preceding steps $s_j = 0$, so (10.12) preserves the class. At the endpoint it decreases the class by one unit. The selector (10.13) assigns distinct values to the three classes; it therefore preserves the same plateaus and their endpoints. $\square$

The initial segment $1 \leq j \leq q_1$ depends on the chosen origin and is an initial restriction of a plateau. At the present origin it has length $q_1 = 6$. The first twenty classes are

$$\underbrace{2, \dots, 2}_6, \quad \underbrace{1, \dots, 1}_7, \quad \underbrace{0, \dots, 0}_7$$

with signatures 0, -1 , $+1$, respectively. Each short interval triggers the next step of the cycle of regimes.

The two phases of interest are defined as

$$g_t^{\text{cron}} = [t]_9, \quad g_t^{\text{cap}} = [\mathcal{C}_t]_9, \quad g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

A vacancy retains the capacity phase while block time advances. The transition of the enriched state remains the TPK composition of selection, transport, and memory update. Thus $\mathcal{C}_t = \mathcal{C}_{t-1}$ neither resets the state nor replaces the holonomic winding Γ_9 by a pause. The two induced scales likewise retain this distinction: between consecutive short events in (10.11), the capacity increment is $21d_m - 1$, that is, 125 or 146, whereas the block increment is 131 or 153.

10.7.5 Rationally certified sample and provenance

The table presents a sample at the canonical origin. h_j is the gap *following* p_j ; thus vacancy $j = 6$, at block 131, begins the first short interval.

j	1	2	3	4	5	6	7	8	9	10
p_j	21	43	65	87	109	131	152	174	196	218
h_j	22	22	22	22	22	21	22	22	22	22
v_j	20	41	62	83	104	125	145	166	187	208
$[v_j]_3$	2	2	2	2	2	2	1	1	1	1

The first short-interval indices and their gaps are

$$(q_m)_{m=1}^{10} = (6, 13, 20, 27, 34, 41, 48, 54, 61, 68),$$

$$(q_{m+1} - q_m)_{m=1}^9 = (7, 7, 7, 7, 7, 7, 6, 7, 7).$$

The sample exhibits the actual order, while the preceding proofs establish its law at arbitrary depth.

The attached file `ampliacion/verificar_vacancias.py` computes rational brackets using the series and remainders of Lemma 10.6. Each floor is accepted only when both endpoints of the bracket yield the same integer. At order 16, it reproduces 512 events up to block 11189, and 64 secondary gaps, together with the balance identities, residues, and plateaus. This finite check retains a scope distinct from that of the universal theorems.

The capacity index, its complementarity with vacancies, and induction of the TRIT selector are recovered from Article I and from the chapters on the schedule and induced clock in the monograph on aperiodic topological–temporal order. Their definitions and the proofs of both scales are assembled here with a single indexing convention. The rational certificate provides an additional check of those results and preserves their provenance.

10.8 The factor 8/81 as a balance between representation and complement

In the double-circle development, the uniform inclusion of nine phases is $Jv = (v, \dots, v)/3$. With the seam

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7),$$

one obtains $T = J^*S_CJ = (8I + C)/9$. If $z = (C - I)v$, the complement of the representation is exactly

$$\eta v = (I - JJ^*)S_CJv = \frac{1}{27}(8z, -z, \dots, -z).$$

Hence,

$$\|\eta v\|^2 = \frac{64 + 8}{729}\|z\|^2 = \frac{8}{81}\|(C - I)v\|^2.$$

Here appears the 8/81 mentioned by the author: one transported phase against eight others, with the uniform normalization of nine phases. The result does not depend on which of the nine positions is marked. The complete operator balance is

$$I - T^*T = \frac{8}{81}(I - C)^*(I - C) + \frac{1}{9}(I - C^*C).$$

If C is isometric, the energy omitted by representation lies exactly in its complement. For an arbitrary seam, its explicit defect is also preserved. The integer $72 = 64 + 8$ counts quadratic weights here; in the local crown it counts a sum of values; in $72/27$ it belongs to an ordered decimal readout. Specifying these types makes it possible to relate the constructions without assuming that a numerical coincidence already provides all maps between them.

This balance remains that of the double-circle transport development. When applied to the arithmetic form $\mathcal{W}$, it produces a signed identity; proving global positivity requires global control of its terms, not merely recovery of the center or positivity of an auxiliary norm. The results already proved concerning memory, transport, and the coercive window are preserved in full.

11 Native product, factorization, and irreducibility

This chapter develops the arithmetic sector of normal forms over the common kernel and the correlative construction of the continuum in Section 9. The local operation extends to words, its fibers distinguish the irreducibles, and evaluation makes it possible to recognize the primes. The development recovers the native product of the corpus; its documentary provenance is collected in Appendix 21.

11.1 The information required for multiplication

Selecting prime numbers requires a multiplication to have been constructed. In Triadic Modular Holography, this operation must be presented from the two APP sheets, preserving both the residual coordinate and its quotient. The residue alone makes it possible to recognize a congruence class. The residue–quotient pair makes it possible to reconstruct the integer result of the local operation. This distinction determines which information must be transported when composing the cells.

Let $D_9 = \{1, \dots, 9\}$. For each pair $(a, b) \in D_9^2$, the additive and multiplicative sheets are written as

$$a + b = R_+(a, b) + 9Q_+(a, b), \quad ab = R_\times(a, b) + 9Q_\times(a, b),$$

with $R_+, R_\times \in D_9$. The choice of the positive representative uniquely fixes the four data. In particular, a result divisible by nine has its residue represented by 9, and its quotient is one less than that of Euclidean division with zero remainder. This is an exact representation convention, which must be maintained throughout the construction.

For example,

$$2 \cdot 5 = 1 + 9 \cdot 1, \quad 9 \cdot 9 = 9 + 9 \cdot 8.$$

The first result preserves the information that distinguishes 10 from 1; the second preserves that which distinguishes 81 from 9. If the quotient is omitted at one of these steps, a subsequent multiplication acts on a different object. Arithmetic memory here is not an interpretation added to the algorithm: it participates in reconstructing its result.

The two sheets are coupled even during multiplication. The product of two digits generates a pair $(R_\times, Q_\times)$; when the quotient from a preceding position is incorporated into the current position, the additive sheet is needed again. The global operation composes both sheets.

Proposition 11.1. Local compatibility of the quotients

The quotients satisfy the identities required to associate and distribute the operations without changing the reconstructed integer. For example, for addition,

$$Q_+(a, b) + Q_+(R_+(a, b), c) = Q_+(b, c) + Q_+(a, R_+(b, c)).$$

For multiplication,

$$Q_\times(R_\times(a, b), c) + cQ_\times(a, b) = Q_\times(a, R_\times(b, c)) + aQ_\times(b, c).$$

Proof. Substituting the two decompositions of $a + b + c$, the final residues are the same positive representative of its class modulo nine. Subtracting them and dividing by nine gives the first equality. For abc , the same argument is applied to

$$(R_\times(a, b) + 9Q_\times(a, b))c = a(R_\times(b, c) + 9Q_\times(b, c)).$$

The distributive identity is obtained by reconstructing $a(b + c) = ab + ac$ with the two sheets. In all three cases, uniqueness of the residual representation determines exactly the quotient that must accompany it. $\square$

These identities make it possible to organize the operation record: two parenthesizations produce the same integer without requiring their execution histories to be identified. Equality of the result and identity of the history are different assertions.

11.2 Nonadic words and normal form

The extension to arbitrarily large integers uses finite sequences of digits from D_9 . The orientation adopted runs from the least significant to the most significant position. For $u = (u_0, \dots, u_{r-1})$, let

$$\text{ev}_9(u) = \sum_{j=0}^{r-1} u_j 9^j.$$

The empty sequence represents zero. In this bijective numeration, the symbol 9 remains a digit; it does not become a zero without a quotient.

Proposition 11.2. Existence and uniqueness of the normal form

The preceding evaluation is a bijection between finite words over D_9 , including the empty word, and nonnegative integers.

Proof. If $n > 0$, define

$$d(n) = 1 + ((n - 1) \bmod 9), \quad q(n) = \frac{n - d(n)}{9}.$$

Then $d(n) \in D_9$, $q(n) \geq 0$, and $q(n) < n$. Iteration terminates at zero and produces a word whose evaluation is n . For uniqueness, the first digit of any word evaluating to n must represent the class of n modulo nine within D_9 ; it must therefore be $d(n)$. Removing it and dividing by nine reduces the question to $q(n)$. Induction proves uniqueness. $\square$

Some normal forms are

$$9 \leftrightarrow (9), \quad 10 \leftrightarrow (1, 1), \quad 81 \leftrightarrow (9, 8), \quad 1000 \leftrightarrow (1, 3, 3, 1).$$

The representation is checked directly: $1000 = 1 + 3 \cdot 9 + 3 \cdot 9^2 + 9^3$. This last equality does not by itself identify the word of an integer with the cubic completion of a geometric domain. The two occurrences of 1000 have different objects and operations; an identification would require its map.

The normal form is an arithmetic coordinate. The TPK state additionally preserves orientation, sheet, route, boundary, and transport memory. When different histories produce the same normal form, the evaluation map brings their values together, but does not establish a bijection between those histories. The development of the article must preserve both levels and the map relating them.

11.3 Construction of the product without prior evaluation of the operands

Word addition is performed position by position. If a position contains two digits, the additive sheet is consulted. If it contains only one, that digit is retained. The quotient arriving from the preceding position is incorporated through another additive lookup, and the quotients of both lookups are added. If $a + b = r + 9q$ and $r + c = r' + 9q'$, then $q + q'$ is transported. For example, $9 + 9 + 1 = 1 + 9 \cdot 2$. The incoming quotient lies between zero and two; when it is zero, no additional cell is consulted. An absent position is not an APP cell containing the digit zero.

Multiplication by a digit $d \in D_9$ acts analogously: at each position, the multiplicative sheet is first applied to the digit of the word and to d ; the pending quotient is then incorporated through the additive sheet. The new quotient is the sum of the quotient of the local product and that of this additive lookup, and lies between zero and nine. The procedure terminates by appending the final quotient when it is nonzero.

Denote these operations by $u \boxplus v$ and $d \odot u$. If $v = (v_0, \dots, v_{m-1})$, the native product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot a_{j+1}) \boxplus (v_j \odot u), \quad j = m-1, \dots, 0,$$

and $u \boxtimes v = a_0$ is defined. Every operation on the right-hand side uses the two local sheets already defined. Word evaluation does not intervene as an instruction for computing the product.

Theorem 11.3. Exactness of the native product

For all finite words u, v ,

$$\text{ev}_9(u \boxtimes v) = \text{ev}_9(u) \text{ev}_9(v).$$

Proof. The cell reconstruction identities prove, by induction over positions, that

$$\text{ev}_9(u \boxplus v) = \text{ev}_9(u) + \text{ev}_9(v), \quad \text{ev}_9(d \odot u) = d \text{ev}_9(u).$$

The product recurrence preserves the invariant

$$\text{ev}_9(a_j) = \text{ev}_9(u) \sum_{k=j}^{m-1} v_k 9^{k-j}.$$

The case $j = m$ is immediate. If it holds for $j + 1$, multiplying by nine and adding $v_j \text{ev}_9(u)$ yields the case j . For $j = 0$, the asserted equality follows. $\square$

Corollary 11.4. Multiplicative structure of the normal forms

The nonempty words, with the product $\boxtimes$, form a commutative monoid whose unit is (1). Upon adjoining the empty word, it is absorbing.

Proof. Evaluation is multiplicative by the theorem and injective by Proposition 11.2. Associativity, commutativity, and the identity are verified after evaluation and recovered by injectivity. $\square$

The proof makes it possible to recognize ordinary multiplication in the constructed operation. The logical order matters: the cell transducer is first defined and its invariant is proved; its integer realization is then identified.

11.4 Multiplicative openings and selection of irreducibles

Once the product has been constructed, factorization can be defined as its fiber. For each nonempty word w , let

$$\text{Fac}(w) = \{(u, v) : u \boxtimes v = w\}.$$

This fiber is finite. Indeed, if $n = \text{ev}_9(w)$, exactness implies $\text{ev}_9(u) \text{ev}_9(v) = n$, with both factors positive and no greater than n . Finiteness justifies the algebraic sum

$$\Delta e_w = \sum_{(u,v) \in \text{Fac}(w)} e_u \otimes e_v.$$

The order in the factor pair is preserved. For $n = 12$, for example, the evaluations of the fiber are

$$(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1).$$

The reduced coproduct removes the two decompositions containing the unit, for $w \neq (1)$:

$$\tilde{\Delta}e_w = \sum_{\substack{u \boxtimes v = w \\ u \neq (1), v \neq (1)}} e_u \otimes e_v.$$

Set $\tilde{\Delta}e_{(1)} = 0$. This convention allows the unit to be treated separately, without confusing it with an irreducible.

Definition 11.5. Native irreducible

A word $w \neq (1)$ is multiplicatively irreducible if $\tilde{\Delta}e_w = 0$.

The definition receives no table of primes and uses no decimal period. It asks whether the operation already constructed admits a nontrivial opening.

Theorem 11.6. Recognition of the primes

Evaluation establishes a bijection between native irreducibles and positive prime numbers.

Proof. If $w = u \boxtimes v$ with factors different from the unit, its evaluation is the product of two integers greater than one; it is therefore not prime. Conversely, if $\text{ev}_9(w) = ab$, with $a, b > 1$, their normal forms u, v satisfy

$$\text{ev}_9(u \boxtimes v) = ab = \text{ev}_9(w).$$

Injectivity gives $u \boxtimes v = w$, so that w is not irreducible. The bijection of normal forms completes the proof. $\square$

This result identifies a property of the operation with the property of its integer realization. Prime values appear upon evaluating the selected objects, after the product and its fibers have been defined.

Proposition 11.7. Coassociativity

On the algebraic space generated by positive words,

$$(\Delta \otimes I)\Delta = (I \otimes \Delta)\Delta.$$

Proof. Both sides enumerate, with coefficient one, the ordered triples (u, v, z) whose product is w . Associativity of the product identifies the two parenthesizations, and all sums are finite. $\square$

11.5 Operator realization of the selector

In $\mathcal{H} = \ell^2(\mathcal{W}_9^+)$, with orthonormal basis $\{e_w\}$, the coproduct is first defined on finite linear combinations. The supports of Δe_w and $\Delta e_{w'}$ are disjoint if $w \neq w'$: an ordered pair has a unique product. Hence,

$$\|\Delta e_w\|^2 = |\text{Fac}(w)|, \quad \|\tilde{\Delta}e_w\|^2 = r(w),$$

where $r(w)$ counts the ordered nontrivial openings.

The closure of $\tilde{\Delta}$ has domain

$$\left\{ x = \sum_w x_w e_w \in \mathcal{H} : \sum_w r(w) |x_w|^2 < \infty \right\}.$$

Its associated positive operator is diagonal:

$$A_{\text{irr}} = \overline{\overline{\Delta}}^* \overline{\overline{\Delta}}, \quad A_{\text{irr}} e_w = r(w) e_w.$$

Its maximal domain is

$$\text{Dom}(A_{\text{irr}}) = \left\{ x \in \mathcal{H} : \sum_w r(w)^2 |x_w|^2 < \infty \right\}.$$

The projector onto the irreducibles is then

$$P_{\text{irr}} = (I - P_{(1)}) \mathbf{1}_{\{0\}}(A_{\text{irr}}).$$

The formula incorporates two distinct exclusions: the kernel removes all words possessing a nontrivial opening, and $I - P_{(1)}$ removes the unit. Positivity comes from a squared norm and has a specific meaning here: it counts decompositions. It is not yet an assertion about the positivity of another form, such as the completed Weil form.

11.6 Normalized readout of the TPK prefix and preservation of genealogy

The normal form admits an explicit composition with the integer prefix produced by the TPK. Denote by $\beta_9 = \text{ev}_9^{-1}$ the normalizer of Proposition 11.2. Here K is the block depth of the TPK state, distinct from the length k of an elementary history in the emission tree. The construction applies to a regional coordinate of the block state; the same calculation applies to each of the three coordinates with its own record.

For the deep row vector $x_K \in \mathbb{Z}^6$, the integer TPK step uses the matrix

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}$$

and produces

$$y_K = x_K A_H, \quad u_{K+1} = y_K \bmod 3, \quad x_{K+1} = \frac{y_K - u_{K+1}}{3}.$$

The block $u_{K+1} \in \mathbb{F}_3^6$, with representatives $\bar{u}_j \in \{0, 1, 2\}$, has index

$$I(u_{K+1}) = \sum_{j=1}^6 \bar{u}_j 3^{6-j}, \quad 0 \leq I(u_{K+1}) < 729.$$

The cylindrical readout updates the prefix through

$$N_{K+1} = 729 N_K + I(u_{K+1}).$$

The matrix A_H belongs to the TPK kernel update. In the following composition it is not replaced by an external arithmetic rule: the block and its index come from the step just executed.

Proposition 11.8. Compatibility of the native prefix and its normal form

If $w_K = \beta_9(N_K)$, then

$$w_{K+1} = (\beta_9(729) \boxtimes w_K) \boxplus \beta_9(I(u_{K+1})).$$

The typed truncation of this readout is

$$\varrho(w) = \beta_9\left(\left\lfloor \frac{\text{ev}_9(w)}{729} \right\rfloor\right); \quad \varrho(w_{K+1}) = w_K.$$

Proof. Compatibility of word addition and multiplication proves that the evaluation of the right-hand side is

$$729 \text{ev}_9(w_K) + I(u_{K+1}) = 729N_K + I(u_{K+1}) = N_{K+1}.$$

Uniqueness of the normal form gives the first equality. Since the block index lies between zero and 728, the integer quotient of N_{K+1} by 729 is N_K . Applying β_9 proves the truncation identity. $\square$

Truncation does not consist of deleting three digits from a bijective word. For example, $\beta_9(729) = (9, 8, 8)$, whereas the integer quotient of 729 by 729 is one; deleting its three digits would produce the empty word. The rule ϱ preserves the residue and quotient convention defining the numeration.

The proposition normalizes an already generated prefix and preserves its compatibility under refinement. Its application to the state retains, in the original record, the block u_{K+1} , the deep quotient x_{K+1} , the carries of the decimal chart, the phase, and the route. It does not assert that w_K alone reconstructs those coordinates or that any word is expressed by a fixed regional section. Equality of normal forms concerns the numerical prefix; identity of histories is decided in the complete state.

The record accompanying the product contains the lookups of both sheets and their residues and quotients. It makes it possible to explain why an output was obtained and to reproduce the operation. The transducer result does not, however, exhaust the state of a TPK prolongation: a normal form may be reached by different histories.

Integration with the dynamics therefore preserves the direction of each map: the TPK state expresses its prefix, which admits the normalized readout just constructed. Compatibility of a product on histories additionally requires specifying its domain, its transport, and its survival. It does not require inverting a numerical readout that does not individualize the history. Identity of values does not replace this compatibility square; each application must be accompanied by the concrete operation and the information it preserves.

The starting point of the arithmetic is already specified: two local operations, a normal form, a global transducer, its fibers, and a selector. The spectral extension must use these results without retrospectively changing their origin.

11.7 Correspondence with the preserved implementation

The code `a9_native.py` implements the operations in this order:

1. Local addition and multiplication cells.
2. Bijective nonadic encoding and separate evaluation.
3. Word addition and multiplication by a digit.
4. Word product recurrence.
5. Enumeration of product fibers.
6. Selection of irreducibles by absence of a nontrivial opening.

The checker uses integer multiplication and a conventional primality test after generating the outputs. These operations check the implementation; they are not incorporated into the body of the word producer. Checks that suppress a quotient or alter a cell document which dependency is broken. The reproduction receipts state the size of the checked window; that computational scope is kept separate from the preceding general proofs.

12 Factorization, arithmetic clocks and hierarchy of traces

The native product distinguishes two properties: irreducibility of a normal form and the period of a multiplicative action on it. This chapter constructs their spectral readouts, preserves the normalizations of each functional and prepares the completed form developed in Section 15.

12.1 From the irreducible to its clock

Product and coproduct determine which objects are irreducible. The arithmetic clock describes another property of those objects: return of a finite multiplicative orbit. For an integer $n \geq 2$ coprime to ten, define

$$\tau_n = \min\{k \geq 1 : 10^k \equiv 1 \pmod{n}\}.$$

The length τ_n corresponds to the orbit of the unit under multiplication by ten in the group of units modulo n . The decimal base fixes this readout; another coprime base determines another clock. Irreducibility, by contrast, belongs to the product constructed before this periodic representation is chosen.

Proposition 12.1. Synchronization of clocks

If $n = \prod_p p^{a_p}$ and $\gcd(n, 10) = 1$, then

$$\tau_n = \text{mcm}_{p^{a_p} \parallel n} \tau_{p^{a_p}}.$$

Proof. An exponent k satisfies $10^k \equiv 1 \pmod{n}$ if and only if it does so modulo each prime power in the factorization. This is equivalent to k being a multiple of every order $\tau_{p^{a_p}}$. The smallest common positive exponent is the least common multiple. $\square$

Factorization therefore organizes synchronization. Powers of the same prime must be retained as such: their period is not obtained by replacing them, without proof, with the base prime's period.

Proposition 12.2. Nonadic residual closure

If $\gcd(n, 90) = 1$, the integer block

$$M_n = \frac{10^{\tau_n} - 1}{n}$$

is divisible by nine.

Proof. The period definition makes M_n integral. Since $10 \equiv 1 \pmod{9}$, the numerator is divisible by nine. The integer n is invertible modulo nine, so $nM_n \equiv 0 \pmod{9}$ implies $M_n \equiv 0 \pmod{9}$. $\square$

For a prime $p \notin \{2, 3, 5\}$, one also has $\tau_p \mid p - 1$, since its orbit belongs to the multiplicative group of order $p - 1$. The congruence of Proposition 12.2 also holds for composites coprime to ninety. Its meaning is residual closure; irreducibility remains determined by product openings.

The numbers 7, 13, 77, 91, 143 have decimal period six, but only the first two are irreducible. The structural relation is not that a period by itself identifies a prime, but that the product constructs components whose composition organizes returns.

12.2 Factorization as mode occupation

Let $\mathcal{P}$ be the set obtained by evaluating native irreducibles. Each mode occupation is a collection $\mathbf{k} = (k_p)_{p \in \mathcal{P}}$ of nonnegative integers with finite support. Its multiplicative evaluation is

$$N(\mathbf{k}) = \prod_{p \in \mathcal{P}} p^{k_p}.$$

Unique factorization identifies these occupations with the positive integers. The Hilbert spaces of occupations and of positive normal forms then have a unitary correspondence of bases. An occupation register preserves how often each irreducible occurs.

In the total-occupation-one sector, $\sum_p k_p = 1$, with basis $\{e_p : p \in \mathcal{P}\}$, define

$$H_{\mathcal{P}} e_p = (\log p) e_p,$$

and, on occupations,

$$H_{\text{occ}} e_{\mathbf{k}} = \left(\sum_p k_p \log p \right) e_{\mathbf{k}}.$$

With maximal diagonal domains, both are self-adjoint. If $Q e_n = n e_n$, the basis identification satisfies

$$U H_{\text{occ}} U^* = \log Q.$$

The equality follows from multiplicative composition:

$$\sum_p k_p \log p = \log N(\mathbf{k}).$$

The operator is not used to select primes retrospectively. It receives the irreducibles already constructed and converts their composition into an additive quantity. This is the precise place of the occupation representation in the article's genealogy.

12.3 One occupation structure and two trace levels

For $\Re s > 1$, the trace on the total-occupation-one sector is

$$P(s) = \text{Tr}(e^{-s H_{\mathcal{P}}}) = \sum_{p \in \mathcal{P}} p^{-s}.$$

The trace on all occupations, by contrast, is

$$Z(s) = \text{Tr}(e^{-s H_{\text{occ}}}) = \prod_{p \in \mathcal{P}} (1 - p^{-s})^{-1} = \sum_{n \geq 1} n^{-s}.$$

The last equality follows from unique factorization. The stated half-plane guarantees absolute convergence and permits grouping of terms. After identification of the integer realization, Z agrees there with the zeta function. No table of zeta values has been used to define the modes.

Proposition 12.3. Hierarchy of moments

On $\Re s > 1$, taking the logarithm branch determined by expansion of the convergent product,

$$\log Z(s) = \sum_{k \geq 1} \frac{P(ks)}{k}.$$

Proof. The absolutely convergent expansion $-\log(1 - z) = \sum_{k \geq 1} z^k/k$, applied to each $z = p^{-s}$, gives

$$\log Z(s) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}.$$

Absolute convergence permits interchange of the sums. $\square$

The two indices preserve different functions: p identifies the irreducible mode and k its occupation multiplicity. Their relationship does not reduce to a list of constants: it produces a collection of moments on the same irreducible set.

12.4 Meissel–Mertens and removal of the linear contribution

The coordinated extension brings together the finite part

$$B_1 = \gamma + \sum_p (\log(1 - p^{-1}) + p^{-1}).$$

The added term p^{-1} cancels exactly the logarithm's first order. Thus,

$$B_1 = \gamma - \sum_{k \geq 2} \frac{P(k)}{k}.$$

Here γ denotes the finite part γ_0 of the same functional, whose construction and bound are developed in Section 13; it is not a metrological input or a decimal chosen for evaluation. The formula relates two operations on the constructed functional: extraction of the finite part and cancellation of each mode's first moment.

Proposition 12.4. Bound on the moment tail

For an integer $K \geq 1$,

$$0 < \sum_{k > K} \frac{P(k)}{k} \leq \frac{2P(K+1)}{K+1} \leq \frac{2^{-K}}{K+1} \left(1 + \frac{2}{K}\right).$$

Proof. Since $k \geq K+1$,

$$\sum_{k > K} \frac{P(k)}{k} \leq \frac{1}{K+1} \sum_p \frac{p^{-(K+1)}}{1 - p^{-1}} \leq \frac{2P(K+1)}{K+1}.$$

By comparison with all integers at least two,

$$P(K+1) \leq \sum_{n \geq 2} n^{-(K+1)} \leq 2^{-(K+1)} + \int_2^\infty x^{-(K+1)} dx.$$

Evaluating the integral gives the second bound. $\square$

The inequality converts the finite sum into an interval, provided the finite evaluations and γ themselves have controlled bounds. The length of a decimal expansion does not replace this estimate.

12.5 Residual exclusion and the separation-two pattern product

For the two-position pattern $\{0, 2\}$, let ν_p be the number of forbidden classes modulo p . At $p = 2$ the two positions coincide; for $p > 2$, they are two distinct classes. Local normalization relative to two independent choices produces

$$\sigma_p = \frac{1 - \nu_p/p}{(1 - 1/p)^2}.$$

The factor at $p = 2$ equals two, while the others form

$$C_2 = \prod_{p>2} \frac{1 - 2/p}{(1 - 1/p)^2} = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right).$$

The pattern's complete singular series is $2C_2$. It is useful to reserve different symbols for the product over $p > 2$ and for the full factor: normalization distinguishes a specific residual contribution, not a dispensable convention.

Proposition 12.5. The same moment system with a different weight

The product expansion satisfies

$$\log C_2 = - \sum_{k \geq 2} \frac{2^k - 2}{k} (P(k) - 2^{-k}).$$

Proof. For each $p > 2$,

$$\log(1 - 2/p) - 2 \log(1 - 1/p) = - \sum_{k \geq 2} \frac{2^k - 2}{k p^k}.$$

The sum begins at $k = 2$, because the linear term cancels. For $p \geq 3$,

$$-\log \left(1 - \frac{1}{(p-1)^2}\right) \leq \frac{4}{3(p-1)^2}.$$

Comparison with the inverse-square series proves finiteness of the absolute-value sum and permits interchange of order. Removing mode 2 produces $P(k) - 2^{-k}$. $\square$

The same hierarchy $P(k)$ thus describes two different operations: Meissel–Mertens linear cancellation and residual exclusion of the two-position pattern. The difference lies in the weight of each moment and removal of mode 2, not in introducing two independent prime catalogues.

For an integer $K \geq 1$, let

$$S_K = \sum_{k=2}^K \frac{2^k - 2}{k} (P(k) - 2^{-k}),$$

and the coordinated extension gives the bound

$$0 < -\log C_2 - S_K \leq \frac{3}{K+1} \left(\frac{2}{3}\right)^{K+1} \left(1 + \frac{3}{K}\right).$$

It is obtained by substituting $2^k - 2 \leq 2^k$, summing the geometric series for each $p \geq 3$, bounding $(1 - 2/p)^{-1} \leq 3$ and comparing the sum over primes with the sum over integers $n \geq 3$. Exponentiation gives an interval for C_2 . Convergence of this product is a result distinct from existence of infinitely many prime pairs separated by two.

12.6 Irreducible-mode truncation and remainder control

Occupation depth K and the irreducible-mode cutoff Y define distinct approximations of the same arithmetic representation. The preceding bounds retain every irreducible mode and truncate depth. Here the local contributions remain complete and the modes produced by the selector are restricted to $p \leq Y$, where $Y \geq 2$ is an integer. Define

$$B_1(Y) = \gamma_0 + \sum_{p \leq Y} (\log(1 - p^{-1}) + p^{-1}), \quad C_2(Y) = \prod_{2 < p \leq Y} \left(1 - \frac{1}{(p-1)^2}\right).$$

The empty product equals one. The finite part γ_0 is the object denoted by γ in the Meissel–Mertens formula; its evaluation with a remainder is developed in Section 13.

Proposition 12.6. Enclosures from an irreducible-mode cutoff

For every integer $Y \geq 2$,

$$B_1(Y) - \frac{1}{2Y} \leq B_1 \leq B_1(Y), \quad \frac{Y-1}{Y} C_2(Y) \leq C_2 \leq C_2(Y).$$

Proof. For each $p \geq 2$, the contribution subtracted from $B_1(Y)$ is

$$d_p = -\log(1 - p^{-1}) - p^{-1} = \sum_{k \geq 2} \frac{1}{kp^k}, \quad 0 < d_p \leq \frac{1}{2p(p-1)}.$$

The last inequality uses $1/k \leq 1/2$ and the geometric sum. Positive comparison and telescoping yield

$$0 \leq B_1(Y) - B_1 = \sum_{p > Y} d_p \leq \frac{1}{2} \sum_{n > Y} \frac{1}{n(n-1)} = \frac{1}{2Y}.$$

Every factor in the tail of C_2 lies in $(0, 1)$. Including every integer $n > Y$ decreases the product. For $M > Y$,

$$\prod_{n=Y+1}^M \left(1 - \frac{1}{(n-1)^2}\right) = \prod_{n=Y+1}^M \frac{n(n-2)}{(n-1)^2} = \frac{M(Y-1)}{Y(M-1)}.$$

Its limit is $(Y-1)/Y$. Convergence of the prime product and the preceding comparison give the second enclosure. $\square$

As Y increases, $B_1(Y)$ and $C_2(Y)$ decrease towards their respective values. The guaranteed widths are $1/(2Y)$ and $C_2(Y)/Y$. The depth and irreducible-mode enclosures may be intersected to refine the evaluation; each retains its parameter and proof. The product C_2 corresponds to the sector $p > 2$; the local factor at mode two preserves the normalization $2C_2$ of the complete singular series.

12.7 Triadic realization of the functional

The source for spectral functionals uses cycles $C_m = \mathbb{Z}/3^m\mathbb{Z}$ and their Laplacian

$$(\Delta_m f)(j) = 2f(j) - f(j+1) - f(j-1).$$

Writing $N_m = 3^m$, let $e_{m,n}$ be the normalized characters of the cycle and $\tilde{n}$ the representative of n between $-(N_m-1)/2$ and $(N_m-1)/2$. The signed-orientation operator is defined by

$$D_m e_{m,n} = \frac{N_m}{\sqrt{\pi}} \sin\left(\frac{\pi \tilde{n}}{N_m}\right) e_{m,n}.$$

This fixes the sign, not merely the square. The realization satisfies

$$D_m^2 = \frac{3^{2m}}{4\pi} \Delta_m.$$

Let P_0 be the orthogonal projector onto constant functions on the cycle. The coordinate π belongs to the prior generation of the HMT kernel. This normalization is part of the analytic realization and must preserve its provenance. The positive trace removes the constant mode and counts one contribution per pair of opposite orientations:

$$\mathrm{Tr}_m^+(F(D_m)) = \frac{1}{2} \mathrm{Tr}(F(D_m)P_0^\perp).$$

With the cycle's Fourier modes, the heat trace converges to

$$\Theta_+(x) = \sum_{n \geq 1} e^{-\pi n^2 x}, \quad x > 0.$$

The proof, developed in the first section of the following chapter, preserves a Gaussian majorant independent of refinement level. The Mellin transform of this limit satisfies exactly

$$\int_0^\infty \Theta_+(x) x^{s/2-1} dx = \pi^{-s/2} \Gamma(s/2) Z(s), \quad \Re s > 1.$$

This brings together additive construction of the integer spectrum and multiplicative decomposition into irreducibles. The identity between functionals is proved; it is not inferred from proximity of two numerical evaluations.

Theta inversion and continuation of the completed function are proved in the next chapter. Finite parts, Laurent coefficients and residual classes modulo three require their specific evaluations; they do not follow merely from heat-trace convergence.

Transport defect and uniform component

The extension on sheet folding provides an explicit representation of a one-winding defect. Let $N = 9^m$, with $m \geq 1$, and let T be a unitary operator on a Hilbert fibre $\mathcal{H}$. On $\mathbb{C}^N$, use the usual Euclidean inner product and the unit vector

$$u_N = N^{-1/2}(1, \dots, 1).$$

Define $\iota_N f = u_N \otimes f$, $P_N = \iota_N \iota_N^*$ and $Q_N = I - P_N$. Transport applies T once upon completion of the cycle:

$$U_N(T) = \sum_{r=1}^{N-1} |r+1\rangle\langle r| \otimes I + |1\rangle\langle N| \otimes T.$$

It acts on the uniform state by

$$U_N(T) \iota_N f = \iota_N f + \frac{e_1}{\sqrt{N}} \otimes (T - I) f.$$

The added term has a uniform component and an orthogonal component. The latter, with the source's normalization, is

$$\delta_N(T) = \frac{N}{\sqrt{N-1}} Q_N U_N(T) \iota_N = \eta_N \otimes (T - I),$$

where

$$\eta_N = \sqrt{\frac{N}{N-1}} \left(e_1 - \frac{u_N}{\sqrt{N}} \right), \quad \|\eta_N\| = 1, \quad \eta_N \perp u_N.$$

The equality is checked by substituting the preceding action and applying Q_N . For two unitary transports T, S on the same fibre, one obtains

$$\delta_N(T)^* \delta_N(S) = (I - T)^* (I - S).$$

Commutativity is not required: the products occur in the indicated order. The same decomposition gives

$$\delta_N(T)^* P_N \delta_N(S) = 0,$$

since P_N annihilates the factor η_N .

These identities specify the information preserved by defect normalization and its position relative to the uniform component. They also allow the closures of the images to be transported isometrically between two cycle lengths: the Gram operator is independent of N . Its domain is the constructed defect sector; it is not by itself an identification of all TPK histories or of any subsequent spectral projection.

12.8 The completed form and its boundary identity

Passage to the double circle uses a form on a test-function domain. Its arithmetic side brings together the prime-power, gamma and polar contributions. Writing it with two columns,

$$\mathcal{J}_{\text{GW}} = J_+^* J_+ - J_-^* J_-,$$

exhibits a difference of forms. The prime-index producer, orbital weights, Fourier normalization and completion must appear before it is identified with another form.

The source in Chapter 96 places the central step in an encoder Ξ and an orthogonal projection P , through

$$\langle \Xi f, \Xi g \rangle = \langle J_+ f, J_+ g \rangle, \quad \langle P \Xi f, P \Xi g \rangle = \langle J_- f, J_- g \rangle.$$

If these two identities are obtained from the declared native actions, their difference gives exactly

$$\mathcal{J}_{\text{GW}}(f, g) = \langle (I - P) \Xi f, (I - P) \Xi g \rangle.$$

The next chapter proves the explicit column decomposition and the consequence of the two moments. Application to the native encoder retains as antecedents its action on test functions, the position of its charts relative to P and the proof of both moments. Orthogonality of P explains the last equality but does not by itself determine the preceding two identities.

Transport telescoping, in turn, preserves the form

$$\|x\|^2 = \sum_{r=0}^{N-1} \|DA^r x\|^2 + \|A^N x\|^2,$$

when $D^* D = I - A^* A$. If on the sheet considered $A = qV$, with $q = 5/9$ and V isometric, the final term equals $q^{2N} \|x\|^2$ and tends to zero. The proof of this extinction is preserved together with the identity placing state x on that sheet.

12.9 Spectral circle and memory circle

The first circle is a coordinate of the spectral line. For $s = 1/2 + i\gamma$, with $\gamma \in \mathbb{R}$, the transformation

$$\tau(s) = \frac{s-1}{s} = \frac{\gamma + i/2}{\gamma - i/2}$$

belongs to the unit circle without the point 1 and is invertible through

$$s = \frac{1}{1-\tau}, \quad \gamma = \frac{1}{2} \cot \frac{\theta}{2}, \quad \tau = e^{i\theta}.$$

The second circle is the character collection of integer memory:

$$\chi_\tau(n) = \tau^n, \quad n \in \mathbb{Z}.$$

The nonadic connection increments memory by one upon completion of a winding. Intertwining relates a spectral phase to its action on that memory; it does not identify zero number j with winding number j .

If $C \in \mathcal{B}(\mathcal{H})$ is the phase operator obtained in the spectral realization, the nine-position calendar acts on $\mathbb{C}^9 \otimes \mathcal{H}$ by

$$S_{C,9} = \sum_{r=1}^8 |r+1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C.$$

Each vector traverses the link containing C exactly once in a complete winding. Thus,

$$S_{C,9}^9 = I_9 \otimes C.$$

The identity is exact for any C ; when C is unitary, $S_{C,9}$ is also unitary. The contractive factor $5/9$ belongs to another transport readout and is distinguished from this unitary phase.

The chapter's interest does not lie in representing an arbitrary matrix on nine positions. It lies in bringing together the origin of operator C , its domain, the preceding encoder and compatibility with TPK cylinders. Those data fix what is transported and why this realization is relevant.

12.10 Compatibility with the joint structure of the continuum

The history space, two sheets, memory, incidence and complex are not reconstructed as independent objects for each section. The prior correlative assembly is developed in Section 9, with its operations, realization conditions and naturality squares. The integer, depth, irreducible and spectral-height operators remain typed; coincident notation does not identify them.

Section 15 brings together the boundary identity with its terminal construction, test functions, terms of the completed form and convergence. Matrix approximations preserve their three matrices of form, moment and residual. Digits are compared only after outputs have been produced and frozen.

This organization preserves the requested argument: construction, operation, resulting structure and recognition. The manuscript does not change the causal kernel to facilitate a subsequent calculation, nor use a list of zeros as input to the arithmetic producer.

13 Finite parts, regularization and residual readouts

The functional constructed from integer modes admits operations preserving different aspects of its structure. Evaluating it at a point of convergence, extracting the regular term accompanying a pole and differentiating its meromorphic continuation are three precise operations. This distinction organizes the constants of this chapter and their relationship with those in the moments chapter.

The arithmetic antecedent is representation of the normal forms obtained with the two APP sheets. Their evaluation identifies each positive integer once; the enumeration operator acts on the corresponding basis by

$$Qe_n = ne_n, \quad \text{Dom } Q = \left\{ u : \sum_{n \geq 1} n^2 |u_n|^2 < \infty \right\}.$$

The subsequent spectral readout is

$$Z(s) = \text{Tr}(Q^{-s}) = \sum_{n \geq 1} n^{-s}, \quad \Re s > 1.$$

The chapter on the completed form proves its agreement with the theta–Mellin functional and its continuation. Here we bring together an evaluation procedure with controlled remainders. History memory remains in the source state; the trace is a readout of expressed modes, not a reconstruction of that entire history from a scalar constant.

13.1 A continuation formula with explicit remainder

Define the Bernoulli numbers through the formal identity

$$\frac{t}{e^t - 1} = \sum_{j \geq 0} B_j \frac{t^j}{j!}.$$

Their coefficients are computed without a numerical table:

$$B_0 = 1, \quad B_m = -\frac{1}{m+1} \sum_{j=0}^{m-1} \binom{m+1}{j} B_j.$$

In particular, $B_1 = -1/2$. Write $(s)_r = s(s+1) \cdots (s+r-1)$, with $(s)_0 = 1$, and denote the corresponding polynomial by $B_m(x) = \sum_{j=0}^m \binom{m}{j} B_j x^{m-j}$. Its periodization is $\tilde{B}_{2p}(x) = B_{2p}(\{x\})$. For $N, p \geq 1$, Euler–Maclaurin summation gives

$$\begin{aligned} Z(s) &= \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1} - \frac{1}{2N^s} \\ &\quad + \sum_{k=1}^p \frac{B_{2k}}{(2k)!} (s)_{2k-1} N^{1-s-2k} + R_{N,p}(s), \end{aligned}$$

with

$$R_{N,p}(s) = -\frac{(s)_{2p}}{(2p)!} \int_N^\infty \tilde{B}_{2p}(x) x^{-s-2p} dx.$$

Proposition 13.1. Domain and bound of the remainder

The remainder integral is holomorphic on $\Re s > 1 - 2p$. If $\sigma = \Re s > 1 - 2p$, then

$$|R_{N,p}(s)| \leq \frac{|B_{2p}| |(s)_{2p}|}{(2p)!} \frac{N^{1-\sigma-2p}}{\sigma + 2p - 1}.$$

The preceding formula continues Z to that half-plane, with its simple pole at $s = 1$, and is compatible with the theta–Mellin continuation.

Proof. The summation formula is obtained by successive integration by parts on each unit interval, using $B_j'(x) = jB_{j-1}(x)$ and the endpoint equalities for polynomials of degree other than one. Applying it to x^{-s} on $[N, M]$, the derivatives supply the factors $(s)_r$. On $\Re s > 1$, passage to $M \rightarrow \infty$ removes the upper endpoint terms and produces exactly the identity and remainder sign written above.

To control the remainder, use

$$|B_{2p}(t)| \leq |B_{2p}|, \quad 0 \leq t \leq 1.$$

One proof of this inequality is the absolutely convergent Fourier series

$$B_{2p}(t) = (-1)^{p+1} \frac{2(2p)!}{(2\pi)^{2p}} \sum_{m \geq 1} \frac{\cos(2\pi mt)}{m^{2p}}.$$

The maximum of the absolute-value bound is attained at $t = 0$. This identity follows from the Fourier coefficients by integration by parts of the polynomials themselves; the factor π merely expresses the Fourier convention of this proof and does not enter numerically into the rational evaluator used later. Comparison with $x^{-\sigma-2p}$ proves the remainder bound. On compact subsets of the half-plane, it also dominates derivatives in s , which add powers of $\log x$; the integral is therefore holomorphic there. The identity on the convergence half-plane and uniqueness of continuation prove the stated compatibility. $\square$

Order p and horizon N are computation and error-control parameters. They do not select a different constant: expressions with different N, p represent the same functional on their common domains. Nor must an asymptotic series converge as p increases indefinitely with N fixed. Each finite pair comes with its own proved bound.

13.2 Euler–Mascheroni as the finite part of the same functional

The finite part at the pole is defined by

$$\gamma_0 = \lim_{s \rightarrow 1} \left(Z(s) - \frac{1}{s-1} \right).$$

Setting $s = 1 + z$, the singular term satisfies

$$\frac{N^{-z}}{z} = \frac{1}{z} - \log N + O(z).$$

The preceding proposition allows the finite part to be taken term by term.

Theorem 13.2. Explicit interval for the finite part

Let

$$G_{N,p} = H_N - \log N - \frac{1}{2N} + \sum_{k=1}^p \frac{B_{2k}}{2kN^{2k}}, \quad H_N = \sum_{n=1}^N \frac{1}{n}.$$

Then

$$|\gamma_0 - G_{N,p}| \leq \frac{|B_{2p}|}{2pN^{2p}}.$$

Moreover,

$$\gamma_0 = \lim_{N \rightarrow \infty} (H_N - \log N).$$

Proof. At $s = 1$, one has $(1)_{2k-1} = (2k-1)!$; hence each Bernoulli term becomes $B_{2k}/(2kN^{2k})$. The bound in Proposition 13.1, evaluated at that point, is $|B_{2p}|/(2pN^{2p})$. With $p = 1$, all terms separating $G_{N,1}$ from $H_N - \log N$, and the remainder as well, tend to zero. This proves the limit. This is the subsequent identification with the Euler–Mascheroni constant. $\square$

The antecedent of γ_0 used in the Meissel–Mertens formula is thus incorporated. The number is not received as an independent numerical value to correct the sum over irreducibles: it is a specified operation on the same integer partition function.

The logarithm used in the calculation also admits a rational interval. For $x \geq 1$, let $z = (x - 1)/(x + 1)$. The integrated series of a geometric progression gives

$$\log x = 2 \sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + r_M(x), \quad 0 \leq r_M(x) \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

The inequality follows by replacing the tail denominators with $2M+1$ and summing the geometric progression. First reduce $x = 2^a y$, with $1 \leq y < 2$, and evaluate $a \log 2 + \log y$. The series argument is then at most $1/3$. For $0 < x < 1$, use $\log x = -\log(1/x)$, reversing the interval endpoints.

13.3 Apéry and the regularized negative evaluation

For an integer $r \geq 2$, specialization of Proposition 13.1 produces a rational centre

$$Z_{N,p}^{[r]} = \sum_{n=1}^N \frac{1}{n^r} + \frac{1}{(r-1)N^{r-1}} - \frac{1}{2N^r} + \sum_{k=1}^p \frac{B_{2k}(r)_{2k-1}}{(2k)!N^{r+2k-1}}$$

and the entirely rational bound

$$|Z(r) - Z_{N,p}^{[r]}| \leq \frac{|B_{2p}(r)_{2p-1}|}{(2p)!N^{r+2p-1}}.$$

The simplification uses $(r)_{2p}/(r+2p-1) = (r)_{2p-1}$. At $r = 3$, one obtains Apéry's constant. The procedure determines its value with bounds; irrationality of that constant is a mathematical statement distinct from this evaluation.

On the negative side, the same continuation admits an exact evaluation, without a decimal approximation or fitting parameter.

Theorem 13.3. Regularized evaluation at minus one

The continuation of the integer-mode functional satisfies

$$Z(-1) = -\frac{1}{12}.$$

Proof. In the formula of Proposition 13.1, take $p = 2$, so that $s = -1$ lies in the interior of the validity half-plane. The factor $(s)_4$ vanishes there; its accompanying integral is holomorphic there, so the remainder is zero. The factor $(s)_3$, multiplying the term with B_4 , also vanishes. Only the following remains:

$$Z(-1) = \sum_{n=1}^N n - \frac{N^2}{2} - \frac{N}{2} - \frac{B_2}{2}.$$

Since additive accumulation gives $\sum_{n=1}^N n = N(N+1)/2$, and the Bernoulli recurrence gives $B_2 = 1/6$, the result is $-1/12$, independently of N . $\square$

The evaluated object is the continuation of the functional, not the ordinary limit of the positive partial sums, which grow without bound. This distinction preserves the operation giving rise to the value: the Bernoulli term remains after cancellation of the polynomial boundary terms. Regularization therefore has a structure and genealogy here, in addition to a value.

13.4 Laurent coefficients and preservation of the three classes

Write

$$Z(1+z) - \frac{1}{z} = \sum_{n \geq 0} a_n z^n, \quad \gamma_n = (-1)^n n! a_n.$$

The regular function on the left is holomorphic around zero. For a circle contained in its domain of holomorphy, Cauchy's formula determines

$$a_n = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{Z(1+z) - 1/z}{z^{n+1}} dz.$$

The continuation formula with remainder controls this extraction: a uniform error bound ε on the circle produces a bound $\varepsilon \rho^{-n}$ for the coefficient. This justifies a procedure at arbitrary order; it does not turn agreement of two finite quadratures into an error bound.

The residual readout maintains the three classes of the integer operator:

$$Q_r e_n = \mathbf{1}_{n \equiv r \pmod{3}} e_n, \quad Z_r(s) = \text{Tr}(Q^{-s} Q_r), \quad r = 0, 1, 2.$$

These projections have a common domain in the convergence half-plane. Their sum is the identity; the difference of classes 1 and 2 preserves the orientation of the character modulo three. One has

$$Z = Z_0 + Z_1 + Z_2, \quad Z_0(s) = 3^{-s} Z(s), \quad L_{-3}(s) = Z_1(s) - Z_2(s).$$

The last series has periodic coefficients $(1, -1, 0)$ and zero mean. Grouping by periods allows continuation near $s = 1$. Each Z_r has residue $1/3$ there.

Let

$$Z_r(1+z) - \frac{1}{3z} = \sum_{n \geq 0} c_{r,n} z^n, \quad \lambda = \log 3.$$

Coefficient comparison gives

$$a_n = c_{0,n} + c_{1,n} + c_{2,n}, \quad \frac{L_{-3}^{(n)}(1)}{n!} = c_{1,n} - c_{2,n},$$

$$c_{0,n} = \frac{1}{3} \left[\frac{(-\lambda)^{n+1}}{(n+1)!} + \sum_{k=0}^n a_k \frac{(-\lambda)^{n-k}}{(n-k)!} \right].$$

The third identity comes from multiplying the complete series, including its pole, by $3^{-1}e^{-\lambda z}$. Sum and difference uniquely recover the other two coefficients. The three readouts preserve aggregation, orientation and change of scale, respectively.

At order zero, evaluation can be completed without receiving a decimal value of $L_{-3}(1)$. Indeed,

$$L_{-3}(1) = \sum_{m \geq 0} \left(\frac{1}{3m+1} - \frac{1}{3m+2} \right) = \int_0^1 \frac{1-x}{1-x^3} dx = \int_0^1 \frac{dx}{1+x+x^2}.$$

Complete the square in the denominator and evaluate the antiderivative:

$$L_{-3}(1) = \frac{2}{\sqrt{3}} \left[\arctan \frac{2x+1}{\sqrt{3}} \right]_0^1 = \frac{\pi}{3\sqrt{3}}.$$

The coordinate π is HMT's prior regional representation; this identity is a subsequent analytic evaluation of a class readout, not a procedure for retrospectively selecting that coordinate. Thus, with $C_r = c_{r,0}$,

$$C_0 = \frac{\gamma_0 - \log 3}{3}, \quad C_{1,2} = \frac{\gamma_0}{3} + \frac{\log 3}{6} \pm \frac{\pi}{6\sqrt{3}}.$$

The following identities are recovered:

$$C_0 + C_1 + C_2 = \gamma_0, \quad C_1 - C_2 = L_{-3}(1), \quad C_1 + C_2 - 2C_0 = \log 3.$$

13.5 Exact representation and scope of the certificate

The program `partes_finitas_exactas.py` preserves the following order: word sequence through the additive sheet; evaluation of integer modes; rational construction of Bernoulli numbers; application of the functional's formula; incorporation of the remainder; directed decimal representation. Native multiplication remains available in the same package and its compatibility is proved in the product chapter. Analytic computation on the expressed modes does not modify their rules.

With $N = 81$, $p = 8$ and 120 terms for each logarithm series, the resulting intervals are

$$\begin{aligned}
 &0.577215664901532860606512090082273204041535 \\
 &\quad < \gamma_0 < \\
 &0.577215664901532860606512090082531387279784, \\
 &1.202056903159594285399738161511447311355829 \\
 &\quad < Z(3) < \\
 &1.202056903159594285399738161511452663119342.
 \end{aligned}$$

The decimal endpoints are outward roundings of exact fractions. The program also preserves each interval's rational width and recovers $-1/12$ exactly. No comparison digit appears in the producer. The separate checker opens reference values after obtaining the outputs; this comparison checks execution and does not replace the proved bounds.

Within its domain, this chapter completes the finite parts and evaluations it has just defined. It uses no assumption about the location of zeros or positivity of the Weil form. Prolongation towards the spectral problem preserves the obligations of its own operators: a regularized evaluation does not by itself provide the boundary identity of another functional.

Provenance: development recovered from the triadic functionals and coordinated readout network. Bringing together the Euler–Maclaurin proof, its rational bounds and its program constitutes a formalization and additional certificate of this edition; no new historical priority is claimed for those analytic identities.

14 Composition of readouts and transport of normalization

Normal forms, residue classes and the enumeration operator arise from the APP-TRIT-TPK constructions developed above. The enriched state retains its route, orientation, carry and memory within the joint discrete structure of the continuum. The following operations act on its subsequent spectral realization: they specify the information preserved when readouts are composed, moments are formed or normalization is changed.

Retain $Qe_n = ne_n$ on $\mathcal{H} = \ell^2(\mathbb{N}_{>0})$. Let $a : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ be a fixed residual table, where $q \geq 1$, and let $D_a e_n = a(n)e_n$. The coefficient with index q represents residue class zero. For $t > 0$,

$$K_a(t) = \text{Tr}(e^{-tQ}D_a) = \frac{P_a(e^{-t})}{1 - e^{-qt}}, \quad P_a(r) = \sum_{j=1}^q a(j)r^j.$$

Indeed, writing $n = qk + j$ groups an absolutely convergent series and allows the geometric progression of each class to be summed.

14.1 Moments, derivatives and recovery of the readout

Proposition 14.1. Mellin realization and recovery of the table

For $\Re s > 1$, the operator $Q^{-s}D_a$ is trace class and

$$M_a(s) := \text{Tr}(Q^{-s}D_a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} K_a(t) dt.$$

For every integer $k \geq 0$,

$$M_a^{(k)}(s) = \text{Tr}((- \log Q)^k Q^{-s}D_a).$$

The complete kernel K_a , together with the declared period, uniquely determines the table a .

Proof. For $\sigma = \Re s > 1$,

$$\sum_{n \geq 1} |a(n)| \int_0^\infty t^{\sigma-1} e^{-nt} dt \leq \|a\|_\infty \Gamma(\sigma) \sum_{n \geq 1} n^{-\sigma} < \infty.$$

Termwise integration is therefore valid. The integral of each mode is $\Gamma(s)n^{-s}$. On any compact subset of that half-plane, the series with factors $(\log n)^k$ converge uniformly, justifying the derivatives and their traces. Finally,

$$P_a(r) = (1 - r^q)K_a(-\log r), \quad 0 < r < 1.$$

Two polynomials agreeing on this interval have the same coefficients; these recover the table. $\square$

Recovery uses the complete kernel. A single moment is a functional of the coefficients and may annihilate information. For example, the auxiliary table $a(n) = 1$ on odd integers and $a(n) = -3$ on even integers has $D_a \neq 0$, whereas

$$M_a(2) = \left(\frac{3}{4} - 3\frac{1}{4}\right)Z(2) = 0.$$

This check distinguishes the readout from its moment; it does not alter the readouts selected by the HMT construction. Nor does the recovered table identify the entire enriched history that produced it.

14.2 Diagonal product and tensor product

Proposition 14.2. Two spectral composition laws

Let a, b be two periodic tables, extended to a common period. On $\Re s > 1$, composition on a single index satisfies

$$D_a D_b = D_{ab}, \quad \text{Tr}(Q^{-s} D_a D_b) = \sum_{n \geq 1} \frac{a(n)b(n)}{n^s}.$$

The product of their moments has the different realization

$$M_a(s)M_b(s) = \text{Tr}_{\mathcal{H} \otimes \mathcal{H}}((Q \otimes Q)^{-s} (D_a \otimes D_b)) = \sum_{k \geq 1} \frac{(a * b)(k)}{k^s},$$

$$(a * b)(k) = \sum_{d|k} a(d)b(k/d).$$

Proof. The first identity is verified on e_n . On $e_m \otimes e_n$, the tensor enumeration operator has eigenvalue mn and the readout has coefficient $a(m)b(n)$. The absolute sum is bounded by $\|a\|_\infty \|b\|_\infty Z(\sigma)^2$, where $\sigma > 1$. It may be factored and regrouped by $k = mn$; the pairs corresponding to the divisors of k give the stated convolution. $\square$

The diagonal product preserves a single mode and combines its residual data; in particular, the composite dodecaphase readout retains the classes modulo three and four on the same index. The tensor product retains two indices before grouping them. The former preserves periodicity; the convolution need not: for $a = b = 1$, its coefficients count divisors and are unbounded, since the value at 2^j is $j + 1$. The diagonal table in this example remains constant. This distinction prevents substitution of one operation for the other.

The sum-product distinction also persists in the heat kernel:

$$\text{Tr}(e^{-t(Q \otimes Q)} (D_a \otimes D_b)) = \sum_{m, n \geq 1} a(m)b(n)e^{-tmn},$$

$$K_a(t)K_b(t) = \text{Tr}(e^{-t(Q \otimes I + I \otimes Q)} (D_a \otimes D_b)).$$

For $t > 0$, the first absolute sum is bounded by $\|a\|_\infty \|b\|_\infty \sum_{m \geq 1} e^{-tm}/(1 - e^{-tm})$, which converges; the second factors into two convergent series. Their eigenvalues are respectively mn and $m + n$. The tensor realization expresses these subsequent operations and does not assume statistical independence between the originating histories.

14.3 Affine transport of Laurent coefficients

A positive dimensionless change $Q \mapsto \lambda Q$, with $\lambda > 0$, acts on the subsequent readout through

$$Z_\lambda(s) = \lambda^{-s} Z(s).$$

The identity is first verified on $\Re s > 1$ and continues meromorphically. The residue of Z_λ at one is λ^{-1} . To compare coefficients with unit residue, define

$$\hat{Z}_\lambda(s) = \lambda Z_\lambda(s) = e^{-\ell(s-1)} Z(s), \quad \ell = \log \lambda,$$

$$\hat{Z}_\lambda(s) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n(\lambda)}{n!} (s-1)^n.$$

Theorem 14.3. Stieltjes transport law

For every $n \geq 0$,

$$\gamma_n(\lambda) = \sum_{j=0}^n \binom{n}{j} \gamma_j \ell^{n-j} - \frac{\ell^{n+1}}{n+1}.$$

This transformation is triangular and affine. Two changes of scale compose by multiplying their parameters; the reciprocal scale gives the inverse transformation.

Proof. Put $z = s - 1$ and multiply the complete Laurent series by $e^{-\ell z}$. The polar term contributes $(-\ell)^{n+1}/(n+1)!$ to the coefficient of z^n ; the regular terms contribute

$$\sum_{j=0}^n \frac{(-1)^j \gamma_j}{j!} \frac{(-\ell)^{n-j}}{(n-j)!}.$$

Multiplication of their sum by $(-1)^n n!$ yields the formula. The product $e^{-\ell z} e^{-mz} = e^{-(\ell+m)z}$ proves composition. Setting $m = -\ell$ gives the inverse. $\square$

In particular,

$$\gamma_0(\lambda) = \gamma_0 - \ell, \quad \gamma_1(\lambda) = \gamma_1 + \gamma_0 \ell - \frac{\ell^2}{2}.$$

The affine term records the contribution of the pole and already affects the first coefficient. For $\lambda = 3$, the transformation of residue class zero is recovered: its coefficients are those of $Z_0(s) = 3^{-s} Z(s) = \hat{Z}_3(s)/3$ in Section 13. This specialization preserves the residue factor $1/3$. The general law transports normalization without recalculating or selecting another primary TPK readout.

15 Weil form, double circle, and spectral representation

The factorization and clocks of Section 12 produce two readouts of the same arithmetic. The first decomposes a normal form into irreducibles; the second records the returns of their multiplicative actions. The spectral development must preserve both lines of provenance. Its object is not obtained by assigning a known zero of a function to a return: the functionals are constructed, their compatibility is established, and their analytic realization is identified afterwards.

This chapter brings together the preserved developments of the completed form and the double circle. It presents separately the unconditional normalization results, the exact consequences of a positive boundary identity, and the realizations of that identity. This separation fixes the logical order of the proofs. In particular, computing a representation that uses positivity cannot replace the construction that must establish it.

15.1 From cyclic refinement to the completed form

Let $N = 3^m$, with $m \geq 1$. The quadratic operator of the preceding chapter is retained on the cycle of N positions. If the Fourier indices are chosen in the symmetric interval

$$-\frac{N-1}{2} \leq n \leq \frac{N-1}{2},$$

its quadratic eigenvalues are

$$\lambda_{m,n} = \frac{N^2}{\pi} \sin^2\left(\frac{\pi n}{N}\right).$$

The coordinate π is the prior regional output of the HMT kernel. At this stage it enters the normalization of the analytic realization, not the selection of irreducibles. The constant mode is removed, and the two orientations of a nonzero mode are counted only once:

$$\Theta_m(x) = \frac{1}{2} \sum_{n \neq 0} e^{-x\lambda_{m,n}} = \sum_{n=1}^{(N-1)/2} e^{-x\lambda_{m,n}}, \quad x > 0.$$

Proposition 15.1. Limit of the heat trace

For every $x > 0$,

$$\lim_{m \rightarrow \infty} \Theta_m(x) = \Theta_+(x) := \sum_{n=1}^{\infty} e^{-\pi n^2 x}.$$

Convergence is uniform as x ranges over a compact subset of $(0, \infty)$.

Proof. For a fixed index n , the relation $\sin y/y \rightarrow 1$ gives $\lambda_{m,n} \rightarrow \pi n^2$. Moreover, $\sin y \geq 2y/\pi$ for $0 \leq y \leq \pi/2$, whence

$$\lambda_{m,n} \geq \frac{4n^2}{\pi}.$$

If $x \geq a > 0$, each term is dominated by $e^{-4an^2/\pi}$, whose sum converges. Extending the terms by zero outside the cycle's index interval and applying dominated convergence yields the limit and its uniformity on compact sets. $\square$

Inversion of this trace explains why the completion contains a polar contribution in addition to the positive modes. Define

$$\vartheta(x) = 1 + 2\Theta_+(x).$$

The periodization of the Gaussian

$$F_x(t) = \sum_{n \in \mathbb{Z}} e^{-\pi x(t+n)^2}$$

converges together with all its derivatives. Its Fourier coefficient of index k is $x^{-1/2}e^{-\pi k^2/x}$, obtained by integrating the Gaussian over the line. The Fourier series converges absolutely; evaluating it at $t = 0$ gives

$$\vartheta(x) = x^{-1/2}\vartheta(1/x).$$

In terms of the positive trace,

$$\Theta_+(x) = x^{-1/2}\Theta_+(1/x) + \frac{x^{-1/2} - 1}{2}.$$

The last summand preserves the difference between removing the constant mode before or after inverting the scale.

Proposition 15.2. Mellin transform and continuation

For $\Re s > 1$, the arithmetic partition function $Z(s) = \sum_{n \geq 1} n^{-s}$ satisfies

$$\Lambda(s) := \int_0^\infty \Theta_+(x) x^{s/2-1} dx = \pi^{-s/2} \Gamma(s/2) Z(s).$$

The function

$$\xi(s) = \frac{1}{2} s(s-1) \Lambda(s)$$

admits an entire continuation and satisfies $\xi(s) = \xi(1-s)$.

Proof. Absolute convergence for $\Re s > 1$ permits termwise integration. The substitution $u = \pi n^2 x$ gives the factor $\pi^{-s/2} \Gamma(s/2) n^{-s}$. To continue the integral, separate the intervals $(0, 1)$ and $(1, \infty)$ and apply the preceding inversion to the former. This yields

$$\Lambda(s) = \frac{1}{s(s-1)} + \int_1^\infty \Theta_+(x) (x^{s/2-1} + x^{(1-s)/2-1}) dx.$$

The integral is entire in s : the exponential decay of $\Theta_+(x)$ uniformly dominates every power and every power of $\log x$ on compact subsets of the plane. The factor $s(s-1)$ removes the two poles displayed explicitly. The expression is invariant under $s \mapsto 1-s$, proving the functional equation. $\square$

The Mellin transform of the trace is not simply $Z(s)$: it contains the factors $\pi^{-s/2} \Gamma(s/2)$. Keeping them explicit is indispensable, since their contributions reappear in the completed quadratic form. The product over irreducibles of the preceding chapter and this additive construction agree in the half-plane of convergence by a proved identity of functionals.

15.2 The arithmetic, gamma, and polar contributions

Fix $R > 0$ and consider

$$\mathcal{D}_R = C_c^\infty((-R/2, R/2)).$$

The inner product will be linear in the first variable. We use

$$\hat{f}(z) = \int_{\mathbb{R}} f(x) e^{-izx} dx, \quad (T_a f)(x) = f(x - a),$$

$$C_{f,g}(a) = \langle f, T_a g \rangle_{L^2(\mathbb{R})}, \quad h_{f,g}(z) = \hat{f}(z) \overline{\hat{g}(\bar{z})}.$$

These definitions fix both the orientation of translations and the Fourier normalization. For $f, g \in \mathcal{D}_R$, $C_{f,g}(a) = 0$ if $|a| \geq R$.

The previously constructed irreducibles supply the prime-power indices and their weights:

$$\ell_{p,k} = k \log p, \quad w_{p,k} = (\log p) p^{-k/2}, \quad k \geq 1.$$

The contribution from the gamma completion uses

$$\mu(a) = \frac{e^{-a/2}}{1 - e^{-2a}}, \quad c_\Gamma = \psi(1/4) - \log \pi,$$

where $\psi = \Gamma'/\Gamma$. In particular, $c_\Gamma < 0$. For example, the monotonicity of ψ on the positive axis, obtained from $\psi'(x) = \sum_{n \geq 0} (n+x)^{-2} > 0$, together with $\psi(1) = -\gamma_E < 0$, gives this inequality.

The completed form is

$$\begin{aligned} \mathcal{W}_R(f, g) &= h_{f,g}(i/2) + h_{f,g}(-i/2) + c_\Gamma C_{f,g}(0) \\ &\quad + \int_0^\infty \mu(a) (2C_{f,g}(0) - C_{f,g}(a) - C_{f,g}(-a)) da \\ &\quad - \sum_{\ell_{p,k} \leq R} w_{p,k} (C_{f,g}(\ell_{p,k}) + C_{f,g}(-\ell_{p,k})). \end{aligned}$$

The prime-power sum is finite: only $p^k \leq e^R$ occur. The form has not been replaced by a finite sampling of zeros.

Proposition 15.3. Existence and compatibility of the form

The preceding expression defines a continuous Hermitian sesquilinear form on $\mathcal{D}_R$. If $R < R'$, their restrictions agree:

$$\mathcal{W}_{R'}|_{\mathcal{D}_R \times \mathcal{D}_R} = \mathcal{W}_R.$$

Proof. Since translations are unitary,

$$2C_{f,g}(0) - C_{f,g}(a) - C_{f,g}(-a) = \langle (I - T_a)f, (I - T_a)g \rangle.$$

As $a \rightarrow 0$, each difference has norm $O(a)$, whereas $\mu(a) = O(a^{-1})$. The integrand is therefore $O(a)$. As $a \rightarrow \infty$, the differences have bounded norms and $\mu(a) = O(e^{-a/2})$. The integral converges absolutely. These same estimates, expressed in terms of the norms of f, g and their first derivatives on a fixed compact set, prove continuity. Hermitian symmetry follows by interchanging f, g and conjugating. If R is enlarged, the added prime-power terms have zero correlations on $\mathcal{D}_R$, proving compatibility. $\square$

One can thus write a single form $\mathcal{W}$ on $\mathcal{D} = C_c^\infty(\mathbb{R}) = \bigcup_R \mathcal{D}_R$. Compatibility concerns the complete difference of contributions; it does not require each auxiliary column separately to preserve its norm when the support is enlarged.

15.3 Transport columns and the positive boundary identity

For a cycle of length N , the uniform-defect representation of the preceding chapter gives

$$\delta_N(T_a)^* \delta_N(T_b) = (I - T_a)^*(I - T_b).$$

This equality allows translation differences to be realized on a normalized transport sheet. To present the argument without repeating that realization, we simply write $\delta_a f = (I - T_a)f$. The polar evaluations decompose as

$$b_{\pm}(f) = \frac{\widehat{f}(i/2) \pm \widehat{f}(-i/2)}{\sqrt{2}}.$$

Define two maps on $\mathcal{D}_R$:

$$\begin{aligned} J_{+,R}f &= \left((\sqrt{w_{p,k}} \delta_{\ell_{p,k}} f)_{\ell_{p,k} \leq R}, (\sqrt{\mu(a)} \delta_a f)_{a>0}, b_{+}(f) \right), \\ J_{-,R}f &= \left((\sqrt{2w_{p,k}} f)_{\ell_{p,k} \leq R}, \sqrt{-c_{\Gamma}} f, b_{-}(f) \right). \end{aligned}$$

The target spaces consist of orthogonal sums of copies of $L^2(\mathbb{R})$, a direct integral in the gamma component, and a scalar component. When their ranges are used as effective spaces, the closures of $\text{ran } J_{+,R}$ and $\text{ran } J_{-,R}$ will be taken. The closure of a diagonal range must not be confused with the sum of all coordinates of the ambient space.

Proposition 15.4. Exact decomposition of the form

The following identity holds:

$$\mathcal{W}_R(f, g) = \langle J_{+,R}f, J_{+,R}g \rangle - \langle J_{-,R}f, J_{-,R}g \rangle.$$

Proof. For each prime power, the first column contributes

$$w_{p,k}(2C_{f,g}(0) - C_{f,g}(\ell_{p,k}) - C_{f,g}(-\ell_{p,k}));$$

the second removes $2w_{p,k}C_{f,g}(0)$. The gamma component is precisely the integral in Proposition 15.3. The negative constant component produces $c_{\Gamma}C_{f,g}(0)$. Finally,

$$b_{+}(f)\overline{b_{+}(g)} - b_{-}(f)\overline{b_{-}(g)} = h_{f,g}(i/2) + h_{f,g}(-i/2).$$

Adding these identities gives the stated expression. $\square$

This decomposition does not by itself establish positivity: it is a difference of two positive forms. The boundary identity in the source containing the proof uses a space $\mathcal{K}_R$, an orthogonal projection P_R , and an encoder $\Xi_R : \mathcal{D}_R \rightarrow \mathcal{K}_R$ whose moments must satisfy

$$\begin{aligned} \langle \Xi_R f, \Xi_R g \rangle &= \langle J_{+,R}f, J_{+,R}g \rangle, \\ \langle P_R \Xi_R f, P_R \Xi_R g \rangle &= \langle J_{-,R}f, J_{-,R}g \rangle. \end{aligned}$$

Theorem 15.5. Consequence of the two boundary moments

If the two preceding identities hold on $\mathcal{D}_R$, the map $E_R = (I - P_R)\Xi_R$ satisfies

$$\mathcal{W}_R(f, g) = \langle E_R f, E_R g \rangle.$$

In particular, $\mathcal{W}_R(f, f) \geq 0$.

Proof. The orthogonal decomposition of $\Xi_R f$ by P_R and $I - P_R$ gives

$$\langle \Xi_R f, \Xi_R g \rangle = \langle P_R \Xi_R f, P_R \Xi_R g \rangle + \langle E_R f, E_R g \rangle.$$

Substitute the two moments and apply Proposition 15.4. □

The significance of the theorem lies in identifying the arithmetic form with a boundary norm. The decisive step in its application is the encoder's action and the proof of both moments; the orthogonality of a projection does not impose them on an arbitrary form. This edition's provenance appendix records the preserved source and the point at which it is materially incorporated. That action is not replaced by an operator defined retrospectively from the positivity to be established.

There is a second conservation law, useful once E_R has been constructed. The incidence matrix

$$B = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-$$

has orthogonal projections

$$P_{\pm} = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

Its normalization gives $A = P_+ + qP_-$, with $q = 5/9$. If $D = \sqrt{1 - q^2} P_-$, then $A^*A + D^*D = I$. Iteration gives

$$\|x\|^2 = \sum_{n=0}^{M-1} \|DA^n x\|^2 + \|A^M x\|^2.$$

On the sheet $P_- x = x$, the terminal term is $q^{2M} \|x\|^2$; consequently, the map

$$Lx = (DA^n x)_{n \geq 0}$$

is isometric on that sheet. Telescoping explains conservation through the distribution of losses; membership of $E_R f$ in the sheet must come from the preceding construction. Eliminating a terminal term does not retrospectively select the form from which it started.

Internal history encoder and its two finite moments

The correlative transport of the TPK already allows an encoder to be constructed before its residual is defined. Fix a horizon $N \geq 1$. At each level, let $\mathcal{H}_k$ be the orthogonal sum of the two sheet fibres over the complete histories at that level. If $d(x)$ counts the outgoing emissions from a history x , transport acts by

$$U_k t_x v = \sum_{\alpha: x \rightarrow y} \frac{1}{\sqrt{d(x)}} t_y U_{\alpha} v.$$

Here we consider the isometric realization of sheet transport: $U_{\alpha}^* U_{\alpha} = I$. Distinct histories have distinct descendants because the complete route is preserved. Therefore,

$$U_k^* U_k|_{\mathcal{H}_x} = \frac{1}{d(x)} \sum_{\alpha: x \rightarrow y} U_\alpha^* U_\alpha = I.$$

Consistency of the weights and preservation of memory are developed in Section 9 and applied to spectral transport in Subsection 15.7. The isometry of each sheet transport is an explicit sufficient condition for this equality; normalization of the weights alone does not prove it.

Let R_k, Q_k be the orthogonal sheet projections, with $R_k + Q_k = I$. The sheet-preserving and sheet-changing parts of transport are

$$A_k = R_{k+1} U_k R_k + Q_{k+1} U_k Q_k, \quad \eta_k = Q_{k+1} U_k R_k + R_{k+1} U_k Q_k.$$

Block calculation gives exactly

$$A_k^* A_k + \eta_k^* \eta_k = R_k U_k^* U_k R_k + Q_k U_k^* U_k Q_k = I.$$

The first equality holds for any bounded transport. The second uses the indicated isometric realization. In general, the left-hand side is the sheet-diagonal part of the Gram operator of U_k , not necessarily its full Gram operator.

Define $F_{0,0} = I$, $F_{0,k} = A_{k-1} \cdots A_0$ for $k \geq 1$, and $T_N = F_{0,N}^* F_{0,N}$. The internal encoder has domain and action

$$\begin{aligned} \Xi_N^{\text{int}} : \mathcal{H}_0 &\longrightarrow \mathcal{K}_N^{\text{int}} := \left(\bigoplus_{k=0}^{N-1} \mathcal{H}_{k+1} \right) \oplus \mathcal{H}_N, \\ \Xi_N^{\text{int}} v &= (\eta_k F_{0,k} v)_{0 \leq k < N} \oplus F_{0,N} v. \end{aligned}$$

The last summand is kept separate even when its underlying space coincides with that of an earlier output. Let Π_N^{term} be the orthogonal projection onto that terminal summand. We obtain the two internal moments

$$(\Xi_N^{\text{int}})^* \Xi_N^{\text{int}} = I, \quad (\Xi_N^{\text{int}})^* \Pi_N^{\text{term}} \Xi_N^{\text{int}} = T_N.$$

Proof. The block identity implies

$$F_{0,k}^* F_{0,k} - F_{0,k+1}^* F_{0,k+1} = F_{0,k}^* \eta_k^* \eta_k F_{0,k}.$$

Summing from $k = 0$ to $N - 1$ yields $I = \sum_{k < N} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + T_N$, the first stated Gram operator. The terminal projection retains only $F_{0,N} v$, so its Gram operator is T_N . Thus the complement $(I - \Pi_N^{\text{term}}) \Xi_N^{\text{int}}$ records sheet outputs and has Gram operator $I - T_N$. All these operators come from the same transport collection; the encoder is not defined from a residual whose positivity is assumed in advance. $\square$

This result recovers an effective encoder with its two moments, within the joint constructor that preserves the other four readouts of the continuum. Applying it to the completed columns of this section entails a precise additional requirement: one would have to identify a prior action $R_R : \mathcal{D}_R \rightarrow \mathcal{H}_0$, with the horizon compatibility required by the test domain, for which

$$\begin{aligned} \langle R_R f, R_R g \rangle &= \langle J_{+,R} f, J_{+,R} g \rangle, \\ \langle F_{0,N} R_R f, F_{0,N} R_R g \rangle &= \langle J_{-,R} f, J_{-,R} g \rangle. \end{aligned}$$

The composition $\Xi_N^{\text{int}} R_R$ would then have the moments of Theorem 15.5. This identification of R_R with the prime-power, gamma, and polar columns is not assembled here. The preceding internal construction is assembled here, with its action and proof; it is not presented as a proof of global positivity of $\mathcal{W}$. Nor is T_N suppressed, or its limit asserted to be zero.

Horizon compatibility and the type of the projection

The terminal projection Π_N^{term} selects the last summand of the history encoder. By contrast, the uniform expectation P_R of Theorem 15.5 selects the uniform component of a nine-position calendar. Equality of their roles in a composition must be proved through the concrete map between those spaces; both are orthogonal projections, but that common property does not identify their ranges.

Dependence on the horizon likewise imposes a verifiable identity. Let R be one and the same map and J_- a fixed column. If, for two consecutive horizons, one requires

$$R^* T_N R = J_-^* J_- = R^* T_{N+1} R,$$

then

$$\eta_N F_{0,N} R = 0.$$

Proof. Subtracting the equalities and using the telescoping identity gives

$$0 = R^* (T_N - T_{N+1}) R = (\eta_N F_{0,N} R)^* (\eta_N F_{0,N} R).$$

Evaluation on each vector implies the stated vanishing. If equality of the moments holds with the same R along an unbounded sequence of horizons and T_N tends strongly to zero, one also obtains $J_- = 0$. $\square$

Therefore, an identification with a nonzero negative column must specify its horizon or its compatible collection of maps $R_{R,N}$, or retain the limiting terminal term that actually applies. Without that distinction, contraction of an output residual cannot be transferred to the complete column representing the second moment. This clarification preserves the recovered encoder in full and determines the additional compatibility its analytic realization must satisfy.

15.4 Representation of translations

Suppose that a positive identity has been established on all of $\mathcal{D}$,

$$\mathcal{W}(f, g) = \langle Ef, Eg \rangle$$

compatible with the support domains. The spectral space is constructed from the form itself:

$$\mathcal{N} = \{f \in \mathcal{D} : \mathcal{W}(f, f) = 0\}, \quad \mathcal{H}_{\mathcal{W}} = \overline{\mathcal{D}/\mathcal{N}}.$$

The Cauchy–Schwarz inequality for a positive form proves that the quotient inner product is well defined. The map $[f] \mapsto Ef$ extends to a unitary isomorphism from $\mathcal{H}_{\mathcal{W}}$ onto $\overline{\text{ran } E}$.

Proposition 15.6. Unitary evolution and generator

Translations induce a strongly continuous unitary group

$$U_t[f] = [T_t f].$$

Its self-adjoint generator H , with the convention $U_t = e^{itH}$, acts on classes of test functions by

$$H[f] = [if'].$$

Proof. Correlations are invariant under a common translation of both arguments. Moreover, $\widehat{T_t f}(z) = e^{-izt} \widehat{f}(z)$; in $h_{T_t f, T_t g}(z)$ the factors also cancel for complex z . Hence $\mathcal{W}(T_t f, T_t g) = \mathcal{W}(f, g)$, and translation preserves $\mathcal{N}$. It defines an isometry on the quotient, inverted by U_{-t} , which extends to the complete space.

As $t \rightarrow 0$, $T_t f - f$ tends to zero in every test-function seminorm on a common compact set. The continuity proved for $\mathcal{W}$ implies strong continuity on the dense quotient; unitarity extends it to the whole space. Stone's theorem supplies a self-adjoint generator. The derivative at $t = 0$ is $[-f']$, which agrees with $iH[f]$. $\square$

The test functions also form a core for the generator. This can be seen by regularizing a vector with $\int \chi(t) U_t dt$, where $\chi \in C_c^\infty(\mathbb{R})$. This regularization and its derivative are bounded operators, controlled by $\|\chi\|_1$ and $\|\chi'\|_1$. First approximate the regularized vectors by classes of test functions, and then approximate the identity with a collection of regularizers. The resulting approximation is in the graph norm $\|x\| + \|Hx\|$, proving the assertion.

The representation has been constructed from a positive form and its translations. This argument uses neither a list of zero ordinates nor an assignment of zeros to calendar turns as inputs.

15.5 Analytic identification and multiplicities

The preceding explicit form is recognized as the completed Weil form. With

$$\Phi_f(s) = \widehat{f}(i(s - 1/2)), \quad \rho^\# = 1 - \bar{\rho},$$

the explicit formula supplies its spectral expression

$$\mathcal{W}(f, g) = \sum_{\rho} \Phi_f(\rho) \overline{\Phi_g(\rho^\#)},$$

in terms of the nontrivial zeros of the completed function and their multiplicities. The global formulation of Weil's criterion states that positivity for all test functions is equivalent to $\Re \rho = 1/2$ for all those zeros. We use the criterion in the formulation of Bombieri [4]. The global condition cannot be replaced by positivity of finitely many matrices.

Mellin–Fourier test-function dictionary

To make the normalization of that recognition explicit, let $\mathcal{D} = C_c^\infty(\mathbb{R}; \mathbb{C})$. The map

$$(\mathcal{J}_{\log} f)(x) = x^{-1/2} f(\log x), \quad x > 0,$$

is a bijection onto $C_c^\infty((0, \infty); \mathbb{C})$, with inverse

$$(\mathcal{J}_{\log}^{-1} u)(t) = e^{t/2} u(e^t).$$

Indeed, the exponential change of variables takes compact supports to compact sets bounded away from zero and infinity. The formulas are inverses and their factors are smooth. Defining $\hat{u}_{\text{Mel}}(s) = \int_0^\infty u(x) x^{s-1} dx$, the substitution $x = e^t$ gives, for every $s \in \mathbb{C}$,

$$\widehat{\mathcal{J}_{\log} f}_{\text{Mel}}(s) = \int_{\mathbb{R}} f(t) e^{(s-1/2)t} dt = \widehat{f}(i(s - 1/2)) = \Phi_f(s).$$

Conjugation is preserved exactly:

$$\widehat{\mathcal{J}_{\log} g}_{\text{Mel}}(1 - s) = \overline{\Phi_g(1 - \bar{s})}.$$

Thus the two factors in the spectral summand agree with those in the explicit formula in multiplicative coordinates, with no residual factor and with the same multiplicities counted. The sum converges absolutely: integration by parts gives rapid decay of the transforms in vertical strips, and the zero count is $O(T \log T)$. The bijection transfers the quantifier over all test functions, not a selection of them. The explicit formula and the difficult implication of the criterion are the cited antecedents; the preceding change of variables proves the correspondence of their domains and normalizations, not a new positivity theorem. This compact-support domain is not asserted to be invariant under the integral Cayley operator: its subsequent realization uses the exponential space of Subsection 17.10.

Once those antecedents are satisfied, writing $\rho = 1/2 + i\gamma$ gives

$$\mathcal{W}(f, g) = \sum_{\gamma \in \Gamma_\xi} m_\gamma \hat{f}(-\gamma) \overline{\hat{g}(-\gamma)}.$$

Here Γ_ξ contains the distinct ordinates and m_γ is the multiplicity of the corresponding zero. Let

$$\mu_\xi = \sum_{\gamma \in \Gamma_\xi} m_\gamma \delta_\gamma.$$

The map

$$\mathcal{F}_\xi[f](\gamma) = \hat{f}(-\gamma)$$

is an isometry from $\mathcal{H}_\mathcal{W}$ into $L^2(\Gamma_\xi, \mu_\xi)$. In this representation, U_t acts as multiplication by $e^{it\gamma}$ and H as multiplication by γ .

Proposition 15.7. Exhaustiveness of the atomic representation

The map $\mathcal{F}_\xi$ is surjective.

Proof. Its closed range is invariant under multiplication by $e^{it\gamma}$ for positive and negative t ; it is therefore reducing. For a fixed ordinate γ_0 , the averages

$$\frac{1}{2T} \int_{-T}^T e^{-it\gamma_0} U_t dt$$

converge strongly to the projection onto the atom γ_0 . This is checked first coordinatewise and then by dominated convergence in the sequence space. There exists $f \in C_c^\infty$ with $\hat{f}(-\gamma_0) \neq 0$; for example, it suffices to modulate a function with nonzero integral. The range thus contains the vector of that atom. Since it contains every finitely supported vector and is closed, it equals the whole space. $\square$

The measure preserves arithmetic multiplicity as a weight. Its relation to operator multiplicity should be specified: the scalar space $L^2(\Gamma_\xi, \mu_\xi)$ has a one-dimensional eigenspace for each atom, even if $m_\gamma > 1$. Therefore, if a function F produces a trace-class operator,

$$\text{Tr}(F(H)) = \sum_{\gamma \in \Gamma_\xi} F(\gamma),$$

whereas the sum preserving the weights of the zeros is

$$\int F(\gamma) d\mu_\xi(\gamma) = \sum_{\gamma \in \Gamma_\xi} m_\gamma F(\gamma).$$

This second sum can be represented as a weighted trace $\text{Tr}(M_m F(H))$, when the product is trace class, where M_m is multiplication by m_γ . A different construction would place a fibre of dimension m_γ at each atom. These two representations are not interchanged without an explicit declaration.

15.6 The two circles and spectral orientation

The first circle is a compact coordinate for the spectral line. For a real number γ , define

$$\tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2}.$$

Its modulus is one and $\tau_\gamma \neq 1$. The inverse formulas are

$$\gamma = \frac{i \tau_\gamma + 1}{2 \tau_\gamma - 1}, \quad \gamma = \frac{1}{2} \cot \frac{\theta}{2} \quad \text{if } \tau_\gamma = e^{i\theta}.$$

If $s = 1/2 + i\gamma$, also

$$\tau_\gamma = \frac{s-1}{s}, \quad s = \frac{1}{1-\tau_\gamma}.$$

Passing to the phase does not lose the ordinate, provided the point on the circle and the orientation convention are retained.

For the preceding self-adjoint generator, functional calculus defines the unitary operator

$$C = (H + iI/2)(H - iI/2)^{-1}.$$

The equality $C^*C = I$ follows because both factors are functions of the same self-adjoint operator and $|(\gamma + i/2)/(\gamma - i/2)| = 1$. The point 1 does not correspond to a finite ordinate, but it may belong to the spectrum of C as an accumulation point when H is unbounded.

The second circle is the dual of integer memory: a phase τ is associated with the character

$$\chi_\tau(n) = \tau^n, \quad n \in \mathbb{Z}.$$

The nonadic connection returns the visible phase to its initial position and increases the memory of a turn by one unit. The spectral realization assigns the action of C to that unit; on an eigenvector with ordinate γ , the memory of n turns acts by τ_γ^n . The first circle encodes the ordinate; the second expresses how its phase acts on additive memory. These are two related mathematical roles, not two lists to be identified by their ordinal positions.

15.7 Cylindrical refinement with memory

The connection must act on the structure being refined. In the common TPK tree, a vertex is a finite history of admissible emissions; two histories reaching the same terminal state remain distinct when their transport records differ. If $\text{Out}(x)$ is the finite set of outgoing emissions, counted with multiplicity, write

$$d(x) = |\text{Out}(x)| \geq 1.$$

Neutral prolongation preserves nonemptiness of the levels. The local weight specified by the tree is $1/d(x)$. For a history $h = (\epsilon_1, \dots, \epsilon_k)$, with successive states $x_0, \dots, x_k$, its weight is

$$\mu_k(h) = \prod_{j=0}^{k-1} \frac{1}{d(x_j)}.$$

The sum of the weights of all its successors is

$$\sum_{\epsilon \in \text{Out}(x_k)} \mu_{k+1}(h\epsilon) = \mu_k(h) \sum_{\epsilon \in \text{Out}(x_k)} \frac{1}{d(x_k)} = \mu_k(h).$$

Starting from the root of weight one, this identity proves normalization and projective consistency by induction. If a reversible symmetry transports the outgoing emissions bijectively with their multiplicities, it preserves $d(x)$, all these weights, and their cylinder measures. Thus the measure used below comes from the declared branching rule, before an operator is assigned to its fibres. It is not confused with a uniform distribution on terminal states that would identify different histories.

In general, let $\mathcal{Z}_k$ be a finite level of cylindrical states and let $\pi_k : \mathcal{Z}_{k+1} \rightarrow \mathcal{Z}_k$ be its truncation map. Suppose a collection of compatible positive weights is given,

$$\mu_k(z) = \sum_{\pi_k(z')=z} \mu_{k+1}(z'), \quad \sum_{z \in \mathcal{Z}_k} \mu_k(z) = 1.$$

The weights belong to the state measure being transported. The shape of a visible numerical cylinder does not by itself suffice to reconstruct them.

Let $q_k : \mathcal{Z}_k \rightarrow \mathbb{Z}$ be the memory coordinates. The increment of a prolongation is

$$m(z', z) = q_{k+1}(z') - q_k(z).$$

Concatenating prolongations satisfies the cocycle law exactly:

$$m(z'', z) = m(z'', z') + m(z', z).$$

Define

$$w(z', z) = \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}}, \quad \mathcal{K}_k = \ell^2(\mathcal{Z}_k) \otimes \mathcal{H}_W.$$

Transport with memory is

$$\tilde{U}_k(e_z \otimes v) = \sum_{\pi_k(z')=z} w(z', z) e_{z'} \otimes C^{m(z', z)} v.$$

Proposition 15.8. Isometry and composition of transport

Each $\tilde{U}_k$ is isometric. Composition across several levels has the same form, with the weight ratio between the endpoint levels and the total memory increment.

Proof. Two distinct predecessors have disjoint sets of descendants. For a fixed predecessor, powers of C are unitary and $\sum_{z' \mapsto z} w(z', z)^2 = 1$. These are exactly the identities preserving inner products of basis vectors. In a composition, the product of weights telescopes:

$$\sqrt{\frac{\mu_{k+2}(z'')}{\mu_{k+1}(z')}} \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}} = \sqrt{\frac{\mu_{k+2}(z'')}{\mu_k(z)}}.$$

The powers of C add their exponents by the cocycle law. Repetition proves the assertion for any finite number of levels. $\square$

The same identity admits an interpretation through a change of representation. If U_k denotes transport without memory and

$$G_k(e_z \otimes v) = e_z \otimes C^{q_k(z)} v,$$

then

$$\tilde{U}_k = G_{k+1}(U_k \otimes I)G_k^*.$$

The memory coordinate is not lost in formulating the isometry: it appears at both ends of the transport. This equality also shows that the twisted representation uses an already specified operator C . By itself, it does not identify an arbitrary prior native transport with that operator; identification requires composing their maps and checking their actions on the same domain.

Proposition 15.9. Nonadic return on the compatible section

Define

$$\Omega_k = \sum_z \sqrt{\mu_k(z)} e_z, \quad J_k v = \Omega_k \otimes v.$$

If every return trajectory from k to $k + 9$ has memory increment one, then

$$\tilde{U}_{\Gamma,k} J_k = J_{k+9} C.$$

For r consecutive turns,

$$\tilde{U}_{\Gamma,k+9(r-1)} \cdots \tilde{U}_{\Gamma,k} J_k = J_{k+9r} C^r.$$

Proof. Multiplying each descendant's weight by its predecessor's coefficient $\sqrt{\mu_k(z)}$ leaves $\sqrt{\mu_{k+9}(z')}$. The condition on the increment allows the same operator C to be factored out of every summand. This gives the first equality. The second follows by induction. $\square$

The level indices are essential. It is the phase that returns, while the section extends to another level; the formula does not reset the state in the previous space.

The memory condition in this proposition has a concrete action in TPK chronology. For a phase $r \in \mathbb{Z}/9\mathbb{Z}$, with representative $\bar{r} \in \{0, \dots, 8\}$, and a memory depth $w \in \mathbb{Z}_{\geq 0}$, each emission updates

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + [1]_9 \right).$$

After n emissions, the number of crossings from representative 8 to representative 0 is $\lfloor (\bar{r} + n)/9 \rfloor$. Therefore,

$$\sigma_9^n(w, r) = \left(w + \left\lfloor \frac{\bar{r} + n}{9} \right\rfloor, r + [n]_9 \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

Taking $q_k = w_k$ recovers the preceding cocycle and the unit increment of each turn. Even a neutral emission on the APP sheets records an extension in this chronology. The calculation does not by itself impose the spectral operator C : it determines the memory on which the realization acts, once its intertwining has been established.

15.8 Nonadic calendar and contraction over one turn

The finite calendar is a representation of this transport. On $\mathbb{C}^9 \otimes \mathcal{H}_{\mathcal{W}}$, let

$$S_C = \sum_{r=1}^8 |r+1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C.$$

The unitarity of C makes S_C unitary, and the complete turn satisfies

$$S_C^9 = I_{\mathbb{C}^9} \otimes C.$$

Each vector crosses the oriented seam containing C exactly once; this is why C^9 does not appear.

Let $u = 9^{-1/2}(1, \dots, 1)$ and $w = u \otimes v$. The uniform compression of one step is

$$A_C = \iota^* S_C \iota = \frac{8I + C}{9}.$$

A direct calculation gives

$$I - A_C^* A_C = \frac{8}{81} (I - C)^* (I - C).$$

The defect of a compressed unitary transport measures the component leaving the uniform state. It is not a loss of unitarity of the complete calendar.

The contraction $q = 5/9$ of the complementary sheet and the calendar are distinct operations. If q corresponds to one nonadic turn, the operator per turn is

$$M_{\text{vuelta}} = q S_C^9 = q(I \otimes C).$$

It can be distributed uniformly over the nine steps by

$$M_{\text{paso}} = q^{1/9} S_C, \quad M_{\text{paso}}^9 = M_{\text{vuelta}}.$$

By contrast, taking $q S_C$ as the operator at each step produces $q^9(I \otimes C)$ upon completing the turn. The two normalizations describe different contractions and must not be identified. At an ordinate γ , r turns in the first regime produce

$$q^r \tau_\gamma^r.$$

The modulus records extinction of the terminal term; the phase records the action on memory. Knowing only the modulus does not determine the spectral ordinates.

15.9 Moment matrices and spectral approximation

The preceding realization allows approximations to be constructed from test functions, not from assigned spectral values. Choose $f_1, \dots, f_d \in \mathcal{D}$ whose classes are linearly independent in $\mathcal{H}_{\mathcal{W}}$. With $e_j = [f_j]$, define

$$G_{ij} = \langle e_j, e_i \rangle, \quad K_{ij} = \langle H e_j, e_i \rangle, \quad Q_{ij} = \langle H e_j, H e_i \rangle.$$

The index convention ensures that, for $x = \sum_j v_j e_j$, one has $\|x\|^2 = v^* G v$. G is positive-definite Hermitian, and K, Q are Hermitian. If the initial classes were dependent, the kernel of G would have to be removed before inverting it.

Proposition 15.10. Galerkin approximation and residual

The generalized problem

$$K v = \gamma G v$$

is equivalent to the ordinary Hermitian problem for $\hat{H} = G^{-1/2} K G^{-1/2}$. For any $\gamma \in \mathbb{R}$ and $v \neq 0$, the spectral residual of the corresponding vector satisfies

$$\frac{\|(H - \gamma)x\|^2}{\|x\|^2} = \frac{v^*(Q - 2\gamma K + \gamma^2 G)v}{v^* G v}.$$

In particular,

$$\text{dist}(\gamma, \sigma(H)) \leq \frac{\|(H - \gamma)x\|}{\|x\|}.$$

Proof. The change $w = G^{1/2}v$ transforms the generalized equation into $\hat{H}w = \gamma w$. Expanding the residual norm gives the quadratic identity because H is symmetric on the vectors used. The final inequality follows from the spectral measure of H :

$$\|(H - \gamma)x\|^2 = \int |\lambda - \gamma|^2 d\langle E_H(\lambda)x, x \rangle.$$

The integrand is bounded below by the squared distance to the spectrum. □

The phase matrix

$$\hat{C} = (\hat{H} + iI/2)(\hat{H} - iI/2)^{-1}$$

is unitary for the Euclidean inner product. If $G^{-1}K$ is used instead in the original basis, the corresponding unitarity is expressed relative to G . These are two bases for the same problem, not two norms interchangeable without transformation.

A small residual approximates the spectrum of the operator already constructed. By itself, it does not count all its eigenvalues, prove global positivity of $\mathcal{W}$, or certify decimal digits affected by quadrature errors. For a validated bound, the entries of G, K, Q must be accompanied by error bounds and a positive lower bound for the form v^*Gv .

15.10 Compact resolvent and exhaustiveness checks

In the atomic realization, the ordinates form a discrete set and escape to infinity. Therefore,

$$R = (H - iI)^{-1}$$

is compact: in the atomic basis it acts by $(\gamma - i)^{-1}$, which tends to zero. It is also recovered directly from the translation group through

$$R = i \int_0^\infty e^{-t} U_{-t} dt.$$

The integral over a finite interval is defined in the strong sense. Its tail from T has operator norm at most e^{-T} , so the truncated integrals converge in norm. Functional calculus reduces the identity to $i/(1 + i\gamma) = (\gamma - i)^{-1}$.

Proposition 15.11. Norm convergence of compressions

If P_d are finite-rank orthogonal projections with $P_d \rightarrow I$ strongly, then

$$\|R - P_d R P_d\| \rightarrow 0.$$

Proof. The strong convergence of P_d , together with the bound $\|P_d\| \leq 1$, is uniform on every compact set. Apply it to the compact closure of the image of the unit ball under R , obtaining $\|(I - P_d)R\| \rightarrow 0$. Applying the same argument to R^* yields $\|R(I - P_d)\| \rightarrow 0$. The decomposition

$$R - P_d R P_d = (I - P_d)R + P_d R (I - P_d)$$

completes the proof. □

This convergence permits checks along resolvent contours. If a closed curve $\mathcal{C}$ separates a group of nonzero eigenvalues and the norm of the compression error is less than the reciprocal of a bound for the approximate resolvent on $\mathcal{C}$, the Neumann series preserves invertibility along the curve. The Riesz projections of the two operators then have the same rank. This is an exhaustiveness check in a delimited spectral region, additional to a vector residual.

The multiplicity counted by that rank is the multiplicity of the represented operator. In the scalar space of Subsection 15.5, it must not be replaced by the weight m_γ without explicitly introducing the weighted observable.

15.11 From internal identities to the analytic assembly

The proofs of the heat and Mellin normalization, the explicit construction of the completed form, its column decomposition, cylindrical transport with memory, the two circles, the calendar, and the spectral checks have been assembled here. The arithmetic construction of irreducibles precedes their appearance in the prime-power weights. Memory comes from compatible prolongations; it is not replaced by a phase digit.

Subsection 15.3 also assembles a finite internal encoder constructed from history transports, together with the proof of its moments I and T_N in the isometric realization. Applying Theorem 15.5 to the native encoder of the completed form retains a more specific identification: the prior action R_R on test functions and equality of those moments with the moments of $J_{+,R}$ and $J_{-,R}$. The source of the completed form states those identities and develops their consequences. This exposition preserves those consequences with their explicit antecedents and the internal composition actually recovered; it does not replace the remaining link with a merely nominal formulation. The distinction concerns the proof assembled in this article, not an assertion that the result is absent from the corpus as a whole.

The complete link between all APP-TRIT-TPK histories and the word normalization of the first chapter also retains its integration requirement. The five consubstantial constructions of the continuum remain part of the common kernel; this chapter presents their spectral readout and does not redefine them as independent structures selected by conventional problems.

Identifying the global form with the explicit formula and Weil's criterion fixes the exact scope of any eventual terminal conclusion. Finite checks of the native producer and spectral approximations are distinct verifiable results. Accordingly, this working edition does not present the chapter as an already completed, self-contained proof of the Riemann hypothesis.

15.12 Bidirectional recovery and assembly of the readouts

The gamma information recovers the test function that produced it. This property provides an explicit action between the complete columns before the sign of their difference is known. The following development extends the integration status described in Subsections 15.3 and 15.11: the global bounded connector is recovered; its contractivity is studied on the same action and the same range, not on another operator.

The construction follows the prime-power weights, the completed trace, and the transports already defined. It does not change the selection of irreducibles or receive zeros as inputs. The sources of the gamma tail, the basal domain, and the prior folding are preserved in full in the provenance package for this integration.

Proposition 15.12. Left inverse of the tail and global connector

Let $H_R = L^2(I_R)$, identified with its zero extension to the line, and let M_R be orthogonal restriction to I_R . Define

$$a_R = 2 \int_R^\infty \mu(a) da, \quad (B_R f)(a) = \sqrt{\mu(a)}(I - T_a)f \quad (a \geq R).$$

Then B_R is bounded and $B_R^* B_R = a_R I$. If $\pi_{\Gamma, \geq R}$ extracts the gamma tail from the positive column, the map

$$L_R = a_R^{-1} B_R^* \pi_{\Gamma, \geq R}$$

satisfies $L_R J_{+,R} f = f$ for $f \in \mathcal{D}_R$, and $L_R L_R^* = a_R^{-1} I$ on H_R . The first identity extends to the joint closed domain of the readouts; it does not assert that the complete gamma column is defined on all of $L^2(I_R)$. In particular,

$$\mathcal{C}_R = J_{-,R} L_R, \quad \mathcal{C}_R J_{+,R} = J_{-,R}$$

is a bounded connector for every $R > 0$.

Proof. For $a \geq R$, the supports of $f, g \in H_R$ and of their translates are disjoint, apart from endpoints of measure zero. Therefore

$$\langle (I - T_a)f, (I - T_a)g \rangle = 2\langle f, g \rangle.$$

Integration gives the first Gram operator. The adjoint has the explicit action

$$B_R^* F = M_R \int_R^\infty \sqrt{\mu(a)} (I - T_{-a}) F(a) da.$$

The integral converges in norm by Cauchy–Schwarz; the tail of μ is integrable and $T_a^* = T_{-a}$. Applying the adjoint to the tail of $J_{+,R} f$ returns $a_R f$. Extraction is a coisometry onto its summand, whence $L_R L_R^* = a_R^{-2} B_R^* B_R = a_R^{-1} I$. The negative column is bounded on H_R : it has finitely many vector-valued rows and a continuous polar functional on that interval. Composition completes the argument. $\square$

Inversion uses the conjugate translations before publishing the recovered function. The source also provides a finite-band version: with $A_R = [R + 1, R + 2]$ and $\nu_R = \int_{A_R} \mu(a) da$,

$$\mathfrak{R}_R F = \nu_R^{-1} \int_{A_R} \sqrt{\mu(a)} M_R F(a) da, \quad \mathfrak{R}_R D_\Gamma f = f.$$

The identity follows from $M_R T_a f = 0$ on that band. Both inverses precede a Weil inequality. The first allows the ambient norm of its connector to be computed exactly.

Proposition 15.13. Connector norm and basal factorization

Write

$$s_R^2 = 2 \sum_{\ell_{p,k} \leq R} w_{p,k} - c_\Gamma, \quad \beta_R = 2 \sinh(R/2) - R.$$

Then

$$\|\mathcal{C}_R\|^2 = \frac{s_R^2 + \beta_R}{a_R}.$$

Consequently, if $0 < h < \log 2$ and

$$2M_\Gamma(h) \geq -c_\Gamma + 2 \sinh(h/2) - h, \quad M_\Gamma(h) = \int_h^\infty \mu(a) da,$$

there exists a dilation realizing the two moments of Theorem 15.5 on all functions in $\mathcal{D}_h$.

Proof. The negative column has Gram operator $J_-^* J_- = s_R^2 I + b_-^* b_-$. The functional

$$b_-(f) = \sqrt{2} \int_{-R/2}^{R/2} f(x) \sinh(x/2) dx$$

has squared norm $2 \int_{-R/2}^{R/2} \sinh^2(x/2) dx = \beta_R$. The Gram operator has norm $s_R^2 + \beta_R$. The preceding identity $L_R L_R^* = a_R^{-1} I$ gives $\mathcal{C}_R \mathcal{C}_R^* = a_R^{-1} J_- J_-^*$, proving the formula.

For $R = h < \log 2$, no prime-power lengths occur in the sum. The stated inequality proves $\|\mathcal{C}_h\| \leq 1$. One can construct, in this order,

$$D_h^{\text{def}} = (I - \mathcal{C}_h^* \mathcal{C}_h)^{1/2}, \quad \Xi_h f = J_{-,h} f \oplus D_h^{\text{def}} J_{+,h} f.$$

The sum of the Gram operators of $\mathcal{C}_h$ and its defect is the identity. Since $\mathcal{C}_h J_{+,h} = J_{-,h}$, the first moment of Ξ_h is $J_{+,h}^* J_{+,h}$; projection onto the first summand has moment $J_{-,h}^* J_{-,h}$. Theorem 15.5 gives positivity. The square root of the defect is taken after contractivity has been proved, not as a substitute for that proof.

If the window is enlarged to $R \geq h$, each added prime channel satisfies $\|(I - T_\ell)f\|^2 = 2\|f\|^2$ on H_h . The map from that image to $\sqrt{2}f$ is an isometry, extended by zero on the complement of its closed range. The direct sum of those rows with the gamma-polar connector retains contractivity on the same basal domain. $\square$

The basal endpoint is determined by

$$2M_\Gamma(h_*) = -c_\Gamma + 2 \sinh(h_*/2) - h_*.$$

The substitution $t = e^{-a/2}$ gives

$$M_\Gamma(h) = \operatorname{arctanh}(e^{-h/2}) + \operatorname{arctan}(e^{-h/2}).$$

The left-hand side of the equation decreases strictly from infinity to zero; the right-hand side is positive and increases strictly for $h > 0$. There is a unique positive root. The source evaluates it as $h_* = 0.08562601762973687068 \dots < \log 2$. One may work with any $h < h_*$ certified by inequalities, without needing the decimal expansion of that endpoint.

The complete form also preserves translation of test functions. On the polar ports, translation acts by

$$\begin{pmatrix} b_+(T_t f) \\ b_-(T_t f) \end{pmatrix} = \begin{pmatrix} \cosh(t/2) & \sinh(t/2) \\ \sinh(t/2) & \cosh(t/2) \end{pmatrix} \begin{pmatrix} b_+(f) \\ b_-(f) \end{pmatrix}.$$

The matrix preserves the difference of squares, and the other rows are translated unitarily. The basal domain may therefore be placed in any translated interval of length h .

Proposition 15.14. Uniqueness on the effective range

Let $\mathcal{M}_{+,R} = \overline{J_{+,R} \mathcal{D}_R}$. The restriction of the constructed connector to this subspace is unique among bounded maps with that output identity. Moreover,

$$\|\mathcal{C}_R|_{\mathcal{M}_{+,R}}\| \leq 1 \iff \mathcal{W}_R(f, f) \geq 0 \quad (f \in \mathcal{D}_R).$$

Proof. Two bounded operators agreeing on the dense range agree on its closure. The equivalence follows from $\mathcal{C}_R J_{+,R} f = J_{-,R} f$ and the norm identity; the inequality extends to the closure by continuity. $\square$

The global formula is therefore constructed:

$$\mathcal{W}_R(f, g) = \langle J_{+,R} f, (I - \mathcal{C}_R^* \mathcal{C}_R) J_{+,R} g \rangle.$$

Changing the ambient representation may facilitate computation of the norm, but it does not permit changing its action on test functions. Failure of contractivity on the entire ambient space does not refute positivity on the complete range, where the row correlations remain. Projection onto that range does not automatically prove the bound either.

Proposition 15.15. Prior folding and all cross terms

With a basal fibre fixed, let $m = 9^q$ and let ℓ_j be the displacements expressed by its routes. On m copies of $L^2(\mathbb{R})$, the folding and its uniform representation are

$$\mathcal{F}_q x = \sum_{j=1}^m T_{\ell_j} x_j, \quad S_q x = u_q \otimes \mathcal{F}_q x = 3^q P_q T_q x, \quad u_q = m^{-1/2}(1, \dots, 1).$$

For every linear readout J defined on those test functions,

$$\|J\mathcal{F}_q x\|^2 = \sum_{i,j} \langle JT_{\ell_i} x_i, JT_{\ell_j} x_j \rangle.$$

Moreover, $\mathcal{F}_q$ is surjective, with right inverse

$$(\mathcal{E}_q f)_j = m^{-1} T_{-\ell_j} f, \quad \mathcal{F}_q \mathcal{E}_q = I, \quad \|\mathcal{E}_q f\|^2 = m^{-1} \|f\|^2.$$

Proof. Averaging the m routes and multiplying by $\sqrt{m} = 3^q$ gives the expression for S_q . Expanding the inner product retains every cross term. Substituting $\mathcal{E}_q f$ produces m copies of f/m ; unitarity of the translations gives the norm. $\square$

If the successors satisfy $\ell_{jr} = \ell_j + \delta_{jr}$, the operation $(\mathcal{A}_q x)_j = \sum_{r=0}^8 T_{\delta_{jr}} x_{jr}$ satisfies exactly

$$\mathcal{F}_{q+1} = \mathcal{F}_q \mathcal{A}_q, \quad S_{q+1} = J_q S_q \mathcal{A}_q, \quad J_q y = y \otimes u_9.$$

The gamma, prime, and polar columns are thus applied to the same sum before their norms are taken. Unitary mixing after an orthogonal sum preserves diagonal energy; it does not manufacture omitted cross terms.

Surjectivity determines the scope of refinement. On the complete fibre, positivity of $\mathcal{W}(\mathcal{F}_q x, \mathcal{F}_q x)$ for all states is equivalent to positivity of $\mathcal{W}(f, f)$ for all test functions. The reverse implication uses $x = \mathcal{E}_q f$, whose components remain smooth and compactly supported. Folding preserves the problem and its cross terms: merely increasing the routes does not restrict the test functions. Restriction to more specific reachable states requires their construction and its proof.

Assembled result and global extension

The gamma inverse, the complete output link, the basal domain, its translation, and folding prior to the readouts have action, domain, and proof here. They are results recovered from the corpus and assembled formalizations, not discoveries attributed retrospectively.

For the requested global scope, it remains to prove the bound for that same output on the entire effective range, including the cross terms in sums of local test functions, compatibly as the windows are enlarged. The output identity of one connector and the contractivity of another cannot be combined without proving that they represent the same action. Comparison with the internal history encoder retains the two moments specified in Subsection 15.3.

15.13 Irreducible-clock windows and refinement with cross terms

The native product determines irreducibles and their powers before the logarithmic readout. The lengths $\ell_{p,k} = k \log p$ and weights $w_{p,k} = (\log p) p^{-k/2}$ appearing below are those already expressed in Sections 11 and 12. They are not selected from the sign to be checked. On this arithmetic output, a collection of test functions can be evaluated jointly, retaining the channels of the completed form and all their cross terms.

This section assembles the density result and Schur criterion from the source “Surviving readout of two APP routes”, and extends the calculation to translated intervals of equal width. The choice of nine windows certified below is an explicit analytic collection; its cardinality does not by itself identify those windows with the nine canonical successors of a TPK state.

Proposition 15.16. Complete kernel of translated intervals

For $\varepsilon > 0$, let

$$b_{\varepsilon,t}(x) = \varepsilon^{-1/2} \mathbf{1}_{[t-\varepsilon/2, t+\varepsilon/2]}(x), \quad \tau_\varepsilon(x) = \left(1 - \frac{|x|}{\varepsilon}\right)_+,$$

$$a_\varepsilon = \frac{4 \sinh(\varepsilon/4)}{\sqrt{\varepsilon}}, \quad \lambda_n = 2n + \frac{1}{2}, \quad F(x) = \sum_{n \geq 0} \frac{e^{-\lambda_n x}}{\lambda_n^2} \quad (x \geq 0).$$

The complete form on these intervals depends on the difference between their centres:

$$\mathcal{W}(b_{\varepsilon,s}, b_{\varepsilon,t}) = K_\varepsilon(s - t),$$

where

$$K_\varepsilon(d) = 2a_\varepsilon^2 \cosh(d/2) + c_\Gamma \tau_\varepsilon(d) + \frac{2F(|d|) - F(|d - \varepsilon|) - F(|d + \varepsilon|)}{\varepsilon} - \sum_{p,k} w_{p,k} [\tau_\varepsilon(d - \ell_{p,k}) + \tau_\varepsilon(d + \ell_{p,k})].$$

The prime-power sum is finite; only lengths $\ell_{p,k} < |d| + \varepsilon$ contribute.

Proof. The overlap of two normalized intervals is the triangular correlation τ_ε . Inserting the translations of each clock gives the two prime terms. The polar integrals of $b_{\varepsilon,t}$ are $a_\varepsilon e^{t/2}$ and $a_\varepsilon e^{-t/2}$. Their cross product gives $a_\varepsilon^2 (e^{(s-t)/2} + e^{(t-s)/2})$, the polar term in the formula. In particular, dependence on the difference between centres belongs to the complete form, although each polar port separately changes under translation.

The expansion $\mu(a) = \sum_{n \geq 0} e^{-\lambda_n a}$ and integration over the linear pieces of τ_ε give

$$\int_0^\infty e^{-\lambda a} [2\tau_\varepsilon(d) - \tau_\varepsilon(d - a) - \tau_\varepsilon(d + a)] da = \frac{2e^{-\lambda|d|} - e^{-\lambda|d-\varepsilon|} - e^{-\lambda|d+\varepsilon|}}{\varepsilon \lambda^2}.$$

To justify interchanging the series and integral, the triangular function also satisfies

$$|2\tau_\varepsilon(d) - \tau_\varepsilon(d - a) - \tau_\varepsilon(d + a)| \leq \min(4, 2a/\varepsilon).$$

Its product with $\mu(a)$ is integrable both at the origin and at infinity. The sum of the absolute integrals is thus bounded, justifying the interchange. The integrated summand is also bounded in absolute value by $4/(\varepsilon \lambda_n^2)$, whose series converges. The joint contribution of both sides cancels the singularity of μ at the origin. One can therefore sum the integrated identity and obtain the stated gamma term.

□

These indicator functions are elements of the closed domain of the readouts, not smooth functions. Extension from $\mathcal{D}_R$ is justified without modifying the form. For a normalized interval,

$$\|b_{\varepsilon,t} - T_a b_{\varepsilon,t}\|_2^2 = 2 \min(|a|/\varepsilon, 1).$$

The integral against $\mu(a)$ is finite. Let $b_{\varepsilon,t}^{(\delta)} = \rho_\delta * b_{\varepsilon,t}$ be a smooth regularization supported in a fixed window. Take $\rho \geq 0$, smooth and compactly supported, with $\int \rho = 1$, and $\rho_\delta(x) = \delta^{-1} \rho(x/\delta)$. The gamma symbol is

$$\sigma_{\Gamma}(\xi) = \int_0^{\infty} \mu(a) |1 - e^{-ia\xi}|^2 da.$$

In Fourier coordinates, the integrand expressing the gamma norm of the difference contains $\sigma_{\Gamma}|\hat{b}|^2|\hat{\rho}(\delta\xi) - 1|^2$. The first product is integrable, and the last factor is bounded by 4 and tends to zero. Dominated convergence proves gamma convergence. The prime channels are finite and bounded in that window; the polar channels are continuous functionals on its support. This gives convergence in the joint readout norm and, consequently, convergence of every finite Gram matrix.

Corollary 15.17. Two orientations of the same prime window

Set $D_{\varepsilon} = K_{\varepsilon}(0)$. For two centres separated by L , the complete Gram matrix is

$$G_2 = \begin{pmatrix} D_{\varepsilon} & K_{\varepsilon}(L) \\ K_{\varepsilon}(L) & D_{\varepsilon} \end{pmatrix}.$$

Therefore

$$\mathcal{W}(af + bg, af + bg) = \frac{D_{\varepsilon} + K_{\varepsilon}(L)}{2} |a + b|^2 + \frac{D_{\varepsilon} - K_{\varepsilon}(L)}{2} |a - b|^2.$$

Positivity on their span requires both $D_{\varepsilon} + K_{\varepsilon}(L) \geq 0$ and $D_{\varepsilon} - K_{\varepsilon}(L) \geq 0$.

If $L \geq \varepsilon$, define

$$I_{\varepsilon}(L) = \frac{F(L - \varepsilon) - 2F(L) + F(L + \varepsilon)}{\varepsilon} > 0,$$

$$P_{\varepsilon}(L) = \sum_{|\ell_{p,k} - L| < \varepsilon} w_{p,k} \left(1 - \frac{|\ell_{p,k} - L|}{\varepsilon} \right).$$

Strict convexity of each exponential proves positivity of I_{ε} . The condition on both signs is written exactly as

$$2a_{\varepsilon}^2 \cosh(L/2) - I_{\varepsilon}(L) - D_{\varepsilon} \leq P_{\varepsilon}(L) \leq 2a_{\varepsilon}^2 \cosh(L/2) - I_{\varepsilon}(L) + D_{\varepsilon}.$$

Thus the complete cross term controls a window of irreducible-clock lengths with their weights, not merely a diagonal value. The identity follows from Proposition 15.16 and diagonalization of the two-entry matrix; it does not presuppose its sign. $\square$

Proposition 15.18. Joint control of nine test functions

Let $\varepsilon = 1/100$, $t_j = j \log 2$, $0 \leq j \leq 8$, and $G_{ij} = K_{\varepsilon}(t_i - t_j)$. The interval calculation preserved with this chapter certifies

$$\lambda_{\min}(G) > 0.8479.$$

Consequently,

$$\mathcal{W}\left(\sum_{j=0}^8 z_j b_{\varepsilon, t_j}, \sum_{j=0}^8 z_j b_{\varepsilon, t_j}\right) > 0.8479 \sum_{j=0}^8 |z_j|^2 \quad (z \neq 0).$$

Proof and reproduction. For $F_N(x) = \sum_{n=0}^{N-1} e^{-\lambda_n x} / \lambda_n^2$, the summand is decreasing and

$$0 \leq F(x) - F_N(x) \leq e^{-\lambda_N x} \left(\lambda_N^{-2} + \frac{1}{2\lambda_N} \right).$$

Indeed, separate the first term of the tail and bound the remainder by the integral of $(2u + 1/2)^{-2}$, factoring out the largest exponential. We use $N = 1024$ and interval operations with outward rounding. The condition $8 \log 2 + \varepsilon < \log 259$ allows all prime-power pairs up to 259 to be enumerated: this gives 71 rows. The irreducibles and products come from the preserved native producer; recognition by ordinary integers is subsequent. The constant c_Γ is evaluated from the certified intervals of the preceding coordinates, not from a target value of this form.

The formula in Proposition 15.16 gives intervals for the nine entries of the first row. For each row i , these intervals verify

$$G_{ii} - \sum_{j \neq i} |G_{ij}| > 0.8479.$$

The matrix is Hermitian. The inequality $2|z_i||z_j| \leq |z_i|^2 + |z_j|^2$, applied to each cross term, directly gives the stated quadratic bound. The verification preserves the intervals for all nine rows and the complete census. The strict margins and the convergence of Gram matrices proved above also allow simultaneous regularization of the nine test functions, yielding positivity on their smooth span for a sufficiently fine regularization. $\square$

This control covers all complex combinations of nine test functions, not merely their sum. Its dimension remains finite. The computational record states that scope and does not use its result as a certificate of a universal inequality.

Proposition 15.19. Refinement and eight detail components

The geometric subdivision of an interval into nine parts satisfies

$$b_{\varepsilon, t} = \frac{1}{3} \sum_{r=0}^8 b_{\varepsilon/9, t+(r-4)\varepsilon/9}.$$

If $V = (1, \dots, 1)^T/3$, then

$$G_{\text{padre}} = V^* G_{\text{hijos}} V.$$

For several predecessors, take the blockwise isometric inclusion with that same column. In a basis formed by the uniform component and the eight detail components for each predecessor, write

$$G_{\text{hijos}} = \begin{pmatrix} A & B^* \\ B & D \end{pmatrix}, \quad A = G_{\text{padres}}.$$

If $A > 0$, the exact condition is

$$G_{\text{hijos}} \geq 0 \iff D - BA^{-1}B^* \geq 0.$$

Proof. The nine successor intervals partition the predecessor apart from endpoints of measure zero. Normalization of their indicator functions proves the linear identity, and sesquilinearity gives the Gram identity with every cross term. Finally,

$$\begin{pmatrix} x \\ y \end{pmatrix}^* \begin{pmatrix} A & B^* \\ B & D \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (x + A^{-1}B^*y)^* A (x + A^{-1}B^*y) \\ + y^* (D - BA^{-1}B^*) y.$$

Taking $x = -A^{-1}B^*y$ proves necessity, and the same identity proves sufficiency. $\square$

The uniform component reproduces the previous Gram matrix; the details contain the new interference terms. Preserving both parts is the exact requirement for extending a positivity proof under this refinement. The return of nine phases does not by itself make the Gram matrix circulant: in the preceding collection, $K_\varepsilon(\log 2) \neq K_\varepsilon(8 \log 2)$. Phase and distance memory remain distinct data.

The detail matrix and its Schur complement provide a verifiable condition for each enlargement. The prospective readout is not defined through a retrospective factorization of the form; it is the form expressed by the previously constructed readouts that is checked. Completing the global argument requires an estimate for all those details, uniform over cofinal enlargements and over the complete test-function domain.

15.14 Coercive assembly and stability of nonadic details

The basal property allows entire function spaces to be controlled. Assembling several intervals requires comparing that local margin with every interaction between routes. The complete kernel of the preceding section provides an estimate for those interactions before a function is chosen. This yields an infinite-dimensional result on a disconnected domain, not merely a Gram matrix of indicator functions.

The positions are taken after representation of the irreducible clocks. The explicit case uses successive translations of the native clock 2 and a declared positional width. The difference operators preserve both orientations. Subsequent refinement subdivides each interval into nine parts and retains the same function before its two energy readouts. This analytic collection is specified as such: its cardinality does not by itself identify it with a canonical selection of TPK states.

Proposition 15.20. Diagonal margin and bound for cross terms

Let $t_1 < \dots < t_m$ be distinct centres and $I_j = (t_j - \varepsilon/2, t_j + \varepsilon/2)$ disjoint intervals, with $0 < \varepsilon < \log 2$. Define

$$\delta_\varepsilon = 2M_\Gamma(\varepsilon) + c_\Gamma - 2 \sinh(\varepsilon/2) + \varepsilon.$$

For $f_j \in C_c^\infty(I_j)$,

$$\mathcal{W}(f_j, f_j) \geq \delta_\varepsilon \|f_j\|_2^2.$$

If $d_{ij} = |t_i - t_j| > \varepsilon$, set

$$\rho_{ij}(\varepsilon) = \sum_{|\ell_{p,k} - d_{ij}| < \varepsilon} w_{p,k} + \varepsilon \mu(d_{ij} - \varepsilon) + 2\varepsilon \cosh((d_{ij} + \varepsilon)/2).$$

Then

$$|\mathcal{W}(f_i, f_j)| \leq \rho_{ij}(\varepsilon) \|f_i\|_2 \|f_j\|_2 \quad (i \neq j).$$

Proof. On an interval of width ε , the gamma tail contributes at least $2M_\Gamma(\varepsilon) \|f_j\|_2^2$. There is no diagonal prime-power correlation, because its minimum length is $\log 2 > \varepsilon$. In a larger ambient window, each prime difference and its counterterm cancel on that support. After centring the interval, the negative polar port has squared norm $2 \sinh(\varepsilon/2) - \varepsilon$. Discarding the positive polar port and the gamma part preceding the tail gives the bound. Translation invariance of the complete form carries the same bound to any centre.

In the cross term, the scalar contribution vanishes because the supports are disjoint. For each positive length, only one of the two orientations can connect the intervals. Unitarity of translation bounds the prime contribution by the stated sum of weights. The gamma cross term has kernel $-\mu(|x - y|)$ on $I_i \times I_j$. Since μ is decreasing,

$$|\mathcal{W}_\Gamma(f_i, f_j)| \leq \mu(d_{ij} - \varepsilon) \|f_i\|_1 \|f_j\|_1 \leq \varepsilon \mu(d_{ij} - \varepsilon) \|f_i\|_2 \|f_j\|_2.$$

The complete polar kernel is $2 \cosh((x - y)/2)$. The bound $|x - y| \leq d_{ij} + \varepsilon$ and the same inequality between the L^1 and L^2 norms give the polar term. Adding the three estimates proves the result. $\square$

Theorem 15.21. Coercivity of the assembly

If

$$\eta = \delta_\varepsilon - \max_i \sum_{j \neq i} \rho_{ij}(\varepsilon) > 0,$$

then, for $\Omega = \bigcup_j I_j$,

$$\boxed{\mathcal{W}(f, f) \geq \eta \|f\|_2^2 \quad \text{for every } f \in C_c^\infty(\Omega).}$$

The assertion holds in the closure of that domain in the joint readout norm. In particular, for

$$m = 9, \quad t_j = j \log 2 \quad (0 \leq j \leq 8), \quad \varepsilon = 1/1000,$$

the certified evaluation gives $\eta > 1$.

Proof. Writing $f = \sum_j f_j$ on the disjoint supports and $x_j = \|f_j\|_2$ gives

$$\begin{aligned} \mathcal{W}(f, f) &\geq \delta_\varepsilon \sum_j x_j^2 - 2 \sum_{i < j} \rho_{ij}(\varepsilon) x_i x_j \\ &\geq \sum_i \left(\delta_\varepsilon - \sum_{j \neq i} \rho_{ij}(\varepsilon) \right) x_i^2. \end{aligned}$$

We used $2x_i x_j \leq x_i^2 + x_j^2$ for each cross term. The sum of the squared norms is $\|f\|_2^2$. Continuity of the readouts extends the inequality to the closed domain.

In the explicit case, the distances are $d \log 2$, $1 \leq d \leq 8$. For $\varepsilon = 1/1000$,

$$2^d(e^\varepsilon - 1) < \frac{256}{999} < 1, \quad 2^d(1 - e^{-\varepsilon}) < \frac{256}{1000} < 1.$$

The first inequality uses the exponential series and $e^\varepsilon < 1/(1 - \varepsilon)$; the second uses $1 - e^{-\varepsilon} < \varepsilon$. Since the expressed powers are integers, the only prime-power clock in the window at that distance is $(p, k) = (2, d)$. Thus

$$\rho_d = (\log 2) 2^{-d/2} + \frac{1}{1000} \mu(d \log 2 - 1/1000) + \frac{2}{1000} \cosh((d \log 2 + 1/1000)/2).$$

The native producer also verifies the complete census up to 259. The calculation of δ_ε uses the previously proved formula for M_Γ and the identity

$$\arctan t = \frac{\pi}{4} - \arctan \frac{1-t}{1+t} \quad (0 < t < 1).$$

The second argument is small, and its alternating series has an error bound given by the first omitted term. The internal representation of π , the rational interval for gamma, and the subsequent interval operations yield the bounds

$$\delta_\varepsilon > 4.4921, \quad \max_i \sum_{j \neq i} \rho_{ij} < 2.5385, \quad \eta > 1.9536 > 1.$$

The receipt retains the margin for each row. No particular functions are evaluated to infer the universal inequality on $C_c^\infty(\Omega)$: it follows from the operator estimate above. $\square$

Corollary 15.22. All refinements of the assembled domain

Successively subdivide each I_j into nine parts. The normalized indicators of the 9^{n+1} intervals at depth n form an orthonormal collection in L^2 . Their full Gram matrix satisfies

$$G_n \geq \eta I \quad (n \geq 0).$$

The uniform inclusion V_n , with nine components $1/3$ per predecessor, satisfies

$$V_n^* G_{n+1} V_n = G_n.$$

In the uniform–detail decomposition,

$$G_{n+1} = \begin{pmatrix} A_n & B_n^* \\ B_n & D_n \end{pmatrix}, \quad A_n = G_n,$$

one obtains, at any depth,

$$\boxed{D_n - B_n A_n^{-1} B_n^* \geq \eta I.}$$

Proof. The theorem applies to the regularizations of each combination of indicators. The norm convergence of the readouts in subsection 15.13 allows passage to the limit. An indicator reaching an endpoint of I_j is approximated first by slightly contracting its support towards the centre, and then by regularizing within I_j . The triangular correlation and its integrable translation bound prove the same convergence; no extension of the coercivity domain is used. Orthonormality in L^2 turns the functional bound into $G_n \geq \eta I$, and exact subdivision gives the compression identity.

For a detail vector y , take $x = -A_n^{-1} B_n^* y$. The Schur identity gives

$$\begin{aligned} y^* (D_n - B_n A_n^{-1} B_n^*) y &= \begin{pmatrix} x \\ y \end{pmatrix}^* G_{n+1} \begin{pmatrix} x \\ y \end{pmatrix} \\ &\geq \eta (\|x\|^2 + \|y\|^2) \geq \eta \|y\|^2. \end{aligned}$$

The same η works at every depth. □

The corollary controls the eight new details per predecessor at arbitrary depth within the assembly. The function is neither replaced by its mean nor stripped of its zero-sum components. This strengthens the Gram matrix of nine indicators: it covers infinite-dimensional function spaces and gives a depth-independent bound.

Finite collections and extension of the domain

For any finite set of distinct centres, there is a sufficiently small positive width for which theorem 15.21 applies. Indeed, $2M_\Gamma(\varepsilon) \rightarrow +\infty$ as $\varepsilon \downarrow 0$, whereas the local negative polar term tends to zero. The distances between centres are bounded away from zero. The sums of prime weights remain bounded as each window shrinks, because there are finitely many lengths below any fixed bound. The gamma and polar cross terms tend to zero. Thus, δ_ε eventually exceeds the sum of interactions in each row.

The admissible width depends on the centres. The unions of narrow intervals obtained in this way do not, by the construction above alone, form a cofinal covering of all compactly supported functions. Global positivity also requires control of the growth of the domain and its new interactions. The proved uniformity in depth remains distinct from extension to an arbitrary test-function domain, which is not proved here.

15.15 Coercivity of details in an arbitrary fixed window

The preceding estimate retains a uniform margin when the interiors of nine separated intervals are refined. A second construction controls the fine details of any fixed window, including its contiguous cells. The two statements have different and complementary domains: the first estimates the full function on a disconnected support; the second estimates the cellwise zero-mean components in an entire window.

The previously constructed clocks, weights, and readouts are retained. In particular, the lengths $\ell_{p,k}$, the weights $w_{p,k}$, the scalar c_Γ , and the internal representation π retain their provenance. The following argument acts on the functional realization of these objects; it does not modify their generator. Write

$$\beta_R = 2 \sinh(R/2) - R, \quad \epsilon_R = \frac{2}{3} - c_\Gamma + 2 \sum_{\ell_{p,k} \leq R} w_{p,k} + \beta_R.$$

The sum is finite and comes from the native product below the bound e^R . Divide $I_R = (-R/2, R/2)$ into M cells of length at most h , with $0 < h \leq 1$.

Theorem 15.23. Explicit threshold for zero-mean details

Let d be a function in the joint domain of the readouts, supported in I_R , whose integral over each cell is zero. Then

$$\mathcal{W}_R(d, d) \geq [\log(1 + 4/h) - \epsilon_R] \|d\|_2^2.$$

Consequently, the form is strictly coercive on this detail space if

$$0 < h < \min\left(1, \frac{4}{\exp(\epsilon_R) - 1}\right).$$

The threshold depends on the window R , but not on the function or on a finite enumeration of examples.

Proof. On each cell $[a, b]$, define $F(x) = \int_a^x d(t) dt$. The zero-mean condition implies $F(a) = F(b) = 0$. Assembling the primitives yields $F \in H_0^1(I_R)$, extended by zero outside the window, with $F' = d$. The Dirichlet inequality on each cell gives

$$\|F\|_2^2 \leq \frac{h^2}{\pi^2} \|d\|_2^2.$$

Here π is the same internal coordinate recognized in the functional realization. No new constant is selected.

The gamma tail admits the positive expansion already used,

$$\mu(a) = \sum_{n=0}^{\infty} e^{-\lambda_n a}, \quad \lambda_n = 2n + \frac{1}{2}.$$

For $\lambda > 0$, let C_λ denote convolution with $e^{-\lambda|x|}$. The contribution of one summand is

$$E_\lambda(d) = \int_0^\infty e^{-\lambda a} \|(I - T_a)d\|_2^2 da = \frac{2}{\lambda} \|d\|_2^2 - \langle d, C_\lambda d \rangle \geq 0.$$

Its positivity is that of an integral of squared norms. Denote the unitary Fourier transform by $\mathcal{F}_u F = (2\pi)^{-1/2} \hat{F}$, distinguishing it from the unnormalized transform used earlier. The multiplier of C_λ is $2\lambda/(\lambda^2 + \xi^2)$. Since $d = F'$, Plancherel implies

$$\begin{aligned}\langle d, C_\lambda d \rangle &= \int_{\mathbb{R}} \frac{2\lambda\xi^2}{\lambda^2 + \xi^2} |(\mathcal{F}_u F)(\xi)|^2 d\xi \\ &\leq 2\lambda \|F\|_2^2 \leq \frac{2\lambda h^2}{\pi^2} \|d\|_2^2.\end{aligned}$$

By Tonelli, the gamma form is the sum of the E_{λ_n} . The first $N + 1$ terms are retained, and only a nonnegative tail is discarded. This yields

$$\begin{aligned}Q_\Gamma(d) &\geq S_N(h) \|d\|_2^2, \\ S_N(h) &= 4 \sum_{n=0}^N \frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2}.\end{aligned}$$

Choose $N = \lfloor 1/h \rfloor$. Then $\lambda_N h \leq 2 + h/2 < \pi$; the retained bounds are nonnegative. Comparison with the integral of a decreasing function gives

$$4 \sum_{n=0}^N \frac{1}{4n+1} \geq \int_0^{N+1} \frac{4}{4x+1} dx = \log(4N+5) \geq \log(1+4/h).$$

Moreover, $\pi^2 > 9$ and $h \leq 1$ imply

$$\frac{h^2(N+1)(2N+1)}{\pi^2} \leq \frac{(1+h)(2+h)}{\pi^2} < \frac{2}{3}.$$

Therefore, $Q_\Gamma(d) \geq [\log(1+4/h) - 2/3] \|d\|_2^2$. In the full form, the scalar term contributes $c_\Gamma \|d\|_2^2$; the prime part is bounded below by $-2 \sum_{\ell_{p,k} \leq R} w_{p,k} \|d\|_2^2$ by unitarity of each translation; and the polar part is bounded by $-\beta_R \|d\|_2^2$, as proved in subsection 15.12. Their sum proves the stated inequality. The threshold displayed is equivalent to $\log(1+4/h) > c_R$. Passage to the closed domain preserves the estimate by continuity of the readouts and of the cell integrals. $\square$

Corollary 15.24. Finite dimensionality of the possible negative directions

For fixed R and a partition satisfying the threshold above, any subspace of the domain on which $\mathcal{W}_R$ is strictly negative definite has dimension at most M , the number of cells.

Proof. Consider the linear map

$$\mathcal{P}(d) = \left(\int_{I_1} d, \dots, \int_{I_M} d \right) \in \mathbb{C}^M.$$

Its kernel is the detail space of theorem 15.23, on which the form is positive definite. The restriction of $\mathcal{P}$ to a strictly negative subspace is injective: a nonzero vector in its kernel would have both a positive and a negative value. The dimension of that subspace cannot exceed M . $\square$

This reduction retains a finite coupling problem in each window. It is not equivalent to proving that the negative index is zero. In a mean–detail decomposition, the detail block is coercive at sufficient depth, but the mean block and the cross terms must be retained. If the Schur complement is used, its sign requires that full composition. In particular, an exact prime resonance between two cells does not disappear when zero mean is imposed: it may act as a negative multiple of the identity on their corresponding details. The theorem retains such terms through its translation bound.

The result provides a second effective uniformity: in any fixed window, sufficiently fine details have an explicit common bound. Subsection 15.14 also proves the sign of the full assembly, including its cross terms, for the specified union of nine intervals. Extending that sign to every window remains the global step that this edition does not regard as proved.

16 Regularized derivatives and odd spectral values

The continuation and functional equation proved in Proposition 15.2 relate two operations on the same functional Z : differentiation at negative integers and convergent evaluation at positive odd integers. The regional coordinates and enumeration operator retain their preceding genealogy. The symbol π denotes the HMT coordinate already used in the theta–Mellin construction.

Let $H_k = \sum_{j=1}^k 1/j$, with $H_0 = 0$, and let B_j be the Bernoulli numbers constructed by the recurrence in Section 13, with $B_1 = -1/2$. The notation B_j distinguishes these coefficients from the Meissel–Mertens constant. Define the Bendersky–Glaisher–Kinkelin hierarchy by

$$A_k = \exp\left(\frac{H_k B_{k+1}}{k+1} - Z'(-k)\right), \quad k \in \mathbb{N}_0.$$

The points $-k$ are regular. The Bernoulli term fixes the normalization of each level before evaluation.

16.1 Covariance of regularized derivatives

Here the functional is $Z_\lambda = \lambda^{-s}Z$, with residue λ^{-1} , and we define

$$D_k(\lambda) = \exp[-Z'_\lambda(-k)].$$

This object retains the normalization of the scaled operator, distinct from the unit-residue normalization used for $\hat{Z}_\lambda$ in Section 14.3.

Proposition 16.1. Transport of regularized derivatives

For $\lambda > 0$ and $k \in \mathbb{N}_0$,

$$\begin{aligned} Z'_\lambda(-k) &= \lambda^k (Z'(-k) - (\log \lambda)Z(-k)), \\ D_k(\lambda) &= D_k(1) \lambda^k \lambda^{kZ(-k)}, \quad D_k(1) = A_k \exp\left(-\frac{H_k B_{k+1}}{k+1}\right). \end{aligned}$$

Proof. The product rule applied to $\lambda^{-s}Z(s)$ gives the first identity. Its negative exponential gives the second, and the definition of A_k gives the third. $\square$

At the initial levels,

$$\begin{aligned} A_0 &= e^{-Z'(0)}, \\ \log A_1 &= \frac{1}{12} - Z'(-1), \quad \log A_2 = -Z'(-2), \\ \log A_3 &= -\frac{11}{720} - Z'(-3), \quad \log A_4 = -Z'(-4). \end{aligned}$$

For example, $D_1(1) = A_1 e^{-1/12}$. Retaining the normalization prevents two different readouts from being identified merely because they involve the same derivative.

16.2 Recovery of odd values

Theorem 16.2. Correspondence between negative even and positive odd levels

For every integer $n \geq 1$,

$$Z'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}} Z(2n+1),$$

$$\frac{\log D_{2n}(\lambda)}{\lambda^{2n}} = \log A_{2n} = (-1)^{n+1} \frac{(2n)!}{2(2\pi)^{2n}} Z(2n+1).$$

Proof. The functional equation of the completed form implies the meromorphic identity

$$\pi^{-s/2} \Gamma(s/2) Z(s) = \pi^{-(1-s)/2} \Gamma((1-s)/2) Z(1-s).$$

At $s = -2n$, the right-hand side is regular and the gamma factor on the left has a simple pole. Hence $Z(-2n) = 0$. The gamma recurrence and its residue at zero give

$$\Gamma(s/2) = \frac{2(-1)^n}{n!(s+2n)} + O(1).$$

Moreover,

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)! \sqrt{\pi}}{4^n n!}.$$

Comparing constant terms in the meromorphic identity yields

$$\pi^n \frac{2(-1)^n}{n!} Z'(-2n) = \pi^{-n} \frac{(2n)!}{4^n n!} Z(2n+1),$$

which gives the first formula. The series $t/(e^t - 1) + t/2$ is even; therefore $B_{2n+1} = 0$ for $n \geq 1$. The definition of A_{2n} and the preceding proposition complete the proof. $\square$

In particular, the first positive even level satisfies

$$Z(3) = 4\pi^2 \log A_2.$$

The Apéry constant evaluated with a remainder in Section 13 is thus related to a regularized derivative of the same functional. The intervals proved there are transported by the monotone exponential once the bounds for the regional coordinate π are retained.

Level zero has a different evaluation. The functional equation, together with gamma reflection and duplication, is equivalent to

$$Z(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) Z(1-s).$$

Euler's formula

$$\Gamma(1+z) = \lim_{m \rightarrow \infty} m^z \prod_{j=1}^m (1+z/j)^{-1}$$

gives $\log \Gamma(1+z) = -\gamma_0 z + O(z^2)$: the linear coefficient is the limit of $\log m - H_m$, and the quadratic remainder is controlled by the convergent series of j^{-2} . Using $Z(1-s) = -1/s + \gamma_0 + O(s)$ and $\Gamma(1-s) = 1 + \gamma_0 s + O(s^2)$, the degree-one terms involving γ_0 cancel, giving

$$Z(s) = -\frac{1}{2} - \frac{1}{2} \log(2\pi)s + O(s^2).$$

Consequently,

$$A_0 = \sqrt{2\pi}, \quad D_0(\lambda) = \sqrt{\frac{2\pi}{\lambda}}.$$

The condition $n \geq 1$ in the theorem preserves this distinction: level zero is not obtained by substituting $Z(1)$, which has a pole, into the odd-value formula. The gamma identities express subsequent analytic normalization without participating in seed selection.

17 Nonadic compression and circular moments of the Weil form

The relation between memory and positivity requires preserving two levels of readout. A nonadic winding records a history and admits a positive Hilbert-space realization; the arithmetic readouts of that same genealogy express a Hermitian form involving the gamma, prime, and polar channels jointly. The transport problem consists in identifying their energies and correlations, with the same domains and without replacing one of these realizations by the other. The following identities make this comparison explicit before applying the Gelfand–Naimark–Segal (GNS) construction to the arithmetic form.

The complete balance of a nine-phase chart is preserved first, including any norm variation at the seam. Next, the loss register, its moment matrices, and a Cayley operator acting on exponentially decaying test functions are constructed. Local coercivity can be transported to entire collections of test functions with tails; passage to their algebraic sum and the global domain additionally requires the mixed correlations specified below. This development does not assert that global Weil positivity has been established.

17.1 Enriched state and modes of the central chart

The construction used is $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow \text{enriched state} \rightarrow \text{joint discrete structure of the continuum}$. APP preserves sum and product evaluations, their quotients and residues, on the same support. The TRIT preserves orientation and regime; the TPK composes selection, transport, and memory update. Nonadic holonomy returns the phase and increases the depth of the history. The five correlative aspects of the continuum are neither separated nor redefined through the spectral question studied subsequently.

This development begins with the realization of the winding register of that structure. It does not seek to replace the complete construction of the continuum with a sequence space. Integer memory is one coordinate of the state, not its entirety. Nor does it identify the reduced phase register with the TPK generator.

The author’s note on sum and product indicates the provenance of the central matrix. In the preserved positive chart, one obtains

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-, \quad P_{\pm} = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

Thus, after normalization of the uniform mode, the differential mode has ratio $q = 5/9$. The names “sum” and “product” describe the APP operations generating the chart; the projectors P_+ and P_- are its symmetric and antisymmetric modes. Equality between a mode and a particular sheet requires preserving the map relating them.

17.2 Uniform compression of the nonadic chart and exact balance

Let $\mathcal{K}$ be a Hilbert space and C a bounded operator on it. On $\mathcal{K}^9$, the nine-phase chart with seam C acts by

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \quad S_C^9 = I_9 \otimes C.$$

The uniform inclusion is

$$Jv = \frac{1}{3}(v, \dots, v), \quad J^*J = I.$$

Write the visible compression and its complementary component as

$$T = J^* S_C J = \frac{8I + C}{9}, \quad \eta = (I - JJ^*) S_C J.$$

Proposition 17.1 (Exact balance of compression and seam). *Without assuming that C is isometric, one has*

$$\begin{aligned} \eta^* \eta &= \frac{8}{81} (I - C)^* (I - C), \\ I - T^* T &= \eta^* \eta + \frac{1}{9} (I - C^* C). \end{aligned} \tag{17.1}$$

Proof. Orthogonality of JJ^* gives

$$\eta^* \eta = J^* S_C^* S_C J - T^* T.$$

The first term is $(8I + C^* C)/9$. The second is $(64I + 8C + 8C^* + C^* C)/81$. Subtraction proves the first identity. Now subtracting $T^* T$ from I and separating $I - (8I + C^* C)/9$ proves (17.1). $\square$

The formula does not eliminate memory: it displays exactly two contributions. One measures dispersion among the nine phases; the other measures the norm defect of the seam. If the preceding construction has given $C^* C = I$, the positive compression–expression identity of the corpus remains:

$$I - T^* T = \eta^* \eta \geq 0.$$

This result explains what memory positivity contains and where identification of the readout must enter. It does not permit replacing that identification with a drawing of a circle.

17.3 Visible contraction and preservation of the winding register

Now suppose that C is unitary, as for the bilateral shift of integer memory. The complete winding of the differential mode is represented by $M = qC$, not by nine successive applications of the same factor q .

For each integer $N \geq 1$, define the loss-and-terminal register

$$L_N v = (\sqrt{1 - q^2} v, \sqrt{1 - q^2} q C v, \dots, \sqrt{1 - q^2} q^{N-1} C^{N-1} v, q^N C^N v).$$

Proposition 17.2 (Isometric preservation of the winding register). *L_N is isometric. As N increases, the last component splits into a newly expressed memory and a new terminal, without erasing the preceding register. The infinite register*

$$L v = (\sqrt{1 - q^2} q^j C^j v)_{j \geq 0}$$

is also isometric.

Proof. For $v, w \in \mathcal{K}$,

$$\langle L_N v, L_N w \rangle = \left[(1 - q^2) \sum_{j=0}^{N-1} q^{2j} + q^{2N} \right] \langle v, w \rangle = \langle v, w \rangle.$$

In the last component, $q^N C^N v$ is transformed into $(\sqrt{1 - q^2} q^N C^N v, q^{N+1} C^{N+1} v)$; that transformation preserves the inner product. The terminal has norm $q^N \|v\|$, which tends to zero; the infinite sum converges and preserves the norm. $\square$

For two initial states, this proof preserves their cross products; it does not merely sum energies of isolated trajectories. The positive sign of memory follows from orthogonality of its component spaces and from the isometry of the transport already constructed.

17.4 Dual memory circle and arithmetic Cayley chart

The integer winding register admits the bilateral shift $Se_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$. Its characters $n \mapsto z^n$, $|z| = 1$, give the dual circle of memory. The visible phase $r \in \{0, \dots, 8\}$ returns; the integer n is not reduced modulo 12 in the complete state.

If that reduction is imposed, a character descends only when $z^{12} = 1$. Thus memory without reset has a precise consequence: it permits the full circle of characters, whereas the finite calendar retains only a finite selection. This does not determine which arithmetic measure is realized on that circle.

The second circle appearing in the spectral chapter of the integral treatise is a chart of the Cayley realization. Subsection 17.10 constructs a Cayley operator directly on test functions before constructing a Hilbert space through the Weil form. Arithmetic transport can thus be compared with memory transport without introducing zero heights as inputs.

17.4.1 Preservation of the arithmetic form under nonadic compression

Arithmetic invariance of the auxiliary Cayley readout permits an additional composition. For $a > b > 1/2$, let C_a be the operator constructed in Subsection 17.10, and let $\mathcal{T}_a = (8I + C_a)/9$. The equality $\mathcal{W}(C_a f, C_a g) = \mathcal{W}(f, g)$ implies, by expansion,

$$\mathcal{W}(f, g) = \mathcal{W}(\mathcal{T}_a f, \mathcal{T}_a g) + \frac{8}{81} \mathcal{W}((I - C_a)f, (I - C_a)g). \quad (17.2)$$

This identity preserves the gamma channel, prime clocks, and polar term jointly. It does not presuppose that the form is positive. The calculation consists in adding coefficients: the mixed terms have coefficients 8 and -8 , which cancel, while the two diagonal terms contribute $72 + 9 = 81$ times the initial form. Its finite iteration gives

$$\mathcal{W}(f, g) = \mathcal{W}(\mathcal{T}_a^N f, \mathcal{T}_a^N g) + \frac{8}{81} \sum_{j=0}^{N-1} \mathcal{W}((I - C_a)\mathcal{T}_a^j f, (I - C_a)\mathcal{T}_a^j g). \quad (17.3)$$

In the ordinary L^2 representation, where C_a is already unitary, the same calculation is a positive norm decomposition. On the arithmetic form it is, initially, exact conservation of signed energy. The difference between these two assertions is located in a common identity; it is not hidden in a change of representation.

In a circular chart $z = e^{i\theta}$, compression sends z to $t(z) = (8 + z)/9$. Therefore,

$$1 - |t(z)|^2 = \frac{8}{81} |1 - z|^2 = \frac{32}{81} \sin^2(\theta/2).$$

Angular separation thus has an exact expression as complementary energy. This is the image of nonadic compression of a phase chart; the drawing does not identify it with another operator of the corpus. Nor is this map $t(z)$ confused with the radial factor $5/9$, which arises from the two modes of the central APP chart.

The local Schur development gives the arithmetic form a positive sign on a concrete functional domain and its transport images. Identities (17.2)–(17.3) do not by themselves authorize declaring all their summands positive for an arbitrary test function: they locate exactly which energies and correlations an extension of that domain must preserve.

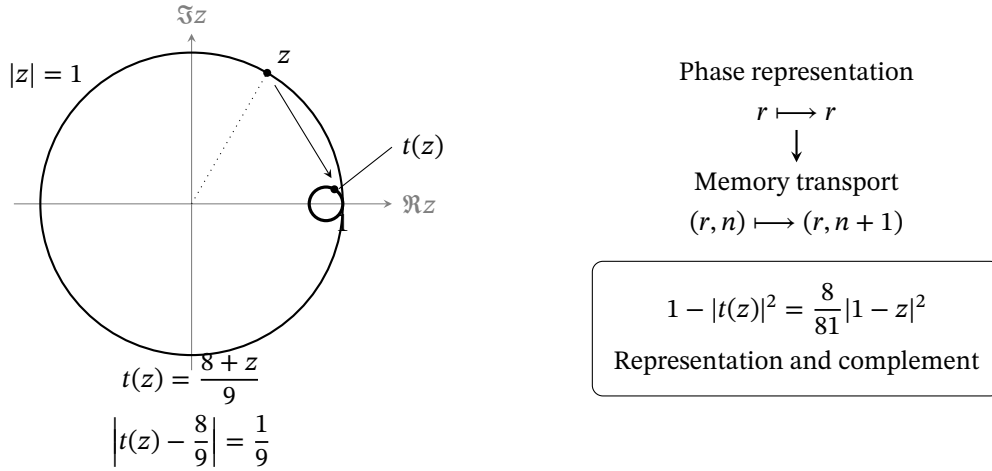


Figure 7: The circular chart of nonadic compression. On the left, the unit circle and its image with center $8/9$ and radius $1/9$, drawn to the same scale; the image of $z = e^{i\pi/3}$ is shown. On the right, phase return is accompanied by memory advancement. The lower balance quantifies the complementary component. This figure represents the map t , distinct from the modal factor $q = 5/9$ of the central chart.

17.5 Criterion for transporting positivity through mixed moments

Let $\mathcal{D}$ be the common domain of the previously constructed readouts. Let $\mathcal{W}$ be the sesquilinear form expressed by their gamma, prime, and polar channels. The analytic formula is received after the native generator; it does not select the seeds or preceding coefficients.

Let C_a be the reversible transport on test functions and S the memory shift. We explicitly use $C_a(\mathcal{D}) = \mathcal{D}$ and the identity $\mathcal{W}(C_a f, C_a g) = \mathcal{W}(f, g)$, proved for the exponential class of test functions in Subsections 17.10 and 17.11. For a collection $\mathcal{E} \subset \mathcal{D}$, suppose a linear map $V : \text{span } \mathcal{E} \rightarrow \mathcal{K}$ has been constructed such that, for every f, g in that collection and every integer n ,

$$\mathcal{W}(C_a^n f, g) = \langle S^n V f, V g \rangle. \quad (17.4)$$

Proposition 17.3 (Transport of positivity by identification of moments). *Identity (17.4) implies positivity of $\mathcal{W}$ on the span of the orbits $C_a^n \mathcal{E}$. If that space is dense in $\mathcal{D}$ for the joint norm of the readouts, and these are continuous for that norm, positivity extends to $\mathcal{D}$.*

Proof. For $F = \sum_j C_a^{n_j} f_j$, invariance of $\mathcal{W}$ under C_a and (17.4) give

$$\begin{aligned} \mathcal{W}(F, F) &= \sum_{i,j} \mathcal{W}(C_a^{n_i - n_j} f_i, f_j) \\ &= \left\| \sum_i S^{n_i} V f_i \right\|^2 \geq 0. \end{aligned}$$

If a combination represents the zero function, the same identity forces the vector combination to have zero norm; the realization is well-defined. Passage to the closure uses continuity of both readouts, not a subsequent choice of sign. $\square$

This is a construction criterion with explicit domains, not an assertion that (17.4) has been proved for all readouts of the corpus. Extension from the initial collection to its orbits requires preserving all cross moments: proving it only for $n = 0$ on that initial collection does not complete the extension. Naturally, a Gram identity already proved for $n = 0$ on all of $\mathcal{D}$ would directly prove positivity on that domain.

17.6 Energy identification and two-column realization

The corpus preserves two columns J_+ and J_- and an effective analytic connector between them. It also describes a native encoding Ξ and an orthogonal projection P . When the two moment identities are calculated on the same domain,

$$\langle \Xi f, \Xi g \rangle = \langle J_+ f, J_+ g \rangle, \quad \langle P \Xi f, P \Xi g \rangle = \langle J_- f, J_- g \rangle,$$

subtraction gives

$$\mathcal{W}(f, g) = \langle (I - P) \Xi f, (I - P) \Xi g \rangle. \quad (17.5)$$

The dossier calculates the uniform expectation and Helmert refinement separately. Their joint conjugation preserves both Grams and cannot, by a mere change of notation, turn a difference operator into an evaluation operator. The recovered gamma connector realizes an additional analytic operation, which must preserve its energy budget. This clarification brings together existing operations; it does not declare the TPK architecture nonexistent.

Formulas (17.4) and (17.5) are two compatible expressions of the route to be completed: the first organizes returns and their correlations; the second organizes the visible part and complementary memory. The Schur-complement estimate in the dossier gives effective control of that energy in a complete window and all its refinements. Its scope is stated with the concrete window, without replacing the global domain by that window.

17.7 Operator provenance and the function of the readouts

The chain used is $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow \text{enriched histories of the joint continuum} \rightarrow \text{arithmetic and memory readouts} \rightarrow \text{circular representation}$. The five continuum operations remain on the same histories; this development studies a readout of their memory and does not construct an isolated fifth branch. The prime readout uses irreducibles produced by the native product; constants already expressed by HMT enter the readouts only subsequently.

Provenance. Cocycle, loss, two sheets, memory groups, and double circle: preexisting authorial architecture and recovered results. The assembled proofs below are a new formalization of these antecedents. No historical priority is claimed for Poisson kernels, Cayley, or Toeplitz identities. The provenance of this exact composition within the corpus has not been exhaustively investigated beyond the source owners indicated.

17.8 Regular memory representation preceding arithmetic GNS

The central APP block, with its symmetric and antisymmetric projectors, is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-.$$

Normalization of the differential mode expresses $q = 5/9$. A complete TPK winding preserves phase and increments integer memory:

$$m(\Gamma_9^n) = n, \quad m(\gamma_2 \gamma_1) = m(\gamma_2) + m(\gamma_1).$$

On a reversible orbit of that memory, its lifted states are identified with e_n , $n \in \mathbb{Z}$, and the winding is the shift $Se_n = e_{n+1}$. Injectivity of this enumeration follows from the memory increment. If one starts only from the semigroup of positive windings, the same bilateral space is the shift extension adding its predecessors; reversibility is not attributed to a transition proved only isometric. In a complete tree, the remaining labels are preserved in a transverse factor: the shift acts on memory and does not identify routes.

This S is not the Weil Cayley operator inserted into the seam. It is the regular realization of the integer memory operation. Its Hilbert-space positivity is known before forming a GNS of the arithmetic form.

17.8.1 Loss vector, correlations, and terminal term

Telescoping over windings produces weights $(1 - q^2)q^{2j}$. Their amplitude in memory is the explicit vector

$$v_q = \sqrt{1 - q^2} \sum_{j=0}^{\infty} q^j e_j, \quad \|v_q\|^2 = 1.$$

With inner product linear in the first variable, for $n \geq 0$,

$$\langle S^n v_q, v_q \rangle = (1 - q^2) \sum_{j \geq 0} q^j q^{j+n} = q^n.$$

Conjugation and unitarity give, for every integer,

$$m_n^{\text{mem}} = q^{|n|}.$$

The truncation $v_{q,N} = \sqrt{1 - q^2} \sum_{j=0}^{N-1} q^j e_j$ preserves the terminal exactly:

$$\|v_{q,N}\|^2 = 1 - q^{2N}, \quad \langle S^n v_{q,N}, v_{q,N} \rangle = q^n (1 - q^{2(N-n)}), \quad 0 \leq n < N.$$

For $n \geq N$, that truncated correlation is zero. Thus, for each fixed n , the difference from the complete moment is q^{2N-n} when $N > n$, and tends to zero. This formula neither erases the last register nor equates a truncation with the whole history.

17.8.2 Triangular factorization and determinant of the memory matrices

For indices $0, \dots, N$, define

$$T_N = (q^{|i-j|})_{0 \leq i,j \leq N}.$$

It is a Gram matrix of $v_q, S v_q, \dots, S^N v_q$, so

$$\sum_{i,j=0}^N \bar{c}_i c_j (T_N)_{ij} = \left\| \sum_{j=0}^N c_j S^j v_q \right\|^2 \geq 0.$$

There is also an exact proof without infinite series. Let

$$L_{ij} = \begin{cases} q^{i-j}, & i \geq j, \\ 0, & i < j, \end{cases} \quad D = \text{diag}(1, 1 - q^2, \dots, 1 - q^2).$$

Then $T_N = LDL^*$. For $i \geq j$, the product entry is

$$q^{i+j} + (1 - q^2) \sum_{k=1}^j q^{i+j-2k} = q^{i-j}.$$

Symmetry gives the full matrix identity. A Cholesky factor is $LD^{1/2}$, and, since $\det L = 1$,

$$\det T_N = (1 - q^2)^N = (56/81)^N > 0.$$

This positive kernel arises from loss and preserved memory. It requires no zeros, positive zeta function, or Weil GNS.

17.9 Phase, nonadic refinement, and weak convergence of states

A fixed-phase representation is obtained from

$$v_{r,\theta} = \sqrt{1-r^2} \sum_{j \geq 0} r^j e^{-ij\theta} e_j, \quad 0 < r < 1,$$

and produces $m_n = r^{|n|} e^{in\theta}$. The Toeplitz matrices are diagonal conjugates of the preceding ones and have determinant $(1-r^2)^N$. Here the phase θ is a coordinate of the character collection; no arithmetic spectral phase has been selected.

On the circle, with normalized Haar measure $d\lambda$, these moments have the explicit measure

$$dv_{r,\theta}(z) = \frac{1-r^2}{|1-re^{-i\theta}z|^2} d\lambda(z).$$

Expanding both geometric series directly proves its Fourier coefficients without using a target measure.

17.9.1 Nonadic roots of the radius and phase compatibility

Taking $r_m = q^{1/9^m}$, one has $0 < r_m < 1$, $r_{m+1}^9 = r_m$, and $r_m \rightarrow 1$. For fixed phase,

$$r_m^{|n|} e^{in\theta} \longrightarrow e^{in\theta} \quad \text{for each } n \in \mathbb{Z}.$$

The measures are probabilities; uniform approximation by trigonometric polynomials extends moment convergence to every continuous function. Consequently,

$$v_{r_m,\theta} \rightharpoonup \delta_{e^{i\theta}}.$$

This shows how an atomic measure can arise as a weak limit of memory states. It is not a strong limit of the vectors in the same regular representation: each coefficient of $v_{r_m,\theta}$ tends to zero, but their norms remain one.

A naturality qualification must be preserved. Under $p_9(z) = z^9$,

$$(p_9)_* v_{r,\theta} = v_{r^9,9\theta}.$$

Thus the nonadic radius with fixed phase does **not** by itself give a projective system of measures under p_9 . This requires retaining lifted phases with $9\theta_{m+1} = \theta_m \pmod{2\pi}$, including their branch choices; alternatively, one must declare that the refinement used keeps the angular coordinate fixed and changes only the radius. These are different operations.

17.9.2 State selection and change of spectral type

Memory produces the group, positivity of its Grams, and these compatible correlation collections. A particular phase or a mixture of phases requires the readout selecting its amplitudes. For example, the vector e_0 produces $m_n = \mathbf{1}_{n=0}$; $(e_0 + e_1)/\sqrt{2}$ produces $m_0 = 1$, $m_{\pm 1} = 1/2$, with all other moments zero. The same memory operation admits both positive states.

The regular representation has vector measures absolutely continuous with respect to Haar measure. A fixed L^2 vector filter cannot produce a nonzero purely atomic measure. The preceding weak limit shows a precise way for spectral type to change under refinement: that limit of states must be preserved and controlled, rather than merely increasing the length of a previously chosen phase orbit. This does not deny the spectral representation of the corpus or joint HMT generation; it distinguishes the domains of two readouts.

17.10 Cayley operator on the exponential test-function domain

A second object is now constructed, without assuming Weil positivity. The native product and its irreducibles already provide $\ell_{p,k} = k \log p$, $w_{p,k} = (\log p)p^{-k/2}$. The gamma and polar readouts complete the form $\mathcal{W}$ written in Subsection 17.11. Memory supplies an integer iteration index; the following operator acts effectively on the functions of that representation.

The operator C_a is an auxiliary analytic readout of this formalization. The choice $a > 1/2$ does not silently replace the canonical seam $a = 1/2$, nor is it declared that this choice has already been selected by the TPK.

Fix $a > b > 1/2$. Let $\mathcal{E}_b$ be the space of smooth functions such that, for each derivative order j , $\sup_x e^{b|x|} |f^{(j)}(x)| < \infty$. It contains all compactly supported test functions. Define

$$(R_a^+ f)(x) = \int_0^\infty e^{-at} f(x+t) dt, \quad (R_a^- f)(x) = \int_0^\infty e^{-at} f(x-t) dt,$$

$$\boxed{C_a f = f - 2aR_a^+ f, \quad C_a^{-1} f = f - 2aR_a^- f.}$$

These integrals are translation readouts constructed on the test functions; they do not use the spectrum of zeros. The parameter $a = 1$, for example, comes from the unit and allows a choice $1/2 < b < 1$; it calibrates no zero.

Domain, inverse, and unitary realization. The inequality $e^{b|x|} |f(x \pm t)| \leq M_0 e^{bt}$ gives the bound $\|R_a^\pm f\|_b \leq \|f\|_b / (a - b)$, also for every derivative. Thus both operators preserve $\mathcal{E}_b$. With the convention $\hat{f}(z) = \int f(x) e^{-izx} dx$, their multipliers are

$$\widehat{C_a f}(z) = c_a(z) \hat{f}(z), \quad c_a(z) = \frac{z - ia}{z + ia},$$

and the inverse operator has multiplier $c_a(z)^{-1}$. The identity on the real Fourier axis proves both inverse compositions. In L^2 , $|c_a(t)| = 1$, so C_a is unitary for the ordinary norm, without using the Weil norm. On test functions it can be written $C_a = (H + ia)(H - ia)^{-1}$, with $H = i\partial_x$; here the integral formula, not a GNS generator, defines the operators.

Convergence condition at the polar boundary. The parameter $a = 1/2$ of the source owner's Cayley operator cannot simply be inserted into these pre-GNS integrals. For a compactly supported test function, $R_{1/2}^+ f$ generally has a tail proportional to $e^{x/2}$ towards $-\infty$; its negative polar evaluation does not converge. Divergence of the assembled form has not been proved; rather, this term-by-term decomposition requires the margin $a > 1/2$. The GNS Cayley operator may subsequently use $a = 1/2$ on its Hilbert-space domain.

17.11 Joint preservation of the gamma, prime, and polar channels

For $f, g \in \mathcal{E}_b$, let

$$C_{f,g}(u) = \int f(x) \overline{g(x-u)} dx, \quad P_f^\pm = \int f(x) e^{\pm x/2} dx.$$

The complete form is defined by

$$\begin{aligned} \mathcal{W}(f, g) &= c_\Gamma C_{f,g}(0) + \int_0^\infty \frac{e^{-u/2}}{1 - e^{-2u}} [2C_{f,g}(0) - C_{f,g}(u) - C_{f,g}(-u)] du \\ &\quad - \sum_{p,k} w_{p,k} [C_{f,g}(\ell_{p,k}) + C_{f,g}(-\ell_{p,k})] \\ &\quad + P_f^+ \overline{P_g^-} + P_f^- \overline{P_g^+}, \quad c_\Gamma = \psi(1/4) - \log \pi. \end{aligned}$$

Existence and continuity of the complete form. The correlation bound $|C_{f,g}(u)| \leq M_f M_g(|u| + 1/b)e^{-b|u|}$ proves absolute convergence of the prime sum, dominated by a series of the form $\sum_{n \geq 2} (\log n)(1 + \log n)n^{-b-1/2}$. The symmetric correlation difference is $O(u^2)$ at zero, whereas the gamma weight is $O(u^{-1})$; at infinity the weight is exponential. Polar evaluations converge because $b > 1/2$. These bounds also prove continuity in the indicated seminorms. Positivity is not used.

Simultaneous invariance of the three channels. Since C_a is unitary on L^2 and commutes with all translations,

$$C_{C_a f, C_a g}(u) = C_{f,g}(u) \quad (u \in \mathbb{R}).$$

By Fubini, the two polar readouts satisfy

$$P_{C_a f}^+ = -\frac{a-1/2}{a+1/2} P_f^+, \quad P_{C_a f}^- = -\frac{a+1/2}{a-1/2} P_f^-.$$

The factors are real and reciprocal, so each cross polar product is preserved exactly. Combining the three channels,

$$\boxed{\mathcal{W}(C_a f, C_a g) = \mathcal{W}(f, g).}$$

The identity holds for all integer powers and all cross terms. It proves invariance of a Hermitian form, not its sign.

17.12 Circular moments of the complete arithmetic form

Define

$$m_n^{\text{ar}}(f, g) = \mathcal{W}(C_a^n f, g), \quad n \in \mathbb{Z}.$$

The complete formula in Subsection 17.11, with $C_a^n f$ in its first argument, calculates these moments from primes, gamma, and polar terms. Each evaluation exists in $\mathcal{E}_b$; it requires neither a list of zeros nor a positive metric. Hermiticity and the preceding invariance give

$$m_{-n}^{\text{ar}}(g, f) = \overline{m_n^{\text{ar}}(f, g)}, \quad \mathcal{W}(C_a^j f, C_a^k g) = m_{j-k}^{\text{ar}}(f, g).$$

For a single test function, the matrix $T_N^{\text{ar}}(f)_{jk} = m_{k-j}^{\text{ar}}(f, f)$ is Hermitian Toeplitz and satisfies the exact identity

$$c^* T_N^{\text{ar}}(f) c = \mathcal{W}\left(\sum_{k=0}^N c_k C_a^k f, \sum_{k=0}^N c_k C_a^k f\right).$$

The matrix has been constructed without assuming positivity. The equality allows its positivity to be compared with a native memory Gram; it does not presuppose it.

17.12.1 Polar growth and a necessary cancellation condition

For $a = 1$, the exact polar contribution to $m_n^{\text{ar}}(f, g)$ is

$$(-1/3)^n P_f^+ \overline{P_g^-} + (-3)^n P_f^- \overline{P_g^+}.$$

For a nonzero real, even, positive test function, $P_f^+ = P_f^- = p > 0$, so $p^2[(-1/3)^n + (-3)^n]$ appears. If the total moment sequence is positive definite, its order-two minors force

$$|m_n^{\text{ar}}(f, f)| \leq m_0^{\text{ar}}(f, f).$$

Thus circular identification must account for the concrete cancellation of this growth with the gamma and prime channels. Formal symmetry of the two sheets does not replace that calculation. This observation does not declare that cancellation fails; it provides an exact falsifier for a candidate composition.

17.12.2 Mixed moment matrices and domain coverage

For a collection of test functions $f_1, \dots, f_d$, the appropriate object is the block Toeplitz matrix

$$\mathsf{T}_{(j,r),(k,s)} = m_{k-j}^{\text{ar}}(f_s, f_r).$$

Its quadratic form is $\mathcal{W}(F, F)$, where $F = \sum_{k,s} c_{k,s} C_a^k f_s$. Controlling only the scalar matrices of some test functions does not control cross terms between different test functions. For a dense collection in each $C_c^\infty([-R, R])$, the continuity proved above permits passage to the complete domain, provided the mixed Grams and compatibilities as R increases have been established. Increasing N along a single orbit does not prove that coverage.

17.12.3 Transport of local coercivity to subspaces with tails

The preceding identity has a positive application that does not yet require identifying all moments with memory. Suppose the Schur complement has proved, on the interval $I = (-1/18, 1/18)$, the bound

$$\mathcal{W}(f, f) \geq \frac{1}{2} \|f\|_{L^2}^2, \quad f \in C_c^\infty(I).$$

This subsection receives that estimate as a result of the corresponding local calculation; it does not deduce it from the existence of C_a . Let $(\tau_t f)(x) = f(x - t)$. Translation preserves correlations and satisfies

$$P_{\tau_t f}^\pm = e^{\pm t/2} P_f^\pm.$$

Its cross polar products are also preserved. Therefore $\mathcal{W}(\tau_t f, \tau_t g) = \mathcal{W}(f, g)$, and the ordinary L^2 norm is invariant. Since C_a and τ_t commute, for every integer n , every $t \in \mathbb{R}$, and every local test function,

$$\boxed{\mathcal{W}(C_a^n \tau_t f, C_a^n \tau_t f) = \mathcal{W}(f, f) \geq \frac{1}{2} \|C_a^n \tau_t f\|_{L^2}^2.}$$

For fixed (n, t) , this is a coercive bound on the entire subspace $C_a^n \tau_t C_c^\infty(I)$, not merely on one example. When $n \neq 0$, transformed test functions may have noncompact exponential tails. Infinitely many prime clocks may enter their arithmetic form; exact equality of the three channels controls their composition, and the bounds in Subsection 17.11 guarantee convergence. That sum has not been replaced by a finite truncation.

Cross terms between subspaces with different exponents or translations, by contrast, retain their complete moments. For example,

$$\mathcal{W}(C_a^n \tau_t f, C_a^m \tau_s g) = \mathcal{W}(C_a^{n-m} \tau_{t-s} f, g).$$

The bound on each individual subspace does not imply positivity of their algebraic sum. This is decided through the mixed Grams in Subsection 17.12.2. An effective positive scope—entire collections of test functions with tails—is thus isolated without promoting it to the whole domain solely through transport invariance.

17.13 Prospective identification of moments and energy

Memory already produces the positive regular representation of Subsection 17.8; the arithmetic channels produce C_a and m_n^{ar} without GNS. Formulating the global step also requires specifying the representation that receives the state limit. Denote that space by $\mathcal{K}_{\text{lim}}$ and its unitary winding operator by U_{mem} , both to be constructed from histories and limit compatibilities. They are not identified in advance with the regular space and its shift S . A prospective identification has the form

$$m_n^{\text{ar}}(f, g) = \langle U_{\text{mem}}^n \mathcal{E} f, \mathcal{E} g \rangle_{\mathcal{K}_{\text{lim}}}$$

for a readout $\mathcal{E}$ constructed from operations on the same histories and not defined as $\mathcal{W}^{1/2}$. In that case, all mixed Toeplitz matrices are Grams; arithmetic GNS follows as a realization of a product already obtained.

It suffices to prove the zeroth-moment identity and the intertwining $\mathcal{E} C_a = U_{\text{mem}} \mathcal{E}$, with the stated domains and compatibilities. The zeroth moment is identification of the complete energy, not a lesser condition: it cannot be regarded as proved merely because an $\mathcal{E}$ with some positive Gram exists.

Why distinguishing the representations matters. The regular shift on $\ell^2(\mathbb{Z}; \mathcal{K})$ has no nonzero eigenvectors. Since S is unitary, any eigenvalue would have modulus 1. Indeed, if $Sx = \lambda x$ with $|\lambda| = 1$, the coordinate relation forces $\|x_n\|$ to be constant; square summability forces $x = 0$. Thus a representation with a nonzero eigenvector admits no isometric intertwiner into that shift: the image would be a nonzero eigenvector of S . This observation applies to atomic arithmetic GNS if its global hypotheses hold. It does not contradict weak convergence of states in Subsection 17.9.2; it shows that the limit requires its own representation and not a limit vector in the original regular space. Construction of $(\mathcal{K}_{\text{lim}}, U_{\text{mem}})$ and the readout identifying the moments is part of the prospective condition, not a consequence already proved from the regular Grams.

In the manuscript's two-column form, the calculation remains

$$\Xi^* \Xi = J_+^* J_+, \quad \Xi^* P \Xi = J_-^* J_-, \quad \mathcal{E} = (I - P) \Xi.$$

If P is an orthogonal projection, these two equalities produce $\mathcal{E}^* \mathcal{E} = J_+^* J_+ - J_-^* J_-$. Verifying only one of them does not suffice, nor does replacing the complete columns with those of an isolated gamma block. The phases and moments in this development make explicit what that composition must transmit to all windings.

The kernel $q^{|n|}$ is not identified here with $m_n^{\text{ar}}(f, g)$. It is an effective positive antecedent and offers a local factorization; the readout $\mathcal{E}$ must also preserve prime labels, weights, polar contributions, and the state limit if spectral type changes. Declaring that equality by definition would introduce the result one seeks to construct.

17.14 Exact checks and scope of the composition

The executable verification preserves 1,046 exact checks with rational fractions: LDL^* factorization and determinants in dimensions 1–13 for $q = 5/9$, truncated correlations through 20 positions, and reciprocal polar factors for four rational values of $a > 1/2$. Ordinary and optimized executions use explicit conditions independent of assertion instructions. Their sources, commands, and receipts are preserved in the reproduction dossier. No interval campaign of Weil moments was run here, nor are these checks converted into a global proof.

The usable advance is twofold: loss memory produces a positive Toeplitz kernel with explicit Cholesky factor and terminal, and the complete form admits an integral Cayley operator preceding GNS that constructs its circular moments and preserves each channel. Nonadic regularization also shows how to pass from regular states to atoms by a weak limit, with its phase condition correctly separated. The union of both constructions retains identification of the two complete moments as a precise equation.

This development neither refutes that union, proves its absence throughout the corpus, nor declares it established by the local identities. It leaves a concrete composition that can be compared and developed in the continuation of this manuscript.

18 Boundary expectations and reconstruction of the analytic channels

18.1 Action of the readouts and content of the moments

The explicit actions of the odometer, the difference readout, the uniform inclusion, the Helmert refinement, the bidirectional boundary, and the polar chart have been recovered and composed. The calculation yields:

1. The moments of the positive and negative columns, considered separately, are determined by explicit operators. They include all cross products between test functions.
2. The uniform expectation does not coincide with the Helmert refinement. On the carry mode it yields an exact factor, calculated below; it does not by itself convert a translation difference into the uniform mode.
3. The gamma-tail resolvent does perform the analytic conversion and exactly reconstructs all negative rows, on every window and before any sign is used.
4. The simultaneous identification of the first moment of the encoder and the moment of its projection does not follow from the three preceding chart identities by a simple direct sum. The specific additional composition that must be verified is identified below. The structural readout is not declared absent, nor is this focal check identified with the full TPK.

The advance consists in calculating the action and cross products, not in reiterating that the projection is positive.

Provenance of the charts. The formulas are assembled from the developments *Polar spectral proof*, *Form and Gram*, *Gamma-tail connector*, *Odometer and carry*, *Pulsation and expectation*, and *Modular polar chart*. The locators and the record of the reading performed are retained in the source dossier accompanying the delivery. Each chart is used with the spaces, normalizations, and identities made explicit below; a documentary reference does not replace these conditions.

18.2 Conventions and genealogical cutoff

The relevant starting point is the APP-TRIT-TPK state, its refinement with memory, and its native readouts. In this formalization:

- APP provides the support and the irreducible selector. The native product, coproductivity, and selector precede the representation of prime-power labels.
- TRIT and sheet transport preserve orientation and the seam.
- The TPK preserves the phase step and the carry register; its odometer induces the translation difference. The logarithmic representation of the clocks produces $\ell_{p,k} = k \log p$ and $w_{p,k} = (\log p)p^{-k/2}$.
- The boundary and polar charts are subsequent realizations of the same readouts. No state is selected by means of zeta zeros or a target value.

This section calculates the arrow between these readouts and the expectation, after their common generation; it does not seek to reconstruct or replace the joint continuum.

We work with orthonormal history coordinates. In these coordinates, $u_N = N^{-1/2}(1, \dots, 1)$. If functions with the uniform probability measure are used, the constant coordinates and averages are conjugated by the corresponding normalization factor; the projections and all the norm ratios below remain the same.

Let $H = L^2(\mathbb{R})$, $T_a f(x) = f(x - a)$, $\mathcal{D}_R = C_c^\infty((-R/2, R/2))$. We write $d_a = T_a - I$; changing simultaneously to $I - T_a$ preserves their Grams but changes the sign of the inverting cotrace.

The inner product is linear in the first variable. We retain the polar evaluations and the scalar of the full form,

$$\begin{aligned} A_f &= \int_{\mathbb{R}} f(x) e^{x/2} dx, & B_f &= \int_{\mathbb{R}} f(x) e^{-x/2} dx, \\ c_{\Gamma} &= -\gamma - \frac{\pi}{2} - 3 \log 2 - \log \pi < 0, & \mu(a) &= \frac{e^{-a/2}}{1 - e^{-2a}}. \end{aligned} \quad (18.1)$$

Here π , γ , and the logarithms retain the internal representation and subsequent realization declared in the preceding chapters. The formula introduces no target values into the generator.

18.3 Action of the odometer on generators

For $N = 9^m$, $m \geq 1$, with the seam located at the first vector,

$$U_{a,N}(e_r \otimes f) = \begin{cases} e_{r+1} \otimes f, & r < N-1, \\ e_0 \otimes T_a f, & r = N-1. \end{cases} \quad (18.2)$$

The uniformization $i_N f = u_N \otimes f$ gives

$$U_{a,N} i_N f = u_N \otimes f + N^{-1/2} e_0 \otimes d_a f. \quad (18.3)$$

Therefore, for $R_N = |u_N\rangle\langle u_N| \otimes I$,

$$\begin{aligned} i_N^* U_{a,N} i_N &= I + \frac{d_a}{N}, \\ (I - R_N) U_{a,N} i_N &= \frac{\sqrt{N-1}}{N} \eta_N \otimes d_a. \end{aligned} \quad (18.4)$$

where

$$\eta_N = \frac{N e_0 - \mathbf{1}_N}{\sqrt{N(N-1)}}. \quad (18.5)$$

This recovers the normalized difference readout:

$$\hat{\delta}_{a,m} f = \eta_N \otimes u_{81} \otimes d_a f, \quad j_m f = u_N \otimes u_{81} \otimes f. \quad (18.6)$$

The action on any combination of generators, without removing their cross terms, is

$$\left\langle \sum_i \hat{\delta}_{a_i, m} f_i, \sum_j \hat{\delta}_{b_j, m} g_j \right\rangle = \sum_{i,j} \langle d_{a_i} f_i, d_{b_j} g_j \rangle. \quad (18.7)$$

In particular,

$$\hat{\delta}_{a,m}^* \hat{\delta}_{b,m} = d_a^* d_b, \quad j_m^* j_m = I, \quad j_m^* \hat{\delta}_{a,m} = 0. \quad (18.8)$$

The last cross product follows from $\langle u_N, \eta_N \rangle = 0$. It is concrete information about the orientation of the modes, in addition to the two isolated norms.

18.4 Two distinct refinements and their expectations

18.4.1 Uniform inclusion and projection of the last digit

Write $M = 9N$ and $\mathbb{C}^M = \mathbb{C}^N \otimes \mathbb{C}^9$. The inclusion and its adjoint are

$$\iota_N x = x \otimes u_9, \quad E_N = I_N \otimes \langle u_9 |, \quad P_N^{\text{unif}} = \iota_N E_N. \quad (18.9)$$

Acting in coordinates,

$$E_N u_M = u_N, \quad E_N \eta_M = a_N \eta_N, \quad a_N = \sqrt{\frac{N-1}{9N-1}}. \quad (18.10)$$

Proof. We have $E_N e_0^{(M)} = e_0^{(N)}/3$ and $E_N \mathbf{1}_M = 3\mathbf{1}_N$. Substitution into $\eta_M = (Me_0 - \mathbf{1}_M)/\sqrt{M(M-1)}$ yields (18.10). $\square$

Consequently,

$$(E_N \otimes I) \hat{\delta}_{a,m+1} = a_N \hat{\delta}_{a,m}, \quad (E_N \otimes I) j_{m+1} = j_m. \quad (18.11)$$

All projected moments are determined:

$$\begin{aligned} \hat{\delta}_{a,m+1}^* (P_N^{\text{unif}} \otimes I) \hat{\delta}_{b,m+1} &= a_N^2 d_a^* d_b, \\ j_{m+1}^* (P_N^{\text{unif}} \otimes I) j_{m+1} &= I, \\ j_{m+1}^* (P_N^{\text{unif}} \otimes I) \hat{\delta}_{b,m+1} &= 0. \end{aligned} \quad (18.12)$$

The complementary fraction of the carry is $1 - a_N^2 = 8N/(9N-1)$. For $N = 9$, the expressed fraction is $1/10$, not 1 , and its output remains a difference.

We do not conflate P_N^{unif} , which averages the last digit, with the projection onto the single global line u_M : the latter annihilates η_M entirely.

18.4.2 Helmert refinement

The operator W_N is fixed by the orthonormal Helmert bases of the two levels, ordered to preserve the uniform and carry modes. It is the isometry that sends the first N vectors of the basis at level N to the corresponding vectors of the basis at level M . In particular, it satisfies

$$W_N u_N = u_M, \quad W_N \eta_N = \eta_M, \quad W_N^* W_N = I. \quad (18.13)$$

Thus, $E_N^H = W_N^*$, $P_N^H = W_N W_N^*$ give

$$E_N^H j_{m+1} = j_m, \quad E_N^H \hat{\delta}_{a,m+1} = \hat{\delta}_{a,m}. \quad (18.14)$$

The uniform–carry cross product remains zero, and that of two differences remains $d_a^* d_b$. The refinement is therefore natural for both modes, but does not transform one into the other. Specifically, $W_N \neq \iota_N$.

18.4.3 Conjugation of expectations and preservation of moments

There is a unitary S_M on $\mathbb{C}^M$ that maps $\iota_N \mathbb{C}^N$ to $W_N \mathbb{C}^N$, with $W_N = S_M \iota_N$. It suffices to map the two orthonormal bases to each other and complete the complements. Then

$$P_N^H = S_M P_N^{\text{unif}} S_M^*. \quad (18.15)$$

Conjugating the state C and its projection simultaneously,

$$(S_M C)^*(S_M P S_M^*)(S_M C) = C^* P C, \quad (S_M C)^*(S_M C) = C^* C. \quad (18.16)$$

Neither moment changes. If only the projection is changed, the fraction in this case goes from a_N^2 to 1, but the analytic operator remains d_a , not I .

More generally, for any projection acting only on the finite history fiber, the moment of a single channel is $\langle \eta, P\eta \rangle d_a^* d_b$. This change of basis does not turn it into $2I$. Any effective coupling between analytic channels is a different operation and must be written down, not inferred from conjugation.

18.5 Metric identities of the boundary and polar charts

18.5.1 Bidirectional boundary

The bidirectional chart preserves the composition

$$\begin{aligned} K_2 &= \mathbb{C}^{81} \otimes u_9^\perp, \\ \Delta_a &= j_{\text{APP}}(\eta_9 \otimes d_a), \\ \beta_0 &= R_{\text{tw}}(L^3 + \varepsilon \nabla_x) F I_{Q_9}|_{K_2}. \end{aligned} \quad (18.17)$$

In this realization, j_{APP} is the isometric inclusion of the indicated fiber; β_0 is injective, and $\Lambda_{\text{tw}}^\#$ is positive definite on the boundary space. These are the conditions of the chart: the factors of the composition are supplied with compatible domains, and that injectivity is not deduced here from an equality of dimensions. With $V_\beta = \beta_0(\beta_0^* \beta_0)^{-1/2}$ and $U_{\text{tw}} = (\Lambda_{\text{tw}}^\#)^{-1/2} V_\beta$,

$$G_a = (U_{\text{tw}} \otimes I) \Delta_a, \quad G_a^*(\Lambda_{\text{tw}}^\# \otimes I) G_b = d_a^* d_b. \quad (18.18)$$

Proof. The polar normalization gives $V_\beta^* V_\beta = I$ and therefore $U_{\text{tw}}^* \Lambda_{\text{tw}}^\# U_{\text{tw}} = I$. Since j_{APP} and η_9 are normalized, $\Delta_a^* \Delta_b = d_a^* d_b$. Composition proves (18.18), including all cross products. $\square$

U_{tw} is isometric for the declared metric $\Lambda_{\text{tw}}^\#$. In ordinary Hilbert-space coordinates, the isometry between the effective spaces is $(\Lambda_{\text{tw}}^\#)^{1/2} U_{\text{tw}} = V_\beta$.

Identity (18.18) contains neither an expectation nor its value on G_a . To transport a projection, one must conjugate it with that same isometry and its adjoint in the correct metric. When this is done, (18.16) preserves the preceding compressed moment: passing to the boundary does not create the second moment.

18.5.2 Effective polar chart

The polar realization by five octads supplies the five-column matrix T , whose Gram and normalization are

$$T^* T = 4I_5 + \frac{4}{3}J_5, \quad \hat{T} = T(T^* T)^{-1/2}, \quad F_{34} = (f_3, f_6, f_9, f_4, f_8). \quad (18.19)$$

The chart, with its spaces made explicit, is

$$U = F_{34} \hat{T}^* : \text{Ran } \hat{T} \subset \mathbb{C}^{24} \longrightarrow \text{span}(f_3, f_6, f_9, f_4, f_8) \subset \mathbb{C}^{12}. \quad (18.20)$$

For $\eta = \text{diag}(1, 1, 1, -1, -1)$,

$$U^* J_{34} U = \hat{T} \eta \hat{T}^*, \quad \hat{T}^* U^* J_{34} U \hat{T} = \eta. \quad (18.21)$$

Here the five vectors of F_{34} are orthonormal, and $F_{34}^* J_{34} F_{34} = \eta$. Substituting $U = F_{34} \hat{T}^*$ and $\hat{T}^* \hat{T} = I_5$ proves both equalities in (18.21). This is the typed formulation of the equality with the diagonal matrix. The five roles are effective; the identification of coordinates must not be erased.

For the two polar evaluations,

$$b_{\pm}(f) = \frac{A_f \pm B_f}{\sqrt{2}}, \quad b_+(f) \overline{b_+(g)} - b_-(f) \overline{b_-(g)} = A_f \overline{B_g} + B_f \overline{A_g}. \quad (18.22)$$

(18.21) transports the signature, and (18.22) transports its cross terms. Neither identifies the isolated positive line as a sufficient source of the negative one: for f real, odd, and nonzero, chosen with $f(x) > 0$ on the positive half-axis of its support, we have $b_+(f) = 0$ and $b_-(f) > 0$. Recovery must involve the common analytic state, as does the gamma-tail connector developed below.

18.6 Difference between channelwise readout and common transfer

To fix all the summands used, let $\mathcal{K}_m = \mathbb{C}^{9^m} \otimes \mathbb{C}^{81} \otimes H$, and let ζ_+, ζ_- be unit vectors of the two polar lines. The indices $\ell \leq R$ range over the prime-power labels (p, k) with $k \log p \leq R$, preserving their weight $w_\ell = (\log p) p^{-k/2}$. The columns are

$$\begin{aligned} J_{+,R} f &= \left((\sqrt{w_\ell} \hat{\delta}_{\ell,m} f)_{\ell \leq R}, (\sqrt{\mu(a)} \hat{\delta}_{a,m} f)_{a>0}, \zeta_+ b_+(f) \right), \\ J_{-,R} f &= \left((\sqrt{-c_\Gamma} j_m f, (\sqrt{2w_\ell} j_m f)_{\ell \leq R}, \zeta_- b_-(f) \right). \end{aligned} \quad (18.23)$$

The continuous component of the first column belongs to $L^2((0, \infty), da; \mathcal{K}_m)$; the remaining components are assembled in the indicated orthogonal sums. The evaluations $b_{\pm}$ are those of (18.22). Their common domain consists of test functions with finite gamma energy. The identity

$$\mathcal{W}_R(f, g) = \langle J_{+,R} f, J_{+,R} g \rangle - \langle J_{-,R} f, J_{-,R} g \rangle \quad (18.24)$$

is verified by expanding $\langle d_\ell f, d_\ell g \rangle = 2\langle f, g \rangle - \langle T_\ell f, g \rangle - \langle f, T_\ell g \rangle$, using the gamma integral and the polar identity (18.22). This preserves the three channels and all cross terms.

Taking the explicit positive column Ξ_+ as the encoder satisfies its first moment by direct calculation. Subsequently applying the uniform expectation produces, in each prime channel,

$$w_\ell a_N^2 \langle d_\ell f, d_\ell g \rangle, \quad (18.25)$$

whereas the required negative row is

$$2w_\ell \langle f, g \rangle. \quad (18.26)$$

These are not the same identity. For example, choose $R > \ell > 0$ and $f \in \mathcal{D}_R$ real and nonnegative, with support overlapping its translation by ℓ . Then

$$\|d_\ell f\|^2 = 2\|f\|^2 - 2\langle f, T_\ell f \rangle < 2\|f\|^2. \quad (18.27)$$

No contractive projection applied solely to that channel can produce $\sqrt{2}f$. The Helmert refinement removes the factor a_N^2 , but (18.27) persists. This is a falsifier of the **channel-by-channel** identification, not a refutation of a common transfer that redistributes energy among the channels.

On the other hand, directly introducing $j_m f$, its prime–scalar copies, and $b_-(f)$ as orthogonal summands of the encoder ensures their outputs but adds their norms to the first moment. Preserving the original first moment requires an explicit isometric redistribution. That redistribution is precisely the operation that the simultaneous identification of the two moments must exhibit.

Invariance of the cyclic subspace under P does not by itself determine the value of PCf either; it only guarantees that this value remains in the subspace.

18.7 Gamma-tail inverse and reconstruction of all rows

Here there is indeed an explicit action on every generator and every linear combination. Let

$$\mu(a) = \frac{e^{-a/2}}{1 - e^{-2a}}, \quad A_R = [R+1, R+2], \quad \nu_R = \int_{A_R} \mu(a) da > 0, \quad M_R = \mathbf{1}_{(-R/2, R/2)}. \quad (18.28)$$

For the sign $d_a = T_a - I$ already fixed, define the cotrace

$$\epsilon_m = \langle \eta_{9m} | \otimes \langle u_{81} | \otimes M_R, \quad \mathcal{R}_{m,R}^- F = -\nu_R^{-1} \int_{A_R} \sqrt{\mu(a)} \epsilon_m F(a) da. \quad (18.29)$$

The minus sign disappears if $I - T_a$ is used, which is the alternative convention for the tail readout. For $f \in \mathcal{D}_R$ and $a > R$, $M_R T_a f = 0$. Consequently,

$$\mathcal{R}_{m,R}^-(a \mapsto \sqrt{\mu(a)} \delta_{a,m} f) = f. \quad (18.30)$$

Moreover, $\|\mathcal{R}_{m,R}^-\| \leq \nu_R^{-1/2}$ by Cauchy–Schwarz. No individual translation is inverted, and Weil positivity is not used.

Write

$$S_R f = (\sqrt{-c_\Gamma} j_m f, (\sqrt{2w_\ell} j_m f)_{\ell \leq R}), \quad B_- f = \zeta_- b_-(f). \quad (18.31)$$

The concrete common row, defined on the full positive ambient space, is

$$\mathcal{C}_R^{\text{tail}} = \begin{pmatrix} 0 & S_R \mathcal{R}_{m,R}^- & 0 \\ 0 & B_- \mathcal{R}_{m,R}^- & 0 \end{pmatrix}. \quad (18.32)$$

From (18.30),

$$\mathcal{C}_R^{\text{tail}} J_{+,R} f = J_{-,R} f \quad \text{for every } f \in \mathcal{D}_R. \quad (18.33)$$

This action contains the scalar, prime–power, and polar rows, with their constants and orientations intact. For any pair f, g ,

$$\begin{aligned} \langle \mathcal{C}_R^{\text{tail}} J_{+} f, \mathcal{C}_R^{\text{tail}} J_{+} g \rangle &= \left(-c_\Gamma + 2 \sum_{\ell \leq R} w_\ell \right) \langle f, g \rangle \\ &\quad + b_-(f) \overline{b_-(g)}. \end{aligned} \quad (18.34)$$

For $f = \sum_i f_i$, $g = \sum_j g_j$, the right-hand side contains the full sum over i, j ; the test functions have not been replaced by orthogonal cells. The prime–gamma cross terms of (18.18) are preserved when the input representations are changed. The final form is

$$J_+^* (I - (\mathcal{C}_R^{\text{tail}})^* \mathcal{C}_R^{\text{tail}}) J_+ = J_+^* J_+ - J_-^* J_- = \mathcal{W}_R. \quad (18.35)$$

This is an effective equality of forms, without yet assuming that the operator in parentheses is positive. The operator is bounded; an explicit bound is

$$\|\mathcal{C}_R^{\text{tail}}\| \leq \nu_R^{-1/2} \sqrt{-c_\Gamma + 2 \sum_{\ell \leq R} w_\ell + 2 \sinh(R/2) - R}. \quad (18.36)$$

The bound fixes the type; it does not prove that the norm is at most one.

18.8 Exact condition for combining both moments in one expectation

Let $\mathcal{M}_+ = \overline{J_+ \mathcal{D}_R}$. The reachable transfer

$$C_{\text{rea}}(J_+ f) = J_- f \quad (18.37)$$

is well defined because the gamma tail proves that J_+ is injective. It is the restriction of (18.32) and is therefore bounded. Any other bounded transfer giving the same outputs coincides with it on $\mathcal{M}_+$.

A change of coordinates does not provide a different reachable transfer. The condition to be established for (18.32) to have an isometric dilation with the native expectation is

$$\|C_{\text{rea}} x\| \leq \|x\|, \quad x \in \mathcal{M}_+. \quad (18.38)$$

Its relation to the two moments can be stated unambiguously. **If** (18.38) is proved from the native operators, $D = (I - C_{\text{rea}}^* C_{\text{rea}})^{1/2}$ exists, and

$$\mathfrak{G}_{\text{dil}} f = u_9 \otimes (J_- f, 0) + \eta_9 \otimes (0, D J_+ f) \quad (18.39)$$

satisfies, for $P = |u_9\rangle\langle u_9| \otimes I$,

$$\mathfrak{G}_{\text{dil}}^* \mathfrak{G}_{\text{dil}} = J_+^* J_+, \quad \mathfrak{G}_{\text{dil}}^* P \mathfrak{G}_{\text{dil}} = J_-^* J_-. \quad (18.40)$$

(18.39) is a normal form of the dilation once (18.38) has been proved, **not** a new prospective proof of (18.38). It identifies exactly what the native colligation must produce: the defect of that common transfer, with all its cross terms, and not merely the defect of each clock separately.

The specific next composition combines the energy budget of the full connector with the recovered local and refinement identities. If an alternative native colligation realizes (18.33) and is contractive on all of $\mathcal{M}_+$, reachable uniqueness proves (18.38). There is no need to invent another readout; its action and common energy do need to be calculated.

18.9 Checks and provenance

The actions (18.8), (18.18), (18.21), and (18.30)–(18.35) assemble pre-existing operators and results. The compositions (18.10)–(18.16) and the falsifier (18.27) are explicit calculations on these formulas; no mathematical priority is claimed for them, nor is any global absence of other operators inferred from them.

We retained 25 rational checks on $v_{9N} = 9Ne_0 - \mathbf{1}$, for $N = 2, 3, 9, 81, 729$. They check the action of the average, the fraction $(N - 1)/(9N - 1)$, the zero mean, the orthogonality of the detail, and the complement $8N/(9N - 1)$. The 25 checks gave the same result in normal and optimized modes, without assertions that can be disabled. They verify the identities (18.10)–(18.12), not global positivity. The programs, commands, receipts, and locators are retained in Appendix 21.

Operational conclusion: the state and the two readouts have not disappeared. The calculation recovers their action and demonstrates precisely the difference between isometric transport of modes, uniform expectation, and analytic recovery of channels. The second moment identity has a concrete verification route through the common connector; conjugation of charts alone does not carry out that verification.

19 Schur reduction and coercivity of the Weil form

19.1 Basis, domain, and result obtained

We use the full columns already constructed from the native arithmetic of APP, the oriented TRIT state, the TPK operations and memory, and the joint structure of the continuum. The native product expresses irreducibles and powers; their lengths and weights subsequently feed the prime, gamma, and polar readouts. The internal coordinate of π and the rational readout of the Euler–Mascheroni finite part enter after that representation. No zero, desired value of the functional, or metrological constant selects these operations.

We retain the full columns and the domains constructed in the chapter on the double circle and spectral transport, together with the support and detail estimates. The columns and the connector are also made explicit in Section 18. The joint construction of the continuum remains the common antecedent: it is not replaced by this functional decomposition, nor are its five constructions replaced by five modules.

The new operation assembled here is the exact elimination of the infinite detail space by means of its coercive form. It produces a finite matrix with all coupling corrections, not merely a matrix of means. A specific case is also settled:

For every function in the closure of $C_c^\infty((-1/18, 1/18))$ in the joint norm of the readouts,

$$\mathcal{W}(f, f) \geq \frac{1}{2} \|f\|_2^2. \quad (19.1)$$

The same assertion holds on any translated interval of length $1/9$, and on all its interior nonadic refinements.

The full window has length $1/9 > 0.085626 \dots$, the endpoint of the preceding baseline criterion. It contains no active prime translation, since $1/9 < \log 2$. This limitation of the example remains explicit.

Provenance. Authorial architecture and readouts: pre-existing. Detail coercivity: a result recovered from the zero-mean threshold theorem, retaining an intermediate estimate stronger than its abbreviated logarithmic bound. Schur assembly, residual certification, and proof for the interval $1/9$: formalization and certificate added in this section; no global historical novelty is claimed.

19.2 Partition, cell means, and details

Fix $R > 0$, $I_R = (-R/2, R/2)$, and $H_R = L^2(I_R)$, with extension by zero. Let V_R be the closure of $\mathcal{D}_R = C_c^\infty(I_R)$ in the joint norm of the full readouts. Equivalently, its norm may be taken to be

$$\begin{aligned} \|f\|_{V_R}^2 &= \|f\|_2^2 + Q_\Gamma(f), \\ Q_\Gamma(f) &= \int_0^\infty \mu(a) \|(I - T_a)f\|_2^2 da, \\ \mu(a) &= \frac{e^{-a/2}}{1 - e^{-2a}}. \end{aligned} \quad (19.2)$$

The prime and polar channels are bounded on each fixed window. The gamma form is closed: it is the form of a positive Fourier multiplier, restricted to the subspace of functions supported in the window; equivalently, this is verified by convergence in the direct integral of differences. The full form is a bounded perturbation of it and defines a semibounded self-adjoint operator L_R .

Partition I_R into contiguous intervals $I_j = (\ell_j, r_j)$, with lengths $h_j \leq h$, and set

$$u_j = h_j^{-1/2} \mathbf{1}_{I_j}, \quad Ua = \sum_{j=1}^M a_j u_j, \quad P = UU^*, \quad Q = I - P. \quad (19.3)$$

In the last sum, $a = (a_1, \dots, a_M)$ is the coefficient vector; the endpoints ℓ_j, r_j are used only in defining I_j .

The indicator functions belong to V_R : their gamma energy is controlled by $\|(I - T_a)u_j\|_2^2 = 2 \min(|a|/h_j, 1)$. Thus P acts continuously on V_R , and every $f \in V_R$ has the unique decomposition

$$f = Ua + d, \quad a_j = \langle f, u_j \rangle, \quad d \in V_D := V_R \cap QH_R. \quad (19.4)$$

Here d has zero integral on each cell. The space V_D is closed in the form norm and dense in QH_R .

The estimate for zero-mean details gives a bound $\mathcal{W}_R(d, d) \geq \eta \|d\|_2^2$ if a sufficiently fine partition is chosen. One may use its expressed logarithmic threshold or retain the exact intermediate estimate

$$\begin{aligned} \eta_N(h, R) &= 4 \sum_{n=0}^N \frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2} \\ &\quad + c_\Gamma - 2 \sum_{k \log p \leq R} (\log p) p^{-k/2} - \beta_R, \\ \beta_R &= 2 \sinh(R/2) - R. \end{aligned} \quad (19.5)$$

Lemma 19.1 (Retained bound for cell details). *For $d \in V_D$ and every integer $N \geq 0$,*

$$\mathcal{W}_R(d, d) \geq \eta_N(h, R) \|d\|_2^2.$$

Proof. On each cell $[\ell_j, r_j]$, define $F(x) = \int_{\ell_j}^x d(t) dt$. The zero mean gives $F(\ell_j) = F(r_j) = 0$. The assembly of these primitives, extended by zero, belongs to $H_0^1(I_R)$, satisfies $F' = d$, and, by the Dirichlet inequality on each cell,

$$\|F\|_2^2 \leq \frac{h^2}{\pi^2} \|d\|_2^2. \quad (19.6)$$

The positive expansion $\mu(a) = \sum_{n \geq 0} e^{-\lambda_n a}$, $\lambda_n = 2n + 1/2$, allows us to write $Q_\Gamma = \sum E_{\lambda_n}$, where

$$E_\lambda(d) = \int_0^\infty e^{-\lambda a} \|(I - T_a)d\|_2^2 da = \frac{2}{\lambda} \|d\|_2^2 - \langle d, C_\lambda d \rangle \geq 0, \quad (19.7)$$

and C_λ is convolution with $e^{-\lambda|x|}$. For the unitary transform $\mathcal{F}_u = (2\pi)^{-1/2} \hat{}$, the multiplier of C_λ is $2\lambda/(\lambda^2 + \xi^2)$. By Plancherel,

$$\begin{aligned} \langle d, C_\lambda d \rangle &= \int_{\mathbb{R}} \frac{2\lambda \xi^2}{\lambda^2 + \xi^2} |\mathcal{F}_u F(\xi)|^2 d\xi \\ &\leq 2\lambda \|F\|_2^2 \leq \frac{2\lambda h^2}{\pi^2} \|d\|_2^2. \end{aligned} \quad (19.8)$$

Retaining $N + 1$ summands and discarding only the nonnegative tail gives

$$Q_\Gamma(d) \geq \left[4 \sum_{n=0}^N \frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2} \right] \|d\|_2^2.$$

The scalar term contributes $c_\Gamma \|d\|_2^2$. Each prime translation contributes at least $-2w_{p,k} \|d\|_2^2$, by Cauchy-Schwarz and the unitarity of $T_{k \log p}$. Only lengths less than or equal to R enter. Finally, the polar contribution is

$$2 \left| \int_{I_R} d(x) \cosh(x/2) dx \right|^2 - 2 \left| \int_{I_R} d(x) \sinh(x/2) dx \right|^2 \geq -\beta_R \|d\|_2^2,$$

since $2 \int_{I_R} \sinh^2(x/2) dx = 2 \sinh(R/2) - R$. Summation proves the bound. $\square$

There is no need to replace the harmonic sum by its lower integral bound. For the certificate on the window $1/9$, one also checks $(2N + 1/2)h < \pi$, so that each retained gamma bound is nonnegative.

For any fixed R , there is a partition that makes this bound positive. Indeed, if $0 < h \leq 1$ and $N = \lfloor 1/h \rfloor$, then

$$4 \sum_{n=0}^N \frac{1}{4n+1} \geq \log(4N+5) \geq \log(1+4/h), \quad \frac{h^2(N+1)(2N+1)}{\pi^2} < \frac{2}{3}.$$

The first term diverges as $h \downarrow 0$; the prime, scalar, and polar terms remain fixed by R . This preserves the distinction between coercivity of the details and the sign of the full form.

19.3 The exact Schur complement has finite size

Let D be the operator associated with the restriction of the form to V_D . If $\eta > 0$, then $D \geq \eta I$ on QH_R , D^{-1} exists, and $\|D^{-1}\| \leq \eta^{-1}$. We do not presume that L_R preserves the detail space.

The indicator functions also belong to the operator domain of L_R . The expression for their gamma image is given in the next section: it has only logarithmic singularities, which belong to $L^2(I_R)$. This allows us to define, without improperly extending a form functional,

$$A = U^* L_R U, \quad B = Q L_R U : \mathbb{C}^M \longrightarrow QH_R. \quad (19.9)$$

The inner product is linear in the first variable, as in the manuscript. The matrices are written so that $\mathcal{W}_R(Ua, Ua) = a^* A a$. The block identity is

$$\mathcal{W}_R(Ua + d, Ua + d) = a^* A a + 2 \operatorname{Re} \langle Ba, d \rangle + \mathfrak{d}[d, d], \quad (19.10)$$

where $\mathfrak{d}$ is the closed form of D .

Theorem 19.2 (Exact elimination of the detail space). *The matrix*

$$S = A - B^* D^{-1} B \quad (19.11)$$

satisfies the exact identity

$$\boxed{\mathcal{W}_R(Ua + d, Ua + d) = a^* S a + \mathfrak{d}[d + D^{-1} B a, d + D^{-1} B a].} \quad (19.12)$$

In particular, $\mathcal{W}_R \geq 0$ on V_R if and only if $S \geq 0$. Its negative index equals that of S .

Proof. For $z = D^{-1} B a \in \operatorname{Dom} D$, $\mathfrak{d}[z, d] = \langle Ba, d \rangle$. Expanding the second term of (19.12) and using $\mathfrak{d}[z, z] = a^* B^* D^{-1} B a$ gives the equality. Sufficiency is immediate. For necessity, choose $d = -D^{-1} B a$. The triangular change $(a, d) \mapsto (a, d + D^{-1} B a)$ is invertible on $\mathbb{C}^M \oplus V_D$. The resulting second form is positive definite; the negative directions are exactly those of the finite matrix. $\square$

This theorem converts the finite-negative-index result into an exact matrix. The matrix still contains D^{-1} : finite size does not mean that its entries have already been calculated. The next section provides a way to enclose them with controlled residuals.

19.4 Explicit images and certifiable bounds for the corrections

Set $M(x) = \int_x^\infty \mu(a) da$, for $x > 0$. For an indicator function $u = h^{-1/2} \mathbf{1}_{(a,b)}$, its gamma image, restricted to I_R , is almost everywhere

$$(L_\Gamma u)(x) = h^{-1/2} \begin{cases} M(x-a) + M(b-x), & a < x < b, \\ -M(a-x) + M(b-x), & x < a, \\ -M(x-b) + M(x-a), & x > b. \end{cases} \quad (19.13)$$

The equality is obtained by integrating the two translations: inside the cell there remains one tail for each endpoint, and outside there remains the integral over the unique interval of translations that meets the cell. It can first be proved with the regulator $a_{\text{traslado}} \geq \epsilon$; convergence in $L^2(I_R)$ of the logarithmic expression justifies its weak form limit. The other parts are

$$L_{\text{primo}} u = - \sum_{k \log p \leq R} w_{p,k} M_R(T_{k \log p} + T_{-k \log p}) u, \quad (19.14)$$

$$L_{\text{polar}} u(x) = 2 \int_a^b h^{-1/2} \cosh((x-y)/2) dy, \quad (19.15)$$

$$L_R u = L_\Gamma u + c_\Gamma u + L_{\text{primo}} u + L_{\text{polar}} u.$$

All discontinuities and singularities of (19.13)–(19.15) lie in a finite list of endpoints and endpoints translated by expressed clocks. Thus, A_{ij} and

$$G_B = B^* B, \quad (G_B)_{ij} = \langle Be_j, Be_i \rangle \quad (19.16)$$

are calculated by explicit one-dimensional integrals. In this last formula, e_j denotes the standard coordinate vector of $\mathbb{C}^M$, not the indicator function in H_R . The initial bound is

$$A - \eta^{-1} G_B \leq S \leq A. \quad (19.17)$$

This is an operator inequality with the cross terms preserved, not a replacement of G_B by its diagonal entries.

Theorem 19.3 (Residual certification of the complement). *For any finite approximation operator $Z : \mathbb{C}^M \rightarrow \text{Dom } D$, define*

$$E = B - DZ, \quad C_Z = Z^* B + B^* Z - Z^* DZ. \quad (19.18)$$

Then

$$B^* D^{-1} B = C_Z + E^* D^{-1} E, \quad (19.19)$$

and therefore

$$\boxed{A - C_Z - \eta^{-1} E^* E \leq S \leq A - C_Z.} \quad (19.20)$$

Proof. Substitute $B = DZ + E$ and expand the product. All terms are defined because the columns of Z belong to $\text{Dom } D$. The bound uses $0 < D^{-1} \leq \eta^{-1} I$. $\square$

An effective choice of Z consists in solving the Galerkin system on step-function details of a nonadic refinement. Its columns are finite combinations of indicator functions and have images (19.13)–(19.15), so E and its Gram are explicitly integrable. The Galerkin system is not declared an exact solution: the term $\eta^{-1} E^* E$ retains its error on the infinite complement.

To certify these integrals without losing the singular endpoints, split

$$M(x) = -\frac{1}{2} \log x + \log 2 + \frac{\pi}{4} + r(x), \quad r(0) = 0, \quad r'(x) = \frac{1}{2x} - \mu(x). \quad (19.21)$$

On $0 < x \leq R$,

$$-\frac{1}{4}e^{R/2} \leq r'(x) \leq \frac{1}{4}, \quad |r'(x)| \leq \frac{1}{4}e^{R/2}. \quad (19.22)$$

Indeed, $\mu(x) = e^{x/2}/(2 \sinh x)$ and $x \leq \sinh x \leq xe^x$ give $e^{-x/2}/(2x) \leq \mu(x) \leq e^{x/2}/(2x)$. The inequalities $1 - e^{-y} \leq y$ and $e^y - 1 \leq ye^y$ prove the bounds. We do not assert that r' has constant sign on an arbitrarily large window. We integrate $\log x$ and $\log^2 x$ analytically on the small endpoint segments; the remainder r is bounded by its Lipschitz constant. Away from the endpoints, subdivided interval quadratures and the exponential series for M are used. For example,

$$\int_0^\delta \log^2 x \, dx = \delta[(\log \delta)^2 - 2 \log \delta + 2]. \quad (19.23)$$

Products of two singularities at distinct points are handled by separating their neighborhoods; for a shared endpoint, the same quadratic integral is used. The resulting intervals are inserted into (19.20), and positivity is verified by an LDL^* decomposition with strictly positive interval pivots or a perturbation bound for a rational matrix. This specifies a finite certificate with control of the infinite complement; no mesh without a tail bound is used.

19.5 Coercivity on a window of length 1/9

Take $R = h = 1/9$, a single cell, and $u = R^{-1/2} \mathbf{1}_{I_R}$. This is a full interval, not nine intervals separated by gaps. The detail section is all of $V_R \cap \{u\}^\perp$.

19.5.1 Margin of the infinite block

In (19.5), take $N = 13$. There are no prime terms because $R < \log 2$. This gives

$$\eta = 4 \sum_{n=0}^{13} \frac{1}{4n+1} - \frac{(1/9)^2 14 \cdot 27}{\pi^2} + c_\Gamma - \beta_{1/9} > 1. \quad (19.24)$$

The interval certificate gives $1.00363976543713133046 < \eta < 1.00363976543713133048$. The condition $(53/2)/9 < \pi$ also guarantees the positivity of each retained gamma bound. The sum is preserved; replacing it by the bound $\log(1 + 4/h) - 2/3$ would lose the necessary margin here.

19.5.2 Mean block

Inside the window, writing $x = R(t - 1/2)$,

$$L_R u(x) = R^{-1/2} g(t), \quad g(t) = c_\Gamma + M(Rt) + M(R(1-t)) + 8 \sinh(R/4) \cosh(R(t - 1/2)/2). \quad (19.25)$$

Thus $A = \int_0^1 g(t) \, dt$, and $\|B\|^2$ is the variance of g on the unit interval. Applying Tonelli to M gives

$$A = c_\Gamma + \frac{2}{R} \sum_{n=0}^{\infty} \frac{1 - e^{-(2n+1/2)R}}{(2n+1/2)^2} + \frac{32}{R} \sinh^2(R/4). \quad (19.26)$$

If the first L terms are retained, the positive tail is at most $\lambda_L^{-2} + (2\lambda_L)^{-1}$, with $\lambda_L = 2L + 1/2$. The first term of the tail is separated, and the remainder is bounded by the integral of a decreasing function. For $L = 1024$, the certificate yields

$$0.9723307136651774 < A < 0.9767284617331572, \quad A > 97/100. \quad (19.27)$$

A tighter approximation is not needed for the proof.

19.5.3 Coupling between means and details

On $0 < x \leq 1/9$, the estimate in (19.21) improves to

$$-9/10 \leq r'(x) \leq 0. \quad (19.28)$$

Indeed, for $0 < x \leq 1/9$,

$$\frac{\sinh x}{x} = \sum_{k \geq 0} \frac{x^{2k}}{(2k+1)!} \leq \frac{1}{1-x^2} \leq 1 + \frac{x}{2} \leq e^{x/2}. \quad (19.29)$$

The intermediate inequality is equivalent to $1 - 2x - x^2 \geq 0$, which holds on that interval. Thus $\mu(x) \geq 1/(2x)$. On the other hand,

$$\mu(x) - \frac{1}{2x} \leq \frac{e^{x/2} - 1}{2x} \leq \frac{1}{4}e^{x/2} \leq \frac{1}{4(1-1/18)} = \frac{9}{34} < \frac{9}{10}. \quad (19.30)$$

We retain the less sharp bound $9/10$ used by the certificate. The function $q(t) = r(Rt) + r(R(1-t))$ is symmetric about $1/2$, and its Lipschitz constant is at most $9R/10$. Its oscillation is at most $9R/20 = 1/20$; hence its standard deviation is at most $1/40$. This last assertion follows from $\text{Var}(X) \leq (\sup X - \inf X)^2/4$: subtract the midpoint of the range and use the fact that the mean minimizes the squared error.

The singular part $v(t) = -\frac{1}{2} \log(t(1-t))$ has mean 1 and variance $1 - \pi^2/12$. To verify this, integrate $\log t$, $\log^2 t$, and the positive series for $\log t \log(1-t)$, which gives $\int_0^1 \log t \log(1-t) dt = 2 - \sum_{n \geq 1} n^{-2}$. The identity $\sum n^{-2} = \pi^2/6$ can be obtained by applying Parseval to the sine series of x on $(-\pi, \pi)$: its coefficients are $2(-1)^{n+1}/n$, and $(1/\pi) \int_{-\pi}^{\pi} x^2 dx = 2\pi^2/3$. This is a subsequent functional recognition of the same π , not a selection of its generator.

The polar part has oscillation $8 \sinh(R/4)[\cosh(R/4) - 1]$, and standard deviation at most half that value. The triangle inequality in the space of centered functions gives, for the full coupling,

$$\|B\|^2 \leq \left[\sqrt{1 - \pi^2/12} + \frac{1}{40} + 4 \sinh(R/4)(\cosh(R/4) - 1) \right]^2 < \frac{1}{5}. \quad (19.31)$$

The scalar term and all constants independent of t are removed by centering, not by a supposed disappearance of the cross terms. The interval evaluation of the bounding expression is $0.19926357337268538427 \dots < 1/5$.

19.5.4 Positive Schur complement and coercivity of the entire form

Theorem 19.4 (Coercivity on the full nonadic window). *On $I_{1/9} = (-1/18, 1/18)$, every function in the joint domain $V_{1/9}$ satisfies*

$$\mathcal{W}_{1/9}(f, f) \geq \frac{1}{2} \|f\|_2^2.$$

The bound holds on every translated interval of the same length and on all its interior refinements.

Proof. From (19.17), (19.24), (19.27), and (19.31),

$$S = A - B^* D^{-1} B > \frac{97}{100} - \frac{1}{5} = \frac{77}{100} > 0. \quad (19.32)$$

Moreover, for $f = au + d$,

$$\mathcal{W}_R(f, f) - \frac{1}{2} \|f\|_2^2 \geq (A - \frac{1}{2})|a|^2 - 2\|B\| \|a\| \|d\|_2 + (\eta - \frac{1}{2})\|d\|_2^2. \quad (19.33)$$

The scalar matrix on the right-hand side has positive diagonal and determinant strictly greater than $\frac{47}{100} \cdot \frac{1}{2} - \frac{1}{5} = \frac{7}{200} > 0$. This proves $\mathcal{W}_R(f, f) \geq \frac{1}{2} \|f\|_2^2$ for every $f \in V_R$. The argument controls all the infinitely many details; it does not replace them by finitely many indicator functions. Translation preserves the correlations and multiplies the polar evaluations by $e^{t/2}$ and $e^{-t/2}$, whose factors cancel in the cross products. It therefore also preserves the full form and the bound. $\square$

19.6 Refinement, double circle, and the two moments

Each subdivision into nine preserves exactly $u_{\text{padre}} = \frac{1}{3} \sum_{r=0}^8 u_r$, and the recomposition of functions occurs before applying J_+ and J_- . Therefore, the Grams of all nonadic indicator functions inside the window satisfy $G_n \geq \frac{1}{2}I$ and $V_n^* G_{n+1} V_n = G_n$. The eight details per predecessor and all their cross terms are preserved. This is uniformity in depth on this window, not a uniform result as R increases.

For a general partition, the exact Schur matrix does not automatically obey the same compression as the raw Gram. If one refines and expresses the fine Schur complement in old-mean/new-detail coordinates, the old Schur complement is obtained by also eliminating that new detail. This is associativity of minimization in (19.12). Asserting $S_{\text{padre}} = V^* S_{\text{hijos}} V$ would omit that second correction.

On the certified window, the connector already constructed, $\mathcal{C}_R = J_{-,R} L_R^{\text{cola}}$, where L_R^{cola} is the cotrace (18.29) applied to the gamma component, satisfies $\mathcal{C}_R J_{+,R} = J_{-,R}$. Its restriction to the closure of the effective image $\mathcal{M}_{+,R}$ is contractive, because

$$\|J_{-,R} f\|^2 \leq \|J_{+,R} f\|^2. \quad (19.34)$$

Contractivity is now proved by means of Schur; it is not defined from a putatively positive Gram. Its defect $\Delta_R = (I - \mathcal{C}_R^* \mathcal{C}_R)^{1/2}$ is then taken on that effective space, and we construct

$$\Xi_R f = J_{-,R} f \oplus \Delta_R J_{+,R} f. \quad (19.35)$$

If P_- is the projection onto the first summand,

$$\Xi_R^* \Xi_R = J_{+,R}^* J_{+,R}, \quad \Xi_R^* P_- \Xi_R = J_{-,R}^* J_{-,R}. \quad (19.36)$$

These are equalities of forms on the joint domain, not products of unbounded operators on all of L^2 . They give the two moments in the dilation codomain. An additional identification with a specific T_N of the double circle requires the intertwining with that operator: an arbitrary projection is not identified with T_N merely because the two moments are available.

19.7 Reproducibility checks and conditions for extension

The verification program retained in Appendix 21 certifies the constants with interval arithmetic and explicit tail bounds. It uses the internal prefix of π with its hashes and the previously retained rational producer of γ and $\log 2$. It does not use `mpmath.pi`, `mpmath.euler`, zero tables, or floating-point arguments as inputs.

Both runs satisfy the same 21 interval checks. The receipts retain exact rational endpoints, precision, library version, provenance, and hashes. The adversarial check $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has two positive diagonal blocks and Schur complement -3 : it rejects using their signs alone.

Substantive falsifiers: omitting the negative polar term; interchanging cell means with point sampling; losing a prime translation on a larger window; replacing the Schur complement by A ; accepting Galerkin without $E^* E / \eta$; or conflating all interior refinements with all windows. None is part of this certificate.

The next concrete extension is a window $R > \log 2$, with a nonadic partition whose detail block is coercive. There, (19.20) must be evaluated, including at least the clock $\log 2$. The present result asserts neither that new sign, nor global Weil positivity, nor RH. Nor does it infer a negation of the corpus from this local scope: it provides an exact elimination, a certifiable procedure, and a first settled case extending the preceding baseline case.

20 Conclusions

On each fixed window, the Weil form admits an exact reduction to a finite problem that preserves its infinite-dimensional complement. The space of zero-mean details can be made coercive by refinement; its elimination produces a Schur complement with an equivalent positivity criterion and the same negative index as the complete form. On the window $(-1/18, 1/18)$, joint control of averages, details and couplings proves

$$\mathcal{W}(f, f) \geq \frac{1}{2} \|f\|_2^2, \quad f \in V_{1/9}.$$

The assertion comprises the closure of smooth compactly supported test functions in the joint readout norm. It is a result on an infinite-dimensional functional domain, stable under translation and interior refinement.

Sign of the form and preservation of the infinite complement

The identity

$$\mathcal{W}_R(Ua + d, Ua + d) = a^*(A - B^*D^{-1}B)a + \mathfrak{d}[d + D^{-1}Ba, d + D^{-1}Ba]$$

explains what the reduction preserves. The second term is positive and contains all details; the first retains their interaction with the averages through $B^*D^{-1}B$. Theorem 19.2 proves that positivity on V_R is equivalent to positivity of this matrix. Its finite size does not replace calculation of its entries: the residual certification in (19.20) supplies bounds retaining the complement's error.

On the window of length $1/9$, the bounds $D > I$, $A > 97/100$ and $\|B\|^2 < 1/5$ produce the stated coercive margin. This local case contains no active prime translation because $1/9 < \log 2$. Invariance of the complete form nevertheless carries the same bound to each subspace $C_a^n \tau_t C_c^\infty((-1/18, 1/18))$, with $a > b > 1/2$. Its elements may have exponential tails, and their prime contribution is preserved with the full convergent sum. The estimate applies to each entire collection, not merely to a selected test function.

The other positive domains retain their own content: the specified assembly of nine intervals controls its cross products and all refinements, while an arbitrary window admits a coercivity threshold for cell details. Distinguishing these domains identifies both the proved margin and the coupling determining the sign of a broader composition.

Arithmetic determines the modes and their readouts

The arithmetic antecedent is constructed before spectral readout. The two APP sheets supply the local operations and their quotients; the native product propagates these data through words with unique normal forms. Product decompositions select irreducibles, and evaluation identifies them with positive primes. The operator realization of the selector counts nontrivial decompositions. Its positivity is a squared norm with this concrete arithmetic meaning.

Factorization subsequently organizes mode occupations and their clocks. The Euler product and prime-power moments arise from traces of that representation. The limit of the cyclic heat trace and its Mellin transform produce the completed function, with its gamma and polar factors and functional equation. Finite parts, regularized evaluations and remainder bounds preserve the normalizations required to compose these readouts.

The unity of this route lies in the enriched APP-TRIT-TPK states and the joint structure of the continuum: route, orientation, sheets, memory and incidences are transmitted through compatible restrictions and refinements. Real evaluation obtains a result of this construction; history and boundary retain the information needed by its subsequent realizations.

The composition laws make this continuity precise. The product of two moments is realized on pairs of modes, and its coefficients are grouped by divisor convolution; diagonal composition retains a single mode. Spectral normalization transports the entire Laurent expansion, including the polar term, through an invertible triangular affine transformation. The functional equation relates negative even derivatives to positive odd values and, in particular, establishes $Z(3) = 4\pi^2 \log A_2$. Finally, irreducible-mode cutoffs provide explicit enclosures complementary to depth cutoffs. Each identity retains the operation producing it and the domain in which it is proved.

Boundary, vacancies and double circle

Reconstruction of the interior from the lifted boundary in the multiaffine chart gives an explicit instance of preservation. The sum-product difference retains both the residual datum and the quotient; the readouts 72, 27, 77, 22 and the completion $729 + 271$ retain their operations and domains. The center (5, 5) and its conjugate vacancies preserve sum and multiplicative residue while the quotient changes. The regional signature 55555 requires three continuation sheets, and the capacity calendar produces lengths 21, 22 and separations 6, 7 with an exact balance.

The double circle carries this preservation into readout transport. Nine-phase compression produces the complement $(8z, -z, \dots, -z)/27$ and the factor $8/81$. The modal ratio $q = 5/9$ determines the memory moments $q^{|n|}$; their triangular factorization proves positivity in every dimension and the determinant $(56/81)^N$. The loss register preserves the terminal and cross products. Refinement may lead to a weak atomic limit of states, whose representation is distinguished from the regular shift in which the initial vectors were constructed.

For the arithmetic readouts, the gamma tail reconstructs the test function and the rows of both columns. The resulting connector preserves exactly $J_+^* J_+ - J_-^* J_-$, with the prime, gamma and polar terms. Its action formulates the energy problem on an explicit operator. Invariance under C_a also preserves all cross moments of the form, not only its diagonal evaluations.

Global condition and relation to the Riemann hypothesis

Weil's global criterion requires positivity on the entire test-function domain. In the two-column formulation, the exact condition is contractivity of the reachable transfer on the closure of the positive image, with compatibility as windows enlarge. In the circular formulation, it is expressed by the identity

$$m_n^{\text{ar}}(f, g) = \langle U_{\text{mem}}^n \mathcal{E}f, \mathcal{E}g \rangle_{\mathcal{K}_{\text{lim}}},$$

for all mixed moments and a collection covering the domain through the stated limits. The energy identity and intertwining must hold jointly; positivity of the regular memory Gram form does not replace them.

The proofs in this article establish reconstruction, invariance and coercivity on the stated domains. The global identification above remains a hypothesis of the spectral results employing it and does not follow from positivity of each transported collection separately. The article therefore does not conclude a resolution of the Riemann hypothesis. Its functional contribution is precise: preservation of arithmetic structure and memory is expressed through recoverable operators, exact reductions and coercive bounds, with the mixed products and domains governing any global extension kept explicit.

21 Provenance of the results and reproduction of the extension

The exposition starts from the APP-TRIT-TPK kernel and maintains its correlative construction of the continuum. Regional readouts, the native product and irreducible clocks precede their analytic realizations. Integration of centre, boundary and memory brings together results of the HMT-MD corpus with the compositions developed for the double circle. Provenance is recorded result by result, so that a consolidated formulation or an additional certificate does not retrospectively make the structure it uses new.

21.1 Dependencies brought together in the main text

The kernel comprises the common definitions and operations, together with its two figures. The definitions, statements and proofs of the five preceding developments are preserved in full.

The central chart, its lifted boundary and transport signatures are developed in Section 10. The distinction between the central point $(5, 5)$, product signature 555555 and the sheets distinguished by its transport is preserved in Definition 10.2 and Proposition 10.3. The capacity calendar and its two return levels are proved in Theorems 10.7 and 10.8. The chart geometry is not identified with the complete history: its recovery uses the multiaffine law and quotients preserved at the boundary.

Double-circle transport and memory moments are brought together in Section 17. The gamma-tail readouts of Section 18 reconstruct the columns with their domains and cross-products. Section 19 establishes the estimate on a complete window and its transport to subspaces with tails. Conservation equalities and positivity of a form are distinct statements: each result of this extension preserves the domain in which it is proved.

The formal antecedents required for these compositions are developed in Appendices A–C: emission rules and compatible histories, regional readouts and nonadic return, and the reversible dodecaphasic register. The equations and proofs used are thereby presented with the domains and representation data required by each result.

21.2 Reproduction order and scope of the checks

The edition preserves the antecedents, the five preceding developments and the focal checks accompanying their identities. Its compilation attests a legible material realization of the text; documentary preservation and computational checks do not thereby certify every result of a project. The extension's contribution consists of incorporating the following compositions with their proofs.

The folder accompanying the PDF preserves the $\text{\LaTeX}$ sources, the five developments in Markdown, the programs and JSON receipts. Reproduction follows the mathematical dependencies of the objects:

1. Construct the centre and boundary readouts while preserving residue and quotient, and check the partition of the 324 seeds under forward and half-turn transports.
2. Evaluate integer parts of the vacancy calendar through rational brackets. Bounds with remainders precede the digits and fix certification of each position.
3. Check seam balances, memory terminals and moment factorizations; preserve cross-terms when forming the joint matrices.
4. Evaluate local bounds for the gamma tail and Schur complement, with the domains of the estimates printed in their statements.
5. Compile the manuscript and compare common sources, transferred equations, references and page layout.

Rational and interval checks verify the finite identities and bounds they declare. Functional proofs establish the corresponding universal statements. File preservation and successful compilation are separate editorial checks; neither is presented as a substitute for a global proof.

Ordinary and optimized execution of the programs uses explicit conditions: disabling the interpreter's assertions does not alter the acceptance criterion. The delivery's instruction file specifies the commands and environment dependencies. Earlier originals are preserved separately and are not overwritten by this extension.

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